Chapter 4: System Modeling & Unified Architecture

Lecture Outline & Key Concepts

- **4.1 Foundations of System Modeling**: Why we model software, the limits of natural language, the Blind Men and an Elephant metaphor, and the 4 fundamental modeling perspectives (External, Interaction, Structural, and Behavioral).
- **4.2 The Unified Modeling Language (UML) & Unified Process**: Historical evolution from the 1990s 'Method Wars', the Three Amigos (Grady Booch, Jim Rumbaugh, Ivar Jacobson), the OMG standard, the 14 UML 2.5 diagrams, and the principles of architecture-centric, use-case-driven iterative design.
- **4.3 Functional & Use Case Models**: High-level system boundary definition, actors vs. stakeholders, active-verb naming semantics, the Relationship Matrix (`<<include>>` mandatory sub-processes vs. `<<extend>>` conditional extension points vs. actor generalization), safe nesting limits, University Enrollment case study, the standard 10-field tabular Use Case Description, seven deadly modeling sins, and AI prompt engineering for use-case synthesis.
- **4.4 Structural & Class Models**: The 3-compartment class anatomy (Name, Attributes, Operations), visibility specifiers (`+`, `-`, `#`, `~`), parameter directionality, the Taxonomy of Relationships (Association, Aggregation, Composition, Generalization, Dependency), navigability and multiplicity constraints, association classes, the Food Delivery Domain Model case study, expert checklists, and AI prompt engineering for domain class models.
- **4.5 Interaction & Sequence Diagrams**: Modeling runtime object collaboration across time, 2D canvas (horizontal participants vs. vertical lifelines), activation focus bars, the 5 messaging types (Synchronous, Asynchronous, Return, Create, Destroy), Combined Fragments (`alt`, `opt`, `loop`, `par`), Hotel Reservation case study, the Boundary-Control-Entity (BCE) architectural pattern, and AI prompt engineering for sequence orchestration.
- **4.6 Process & Activity Diagrams**: Modeling procedural workflows and token flows, initial/final nodes, actions, Object Flows and state pins, deep dive into Fork/Join (parallel concurrency synchronization) vs. Decision/Merge (Boolean mutually exclusive branching), Swimlanes/Partitions across organizational roles, interruptible activity regions and exception handling, Food Delivery Fulfillment case study, and AI prompt engineering for business workflows.
- **4.7 Behavioral & State Machine Models**: State-dependent behavior (same event, different results), the 4 transition triggers (Signal, Call, Time, Change), transition syntax (`Trigger [Guard] / Effect`), Actions (instantaneous, atomic) vs. Activities (ongoing, interruptible), `entry/` and `exit/` mechanics, Composite States and Shallow/Deep History pseudostates (`H`, `H*`), orthogonal concurrency regions, HVAC & Food Delivery Order Lifecycle case study, and AI prompt engineering for state machines.
- **4.8 Text-Based Architecture Modeling with PlantUML**: The Architecture-as-Code paradigm, overcoming GUI drag-and-drop bottlenecks, Git version control and PR diffing, core syntax matrix across all 5 UML diagram types, skinparam styling, holistic architecture synthesis, and modern Generative AI workflows.
- **4.9 Model Selection, Review Quiz & References**: Comprehensive 6-model decision matrix, 10 conceptual review questions, and foundational literature.

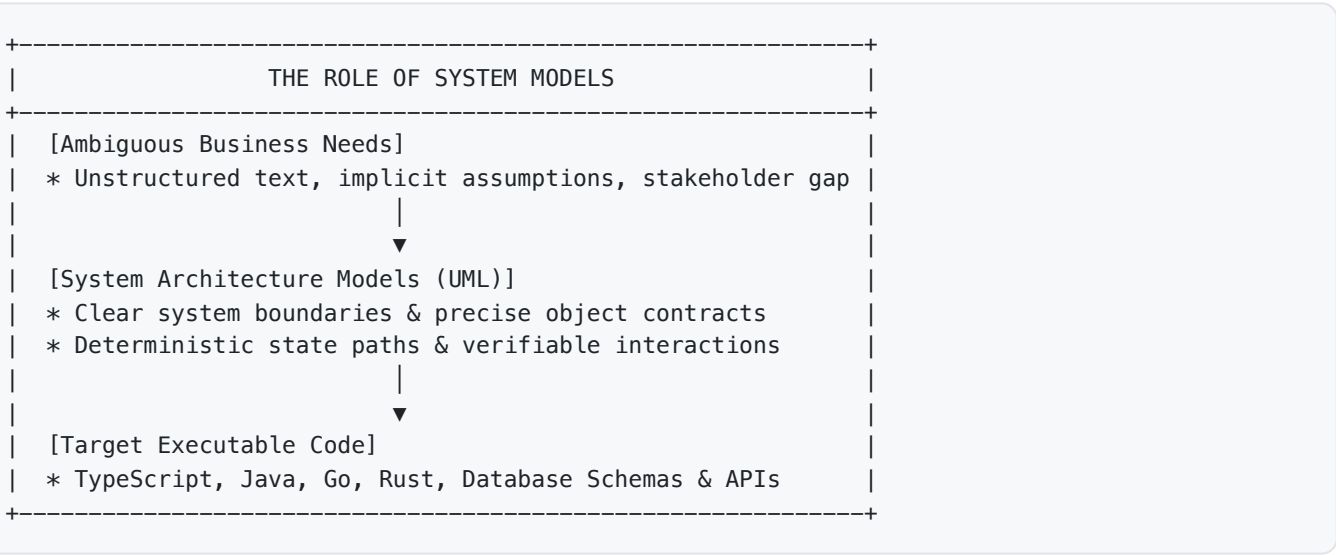
- **Appendix: Solutions & Detailed Explanations to All 8 Interactive Concept Checks (CCQ 1–8).**

4.1 Foundations of System Modeling

In software engineering, constructing a modern, multi-tier system without an architectural model is akin to erecting a forty-story skyscraper without blueprints. While natural language requirements (such as user stories and business paragraphs) capture human intentions, they suffer from unavoidable ambiguity, linguistic inconsistency, and hidden cognitive gaps. System modeling provides the formal, graphical abstractions required to design, communicate, and verify software architectures before committing expensive development resources.

4.1.1 What is a Model?

A **model** is an abstraction of a system that emphasizes relevant architectural details while intentionally suppressing non-essential complexities. Just as an electrical engineer relies on schematic circuit diagrams rather than microscopic physical wire scans, a software engineer relies on architectural diagrams to reason about component interactions, data dependencies, and state lifecycles.



4.1.2 The Blind Men and an Elephant: Four Modeling Perspectives

No single diagram can capture the totality of a non-trivial software system. Attempting to force database schema, network protocols, user interactions, and UI workflows into a single visual diagram results in unreadable architectural chaos. Instead, software engineers view the system through **four distinct, orthogonal perspectives**:

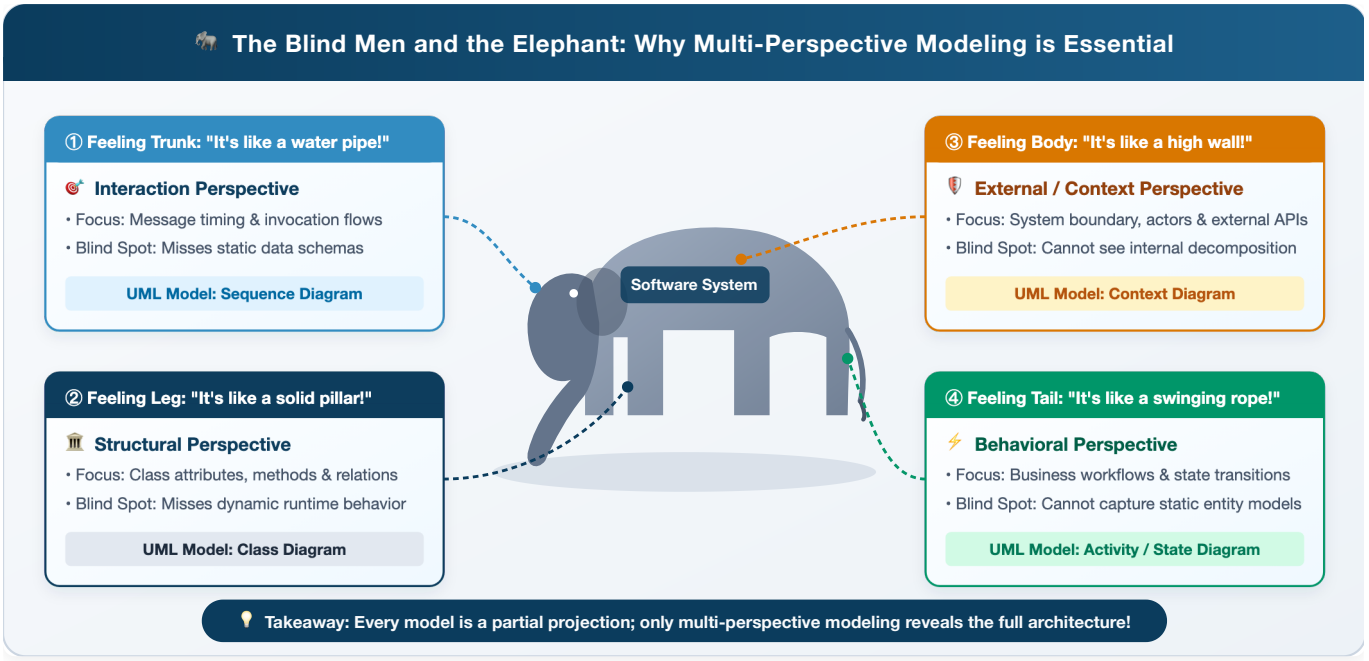


Figure 4.1.1: The Blind Men and the Elephant Metaphor. Just as each observer experiences only one facet of the elephant, each UML diagram captures a specific perspective of the system architecture.

1. **External Perspective (Context & Boundaries):** Models the system in its environment, establishing the perimeter boundary separating internal software responsibilities from external human users, IoT hardware, and third-party SaaS APIs.
2. **Interaction Perspective (Dynamic Collaborations):** Models how external actors communicate with the system, and how internal components pass runtime messages across memory boundaries and network sockets over time.
3. **Structural Perspective (Static Organization):** Models the static layout of domain entities, class inheritance hierarchies, interfaces, data attributes, and relational associations independent of execution time.
4. **Behavioral Perspective (State & Event Dynamics):** Models how the system responds to internal and external events, capturing dynamic state lifecycles, concurrency, and conditional workflow branches.

🤷 Concept Check (CCQ 1) — Four System Modeling Perspectives

Question

A software architect wants to define **how domain entity data is structured and how classes inherit and associate with each other**, completely independent of runtime execution order. Which modeling perspective should the architect adopt?

- A. External Perspective
- B. Interaction Perspective
- C. Structural Perspective
- D. Behavioral Perspective

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.2 The Unified Modeling Language (UML) & Unified Process

Before the mid-1990s, the software industry suffered through what historians call the **'Method Wars'**. Dozens of competing object-oriented notations existed—most notably Grady Booch's Booch Method, Jim Rumbaugh's Object Modeling Technique (OMT), and Ivar Jacobson's Object-Oriented Software Engineering (OOSE). Each methodology had its own proprietary notation for classes, associations, and inheritance, forcing engineers to relearn diagram syntax across employers and projects.

4.2.1 The Three Amigos and the Birth of UML

In 1994, Rational Software brought together the three leading pioneers—**Grady Booch**, **James Rumbaugh**, and **Ivar Jacobson**, affectionately known across the engineering world as **The Three Amigos**.

Pioneer	Original Method	Core Contribution to UML
Grady Booch	Booch Method	Object design, macro-level structural architecture
James Rumbaugh	OMT (Object Modeling Technique)	Rich static relationships, statecharts, data analysis
Ivar Jacobson	OOSE (Object-Oriented Software Eng.)	Use Cases (1986), actor goals, user-driven requirements

By synthesizing the best elements of their respective notations, the Three Amigos published UML 1.0 in 1997, which was subsequently adopted by the **Object Management Group (OMG)** as an open international standard (ISO/IEC 19505).

4.2.2 The 14 UML Diagrams: Structure vs. Behavior

UML 2.5 defines 14 official diagram types divided symmetrically into two overarching taxonomic branches:

Classification Branch	UML 2.5 Diagram Types	Primary Architectural Focus
Structure Diagrams *(Static View)*	• **Class Diagram** (Core) • Object Diagram • Component Diagram • Deployment Diagram • Package Diagram • Composite Structure • Profile Diagram	Static organization of classes, interfaces, modules, namespaces, and runtime hardware nodes.
Behavior Diagrams *(Dynamic View)*	• **Use Case Diagram** (Scope & Actors) • **Activity Diagram** (Workflow & Logic) • **State Machine Diagram** (Lifecycles)	Dynamic execution flow, user business goals, operational processes, and reactive state transitions.
Interaction Diagrams *(Behavioral Sub-branch)*	• **Sequence Diagram** (Timing & Calls) • Communication Diagram • Timing Diagram • Interaction Overview	Inter-object message exchanges over time, invocation protocols, and distributed call chains.

> 📌 **Engineering Pragmatism (The 80/20 Rule)**: In modern enterprise software development, five core diagram types deliver over 80% of architectural value: **Use Case Diagrams**, **Class Diagrams**, **Sequence Diagrams**, **Activity Diagrams**, and **State Machine Diagrams**.

🙋 Concept Check (CCQ 2) — UML Pioneers & The Three Amigos

Question

Which of the "Three Amigos" was specifically renowned for inventing **Use Cases (1986)** to anchor software architecture to tangible user goals?

- A. Grady Booch
- B. James Rumbaugh

- C. Ivar Jacobson
- D. Martin Fowler

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.3 Functional & Use Case Models

Use case modeling is the primary technique for establishing the functional scope of a software system. Invented by Ivar Jacobson, use cases shift the focus from internal algorithmic mechanics to the concrete business goals of external users.

4.3.1 Visual Anatomy of a Use Case Diagram

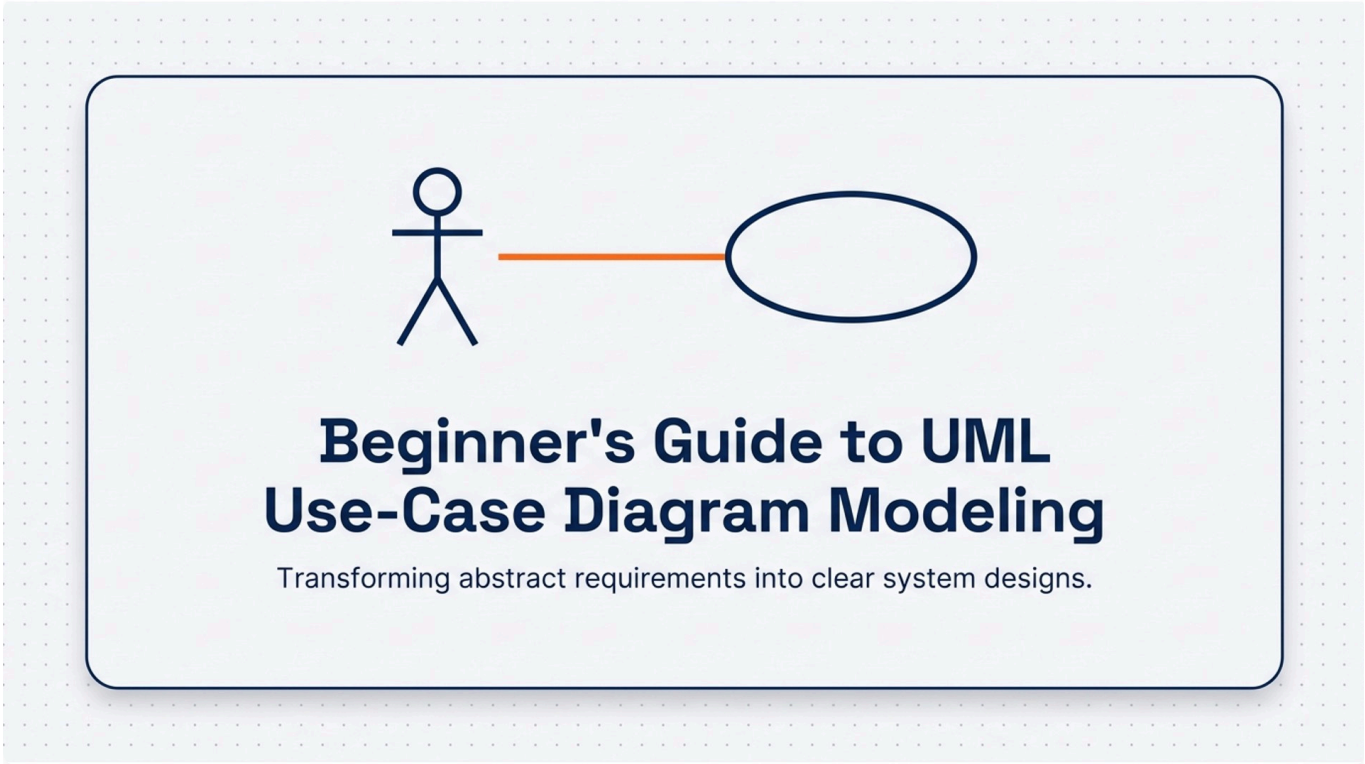


Figure 4.3.1: The Anatomy of Use-Case Diagram Modeling. The diagram clearly separates external actors from internal system capabilities bounded by the system perimeter.

A standard UML Use Case Diagram comprises four foundational visual elements:

1. **System Boundary (Rectangle):** A labeled box demarcating what is inside the software system (scope of development) versus what resides in the external environment.
2. **Actors (Stick Figures):** Entities outside the system that interact with it. Actors can be human users, hardware devices, or external cloud services.
3. **Use Cases (Ovals):** Discrete, goal-oriented sequences of interactions that yield an observable result of measurable value to an actor.
4. **Association Lines (Solid Lines):** Communication paths connecting actors to the use cases they participate in or trigger.

4.3.2 User Goals vs. Technical Decomposition

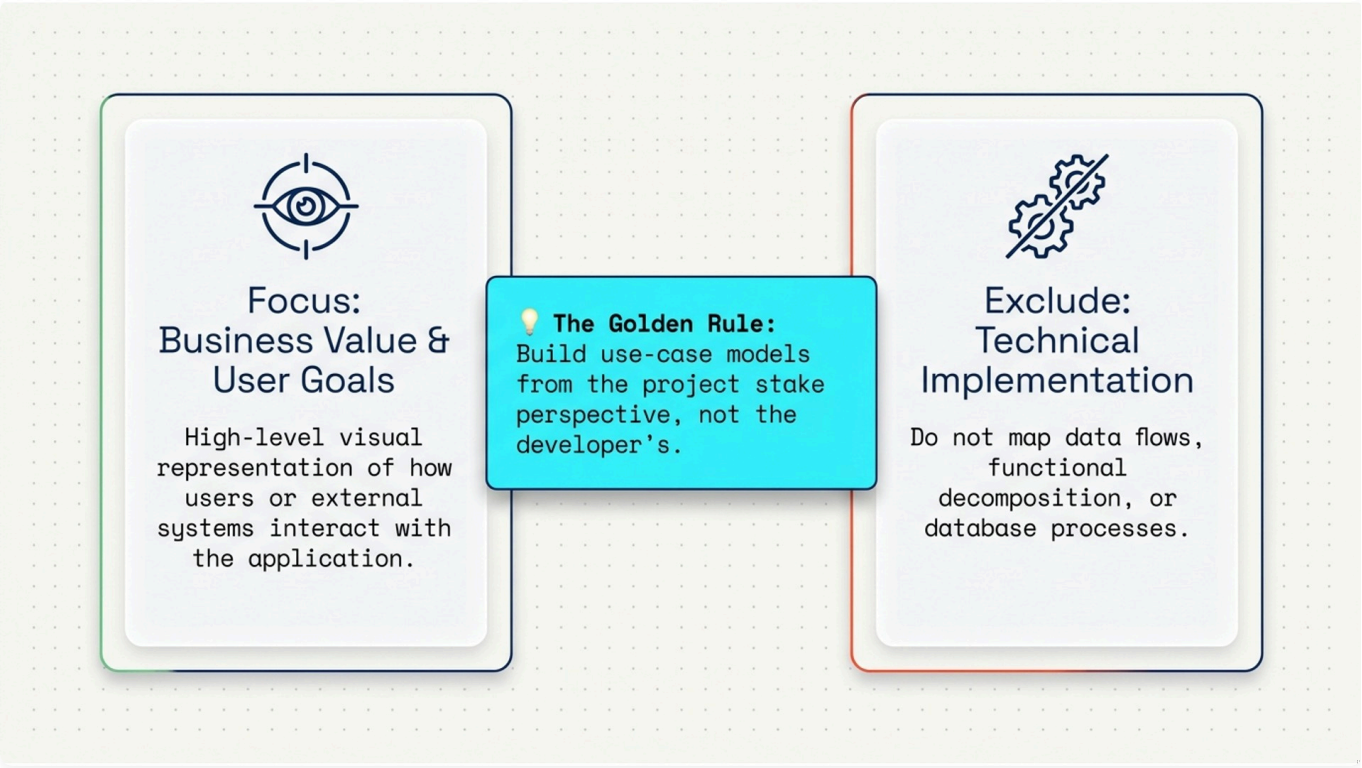


Figure 4.3.2: User Goals vs. Technical Steps. Use cases must represent complete user goals yielding business value, not fine-grained UI clicks or CRUD sub-operations.

A widespread anti-pattern among novice modelers is treating use cases as flowcharts or individual GUI clicks (e.g., creating separate use cases for "Enter Username", "Validate Password", "Click Submit").

In professional engineering:

- **Anti-Pattern (Functional Decomposition):** Creating separate use case ovals for "Input Credit Card Number", "Check Expiration Date", "Query Card Balance".
- **Best Practice (Business Goal):** Encapsulating the entire user goal in a single cohesive use case: **"Process Checkout Payment"**.

4.3.3 Actor Taxonomy: Primary, Supporting, and Automated

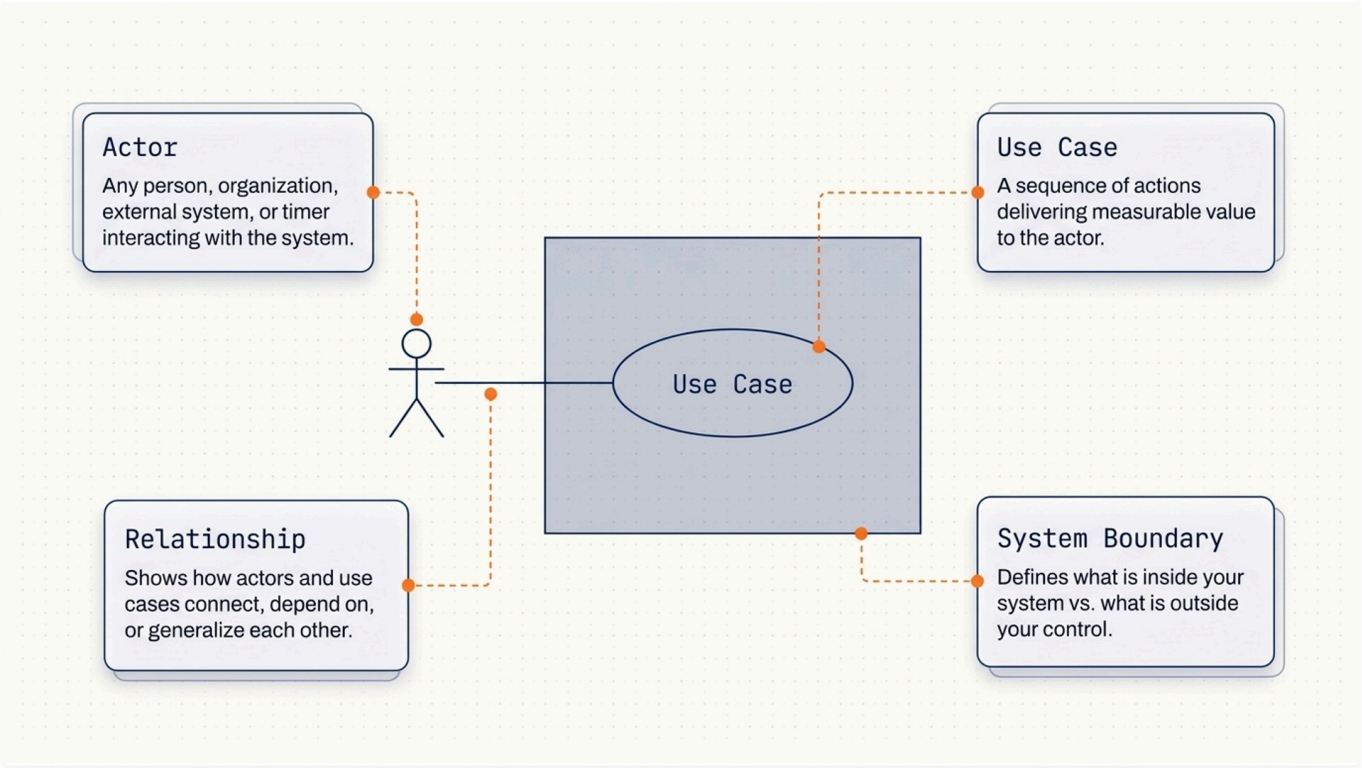


Figure 4.3.3: Actor Taxonomy. Classifying actors based on who initiates value versus who provides supporting services.

Actors are categorized based on their architectural role:

- **Primary Actors:** The principal initiators who trigger use cases to achieve a business goal (e.g., *Customer*, *Student*, *Doctor*). They are positioned on the **left** side of the boundary.
- **Secondary / Supporting Actors:** External systems or personnel that provide services to the system to help fulfill the primary actor's goal (e.g., *Payment Gateway*, *SMS Service Provider*, *Identity Provider*). They are positioned on the **right** side of the boundary.
- **Automated / Time Triggers:** Internal timers or schedulers (e.g., *Cron Daemon*, *Billing Timer*) modeled as specialized actors triggering periodic batch operations.

4.3.4 The Relationship Matrix: Include, Extend, and Generalization

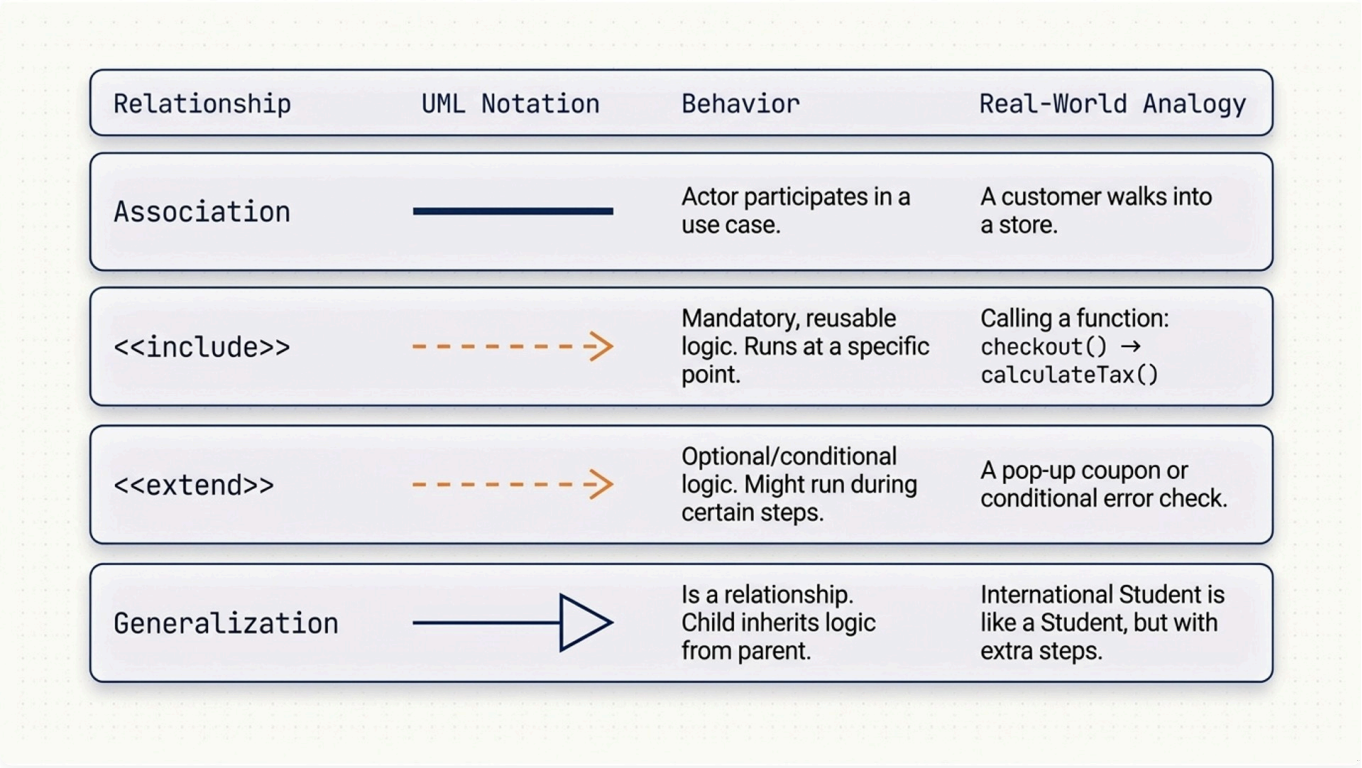


Figure 4.3.4: The Use Case Relationship Matrix. Defining the strict semantic rules governing associations, <>, <>, and generalization.

Relationship	Visual Syntax	Arrow Direction	Architectural Meaning
Association	Solid Line `——`	None / Open	Communication connection between Actor & Use Case.
`<<include>>`	Dashed Arrow `- - - >`	Base → Target	**Mandatory shared logic.** Base cannot complete execution without Target.
`<<extend>>`	Dashed Arrow `- - - >`	Extension → Base	**Optional / conditional logic.** Executed only if Extension Point condition is met.
Generalization	Solid + Hollow Triangle `——▷`	Sub-type → Super-type	**Inheritance.** Sub-actor or specialized use case inherits behavior of Super-type.

4.3.5 Clear Naming Semantics: Strong Verbs & Singular Roles

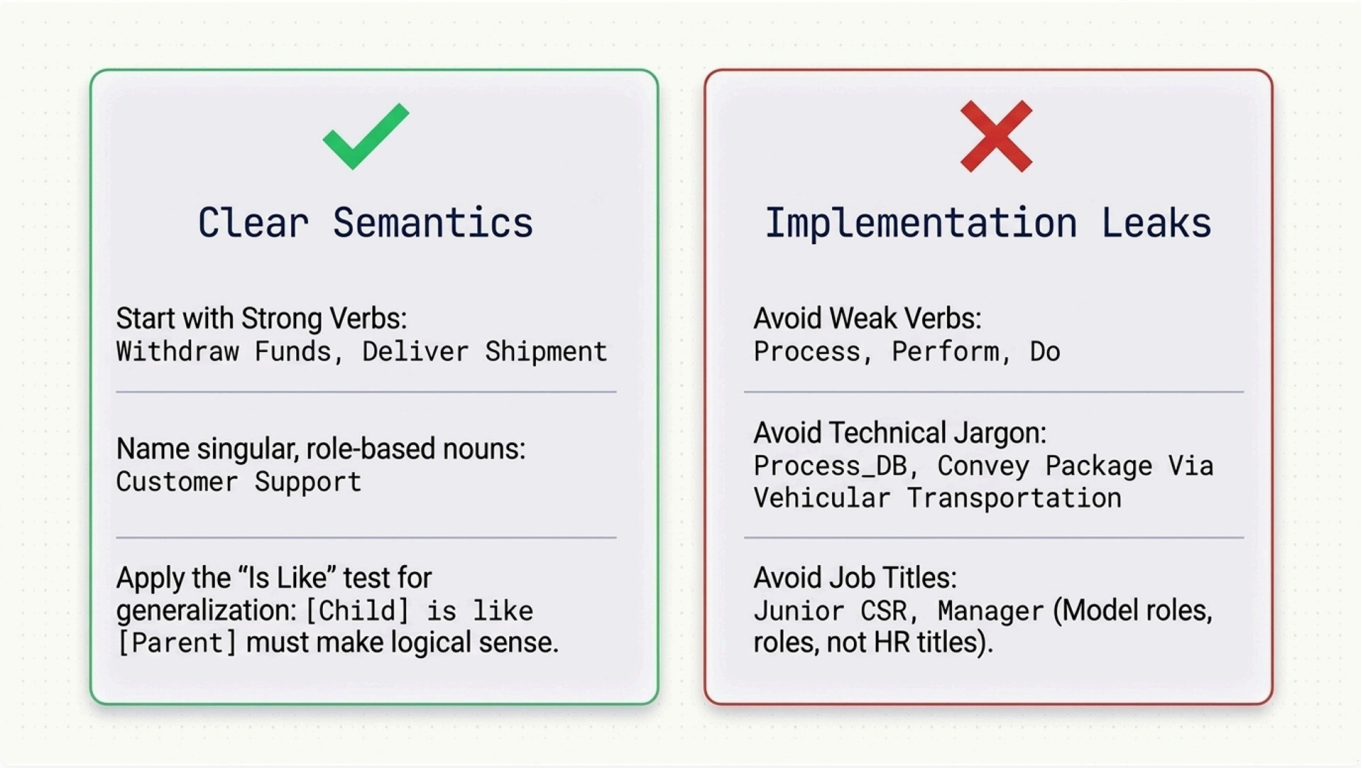


Figure 4.3.5: Clear Naming Semantics. Choosing strong action verbs and singular roles while avoiding vague verbs, technical database jargon, and HR job titles.

Dimension	Recommended Best Practice	Anti-Pattern to Avoid
Use Case Verbs	Strong, goal-oriented verbs: `Withdraw Funds`, `Deliver Shipment`, `Place Order`	Weak or vague verbs: `Process`, `Do`, `Perform`, `Manage`
Domain Terms	Plain business vocabulary: `Authenticate User`, `Generate Invoice`	Technical database jargon: `Process_DB_Transaction`, `SQL_Query_Record`
Actor Roles	Singular business roles: `Customer Support`, `Billing Manager`, `Courier`	Department names or HR job ranks: `Support Dept`, `Junior CSR Grade 2`
Generalization	Passes the "is-a / is-like" common-sense test: `GradStudent is Student`	Arbitrary groupings or procedural steps disguised as inheritance

4.3.6 Safe Nesting Limits (Avoiding Over-Engineering)

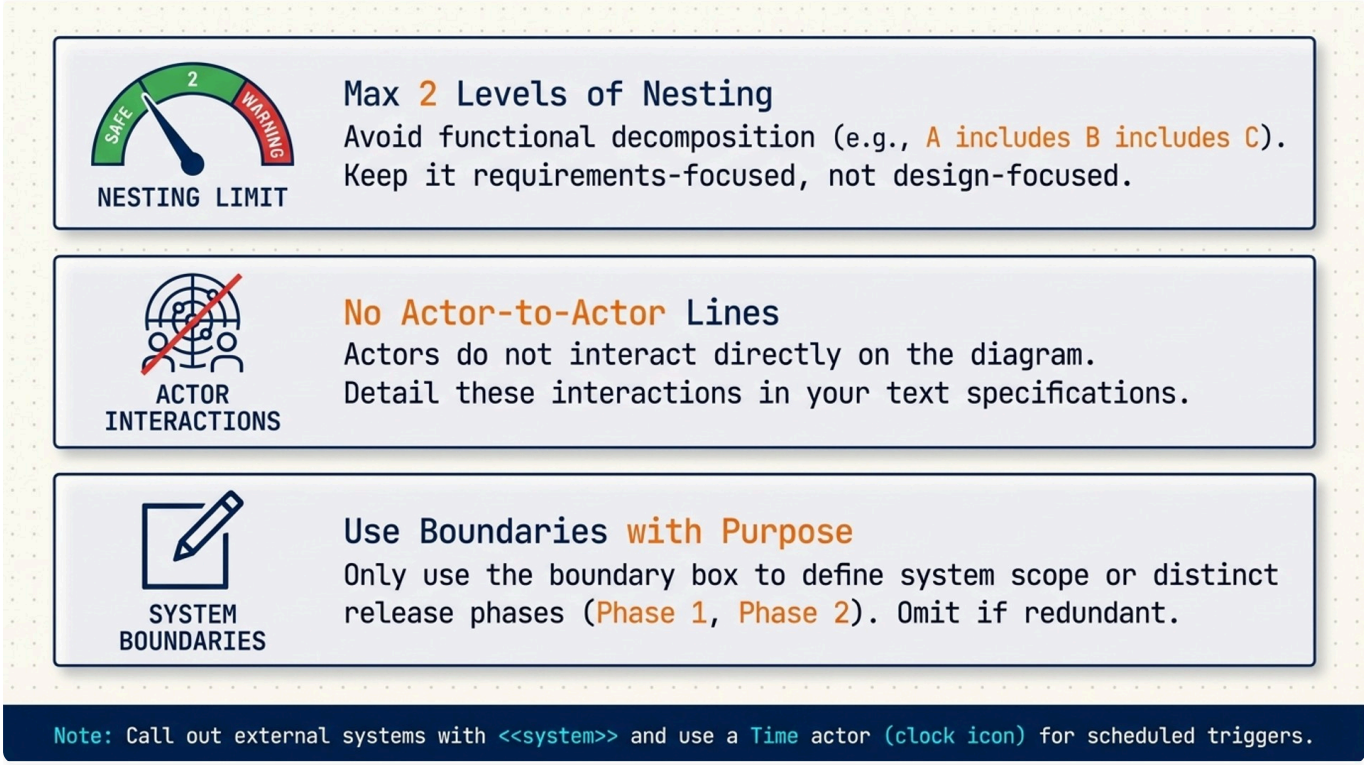


Figure 4.3.6: Safe Nesting Limits. Avoid deep hierarchical chains of <> and <>. Limit nesting to at most 1–2 levels to preserve architectural clarity.

Three golden rules keep use case models clear and maintainable:

1. **Max 2 Levels of Nesting:** Keep `<<include>>` chains shallow. If A includes B, and B includes C, the diagram degrades into a confusing procedural flowchart.
2. **No Actor-to-Actor Lines:** Actors communicate only with use cases across the system boundary, never directly with each other on the diagram. Human conversations belong in textual scenario descriptions.
3. **Use Boundaries with Purpose:** Leverage the boundary rectangle to define clear system scope or delineate release milestones (e.g., Phase 1 MVP vs. Phase 2 enhancements).

4.3.7 Case Study: University Enrollment System

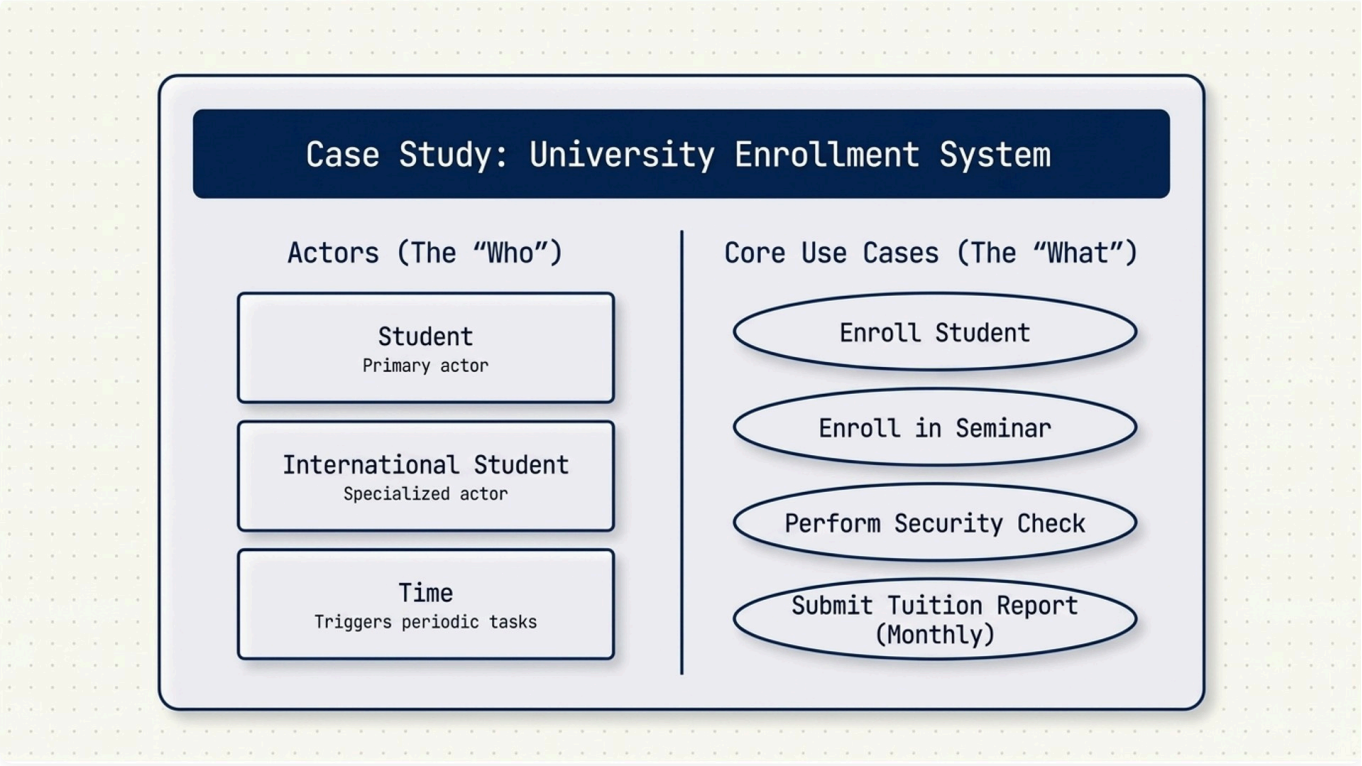


Figure 4.3.7: University Course Enrollment Case Study: Separating Actors ('The Who') from Use Cases ('The What').

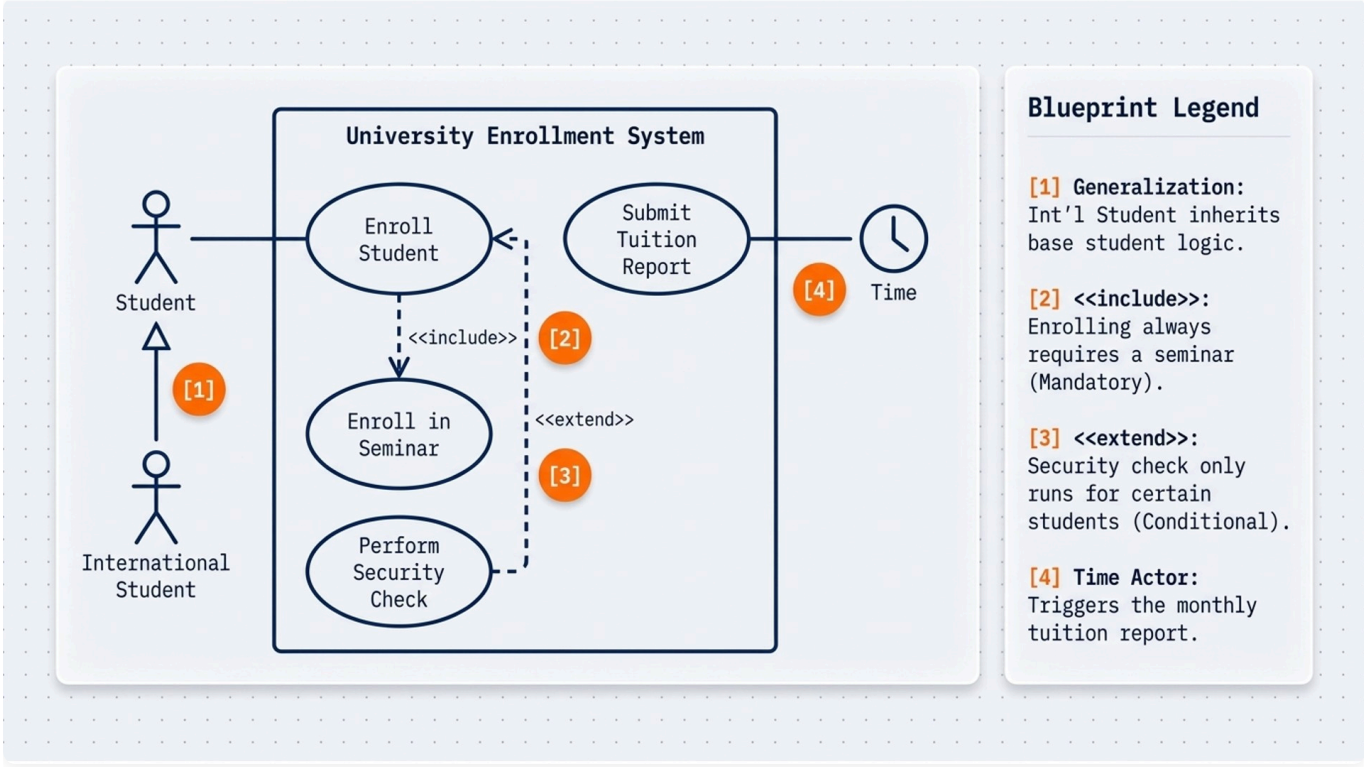
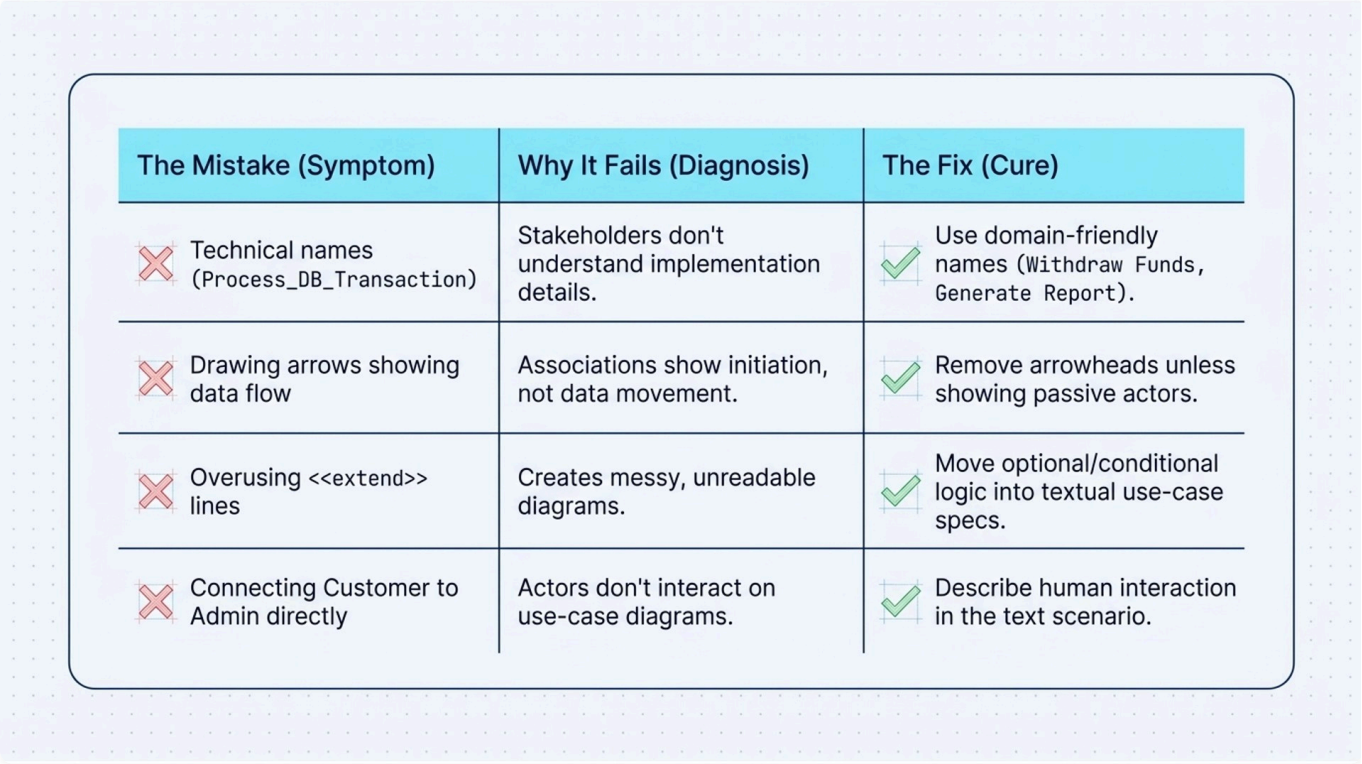


Figure 4.3.8: Complete Enrollment System Model. Combining actor generalization, mandatory <> relationships, conditional <> extension points, and automated timer actors.

Standard Tabular Use Case Description: "Enroll Student in Course"

Specification Field	Engineering Contract & Scenario Details
Use Case ID & Name	**UC-04: Enroll Student in Course**
Primary Actor	Student (Undergraduate or Graduate)
Stakeholders & Goals	• **Student**: Register for degree-required course section before quota fills. • **Registrar**: Ensure capacity limits and academic prerequisite rules hold. • **Academic Dept**: Maintain balanced classroom seat distribution.
Preconditions	1. Student is authenticated and in good academic standing. 2. Course enrollment registration window is open.
Success Guarantee	Student is formally enrolled on official roster, seat quota decremented by 1, tuition charge generated.
Trigger	Student clicks "Confirm Course Enrollment" button in the web/mobile portal.
Main Success Scenario *(Happy Path)*	1. Student browses open course sections in catalog. 2. Student selects desired section and submits enrollment request. 3. System verifies student satisfies all prerequisite course requirements. 4. System checks available seat quota in the section. 5. System reserves seat and updates course roster. 6. System notifies Billing Service to add tuition charge. 7. System displays enrollment confirmation receipt with timestamp.
Extensions *(Alternative & Exceptions)*	**3a. Prerequisite check fails:** 1. System displays missing prerequisite course codes. 2. System offers link to submit Prerequisite Waiver. 3. Use case terminates without enrollment. **4a. Section is full (Quota = 0):** 1. System alerts student that section is currently full. 2. System triggers `<<extend>>` UC-05: Join Waiting List. 3. If student agrees, student added to priority queue. **6a. Billing Service offline:** 1. System queues billing event in transactional outbox. 2. System completes enrollment and logs audit warning.

4.3.8 Common Anti-Patterns in Use Case Modeling



The Mistake (Symptom)	Why It Fails (Diagnosis)	The Fix (Cure)
✗ Technical names (`Process_DB_Transaction`)	Stakeholders don't understand implementation details.	✓ Use domain-friendly names (`Withdraw Funds`, `Generate Report`).
✗ Drawing arrows showing data flow	Associations show initiation, not data movement.	✓ Remove arrowheads unless showing passive actors.
✗ Overusing `<<extend>>` lines	Creates messy, unreadable diagrams.	✓ Move optional/conditional logic into textual use-case specs.
✗ Connecting Customer to Admin directly	Actors don't interact on use-case diagrams.	✓ Describe human interaction in the text scenario.

Figure 4.3.9: Common Anti-Patterns. Avoiding functional decomposition, missing boundaries, reversed extend arrows, and mixing UI implementation details.

1. **Reversed Arrowheads:** Pointing `<<extend>>` from the base use case to the extension. Remember: the base use case should be completely unaware of the extension!
2. **Functional Decomposition Trap:** Treating use case diagrams as flowcharts by breaking an operation into tiny procedural pieces.
3. **Missing System Boundaries:** Leaving ovals floating without a boundary box, making it impossible to determine scope.
4. **Actors Inside the Boundary:** Placing external systems or users inside the box.
5. **Overusing Generalization:** Creating complex inheritance trees for use cases instead of simple parameter variations.
6. **Modeling UI Clicks:** Writing "Click Submit Button" instead of "Confirm Order".
7. **Orphan Use Cases:** Creating use cases that connect to no actors whatsoever.

4.3.9 Best Practices Checklist

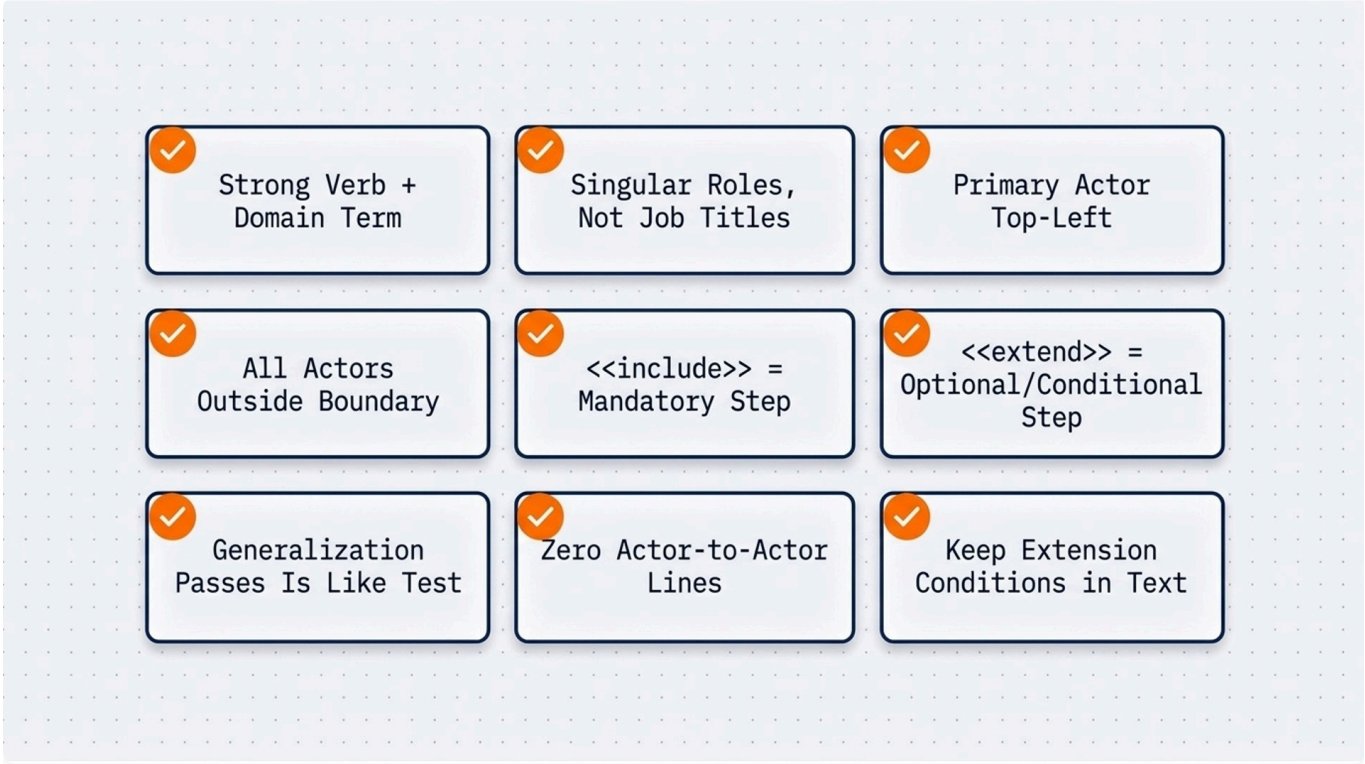


Figure 4.3.10: Use Case Best Practices Checklist. A 9-point architectural quality gate before finalizing diagrams for team review.

Check #	Quality Gate Inspection Item	Verification Rule
1	**Strong Verb + Noun**	Does every use case begin with an active transitive verb and concrete noun?
2	**Singular Roles**	Are actors named as singular operational roles rather than HR job titles?
3	**Primary on Left**	Are primary initiating actors positioned on the left side of the boundary?
4	**Actors Outside Box**	Are all actors placed strictly outside the system boundary rectangle?
5	**Strict `<<include>>`**	Is `<<include>>` reserved exclusively for mandatory, unavoidable sub-routines?
6	**Strict `<<extend>>`**	Is `<<extend>>` reserved exclusively for optional, conditional extensions?
7	**"Is-Like" Generalization**	Does every actor generalization hierarchy pass the common-sense inheritance test?
8	**Zero Actor Interlinks**	Are there zero lines connecting actors directly to each other?
9	**Conditions in Text**	Are detailed extension conditions and exception rules recorded in text specifications?

4.3.10 Communication-First Mindset

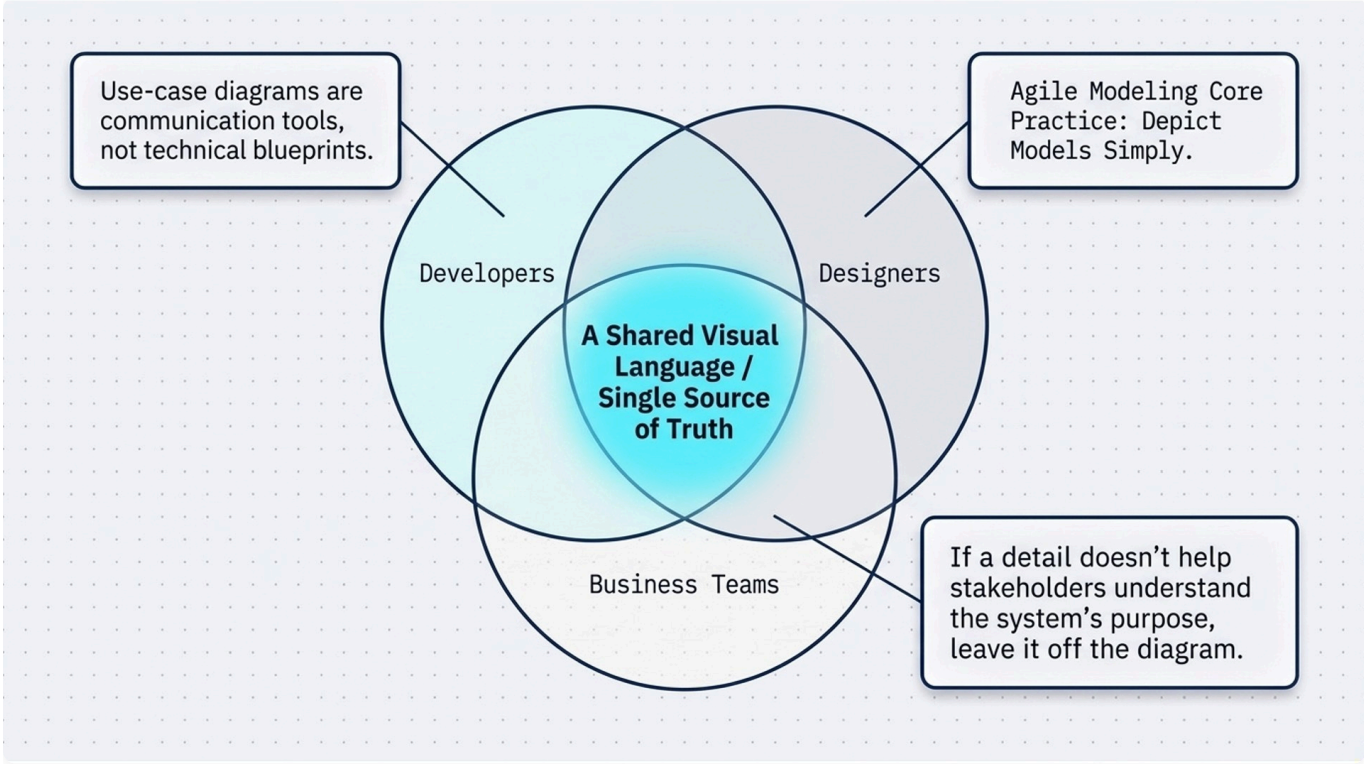


Figure 4.3.11: The Communication-First Mindset. Use case models serve as the vital tripartite bridge uniting non-technical business stakeholders, UX/system designers, and software engineers.

4.3.11 Generative AI Assistance & Food Delivery Case Study

Modern LLMs can rapidly synthesize use case models from raw client transcripts when prompted with strict boundary constraints:

```
[System Role: Lead Software Architect & Requirements Engineer]
Task: Analyze the following requirements and produce:
1. PlantUML Use Case Diagram markup adhering strictly
   to UML 2.5 standards.
2. Structured tabular Use Case Description for the
   primary happy path.

Rules & Semantic Constraints:
- Delineate boundary with 'rectangle "Food Delivery" { ... }'.
- Place Primary Actors on left, Secondary Actors on right.
- Use strict <<include>> for mandatory sub-routines and
  <<extend>> for optional paths.
- Naming format: Active transitive verb + concrete domain noun.

Requirement Text:
"A customer browses restaurants, adds food items to cart,
and proceeds to checkout. During checkout, the customer
must authorize payment through a third-party gateway.
Optionally, the customer may apply a promo voucher code
or request contact-free dropoff. Kitchen staff accept orders
and update cooking status. Couriers accept delivery runs."
```

Generated Food Delivery Use Case Model

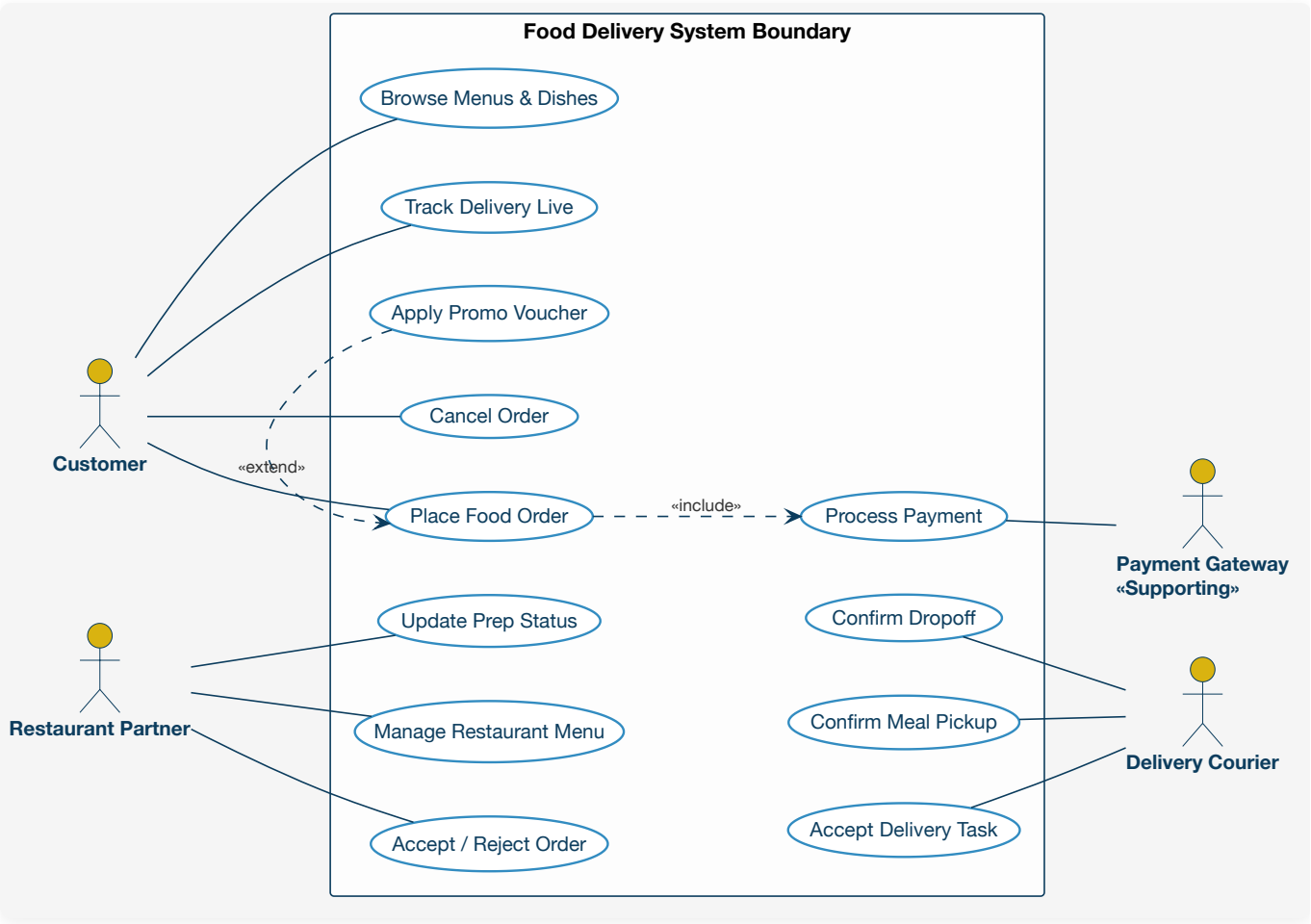


Figure 4.3.12: Generated Food Delivery Use Case Diagram. Delineating system boundaries, primary customer interactions, restaurant kitchen management, courier dispatch, mandatory payment processing, and optional voucher extension.

```
@startuml
left to right direction
actor Customer
actor "Restaurant Partner" as Restaurant
actor "Payment Gateway\n«Supporting»" as Payment
actor "Delivery Courier" as Courier

rectangle "Food Delivery System" {
  usecase "Browse Menus" as UC_Browse
  usecase "Place Food Order" as UC_PlaceOrder
  usecase "Apply Promo Voucher" as UC_Voucher
  usecase "Track Delivery Live" as UC_Track
  usecase "Manage Menu" as UC_Menu
  usecase "Accept / Reject Order" as UC_AcceptOrder
  usecase "Process Payment" as UC_Payment
  usecase "Accept Delivery Task" as UC_AcceptTask
}

Customer -- UC_Browse
Customer -- UC_PlaceOrder
Customer -- UC_Track
Restaurant -- UC_Menu
Restaurant -- UC_AcceptOrder

UC_PlaceOrder ..> UC_Payment : <<include>>
UC_Voucher .> UC_PlaceOrder : <<extend>>
UC_Payment -- Payment
UC_AcceptTask -- Courier
@enduml
```

4.3.12 Diagnostic Case Study: Spotting Flaws in Use Case Models

To build rigorous architectural intuition, software engineers must learn to conduct diagnostic critiques of draft models. Consider the following draft use case diagram created by a junior analyst for an online food ordering workflow:

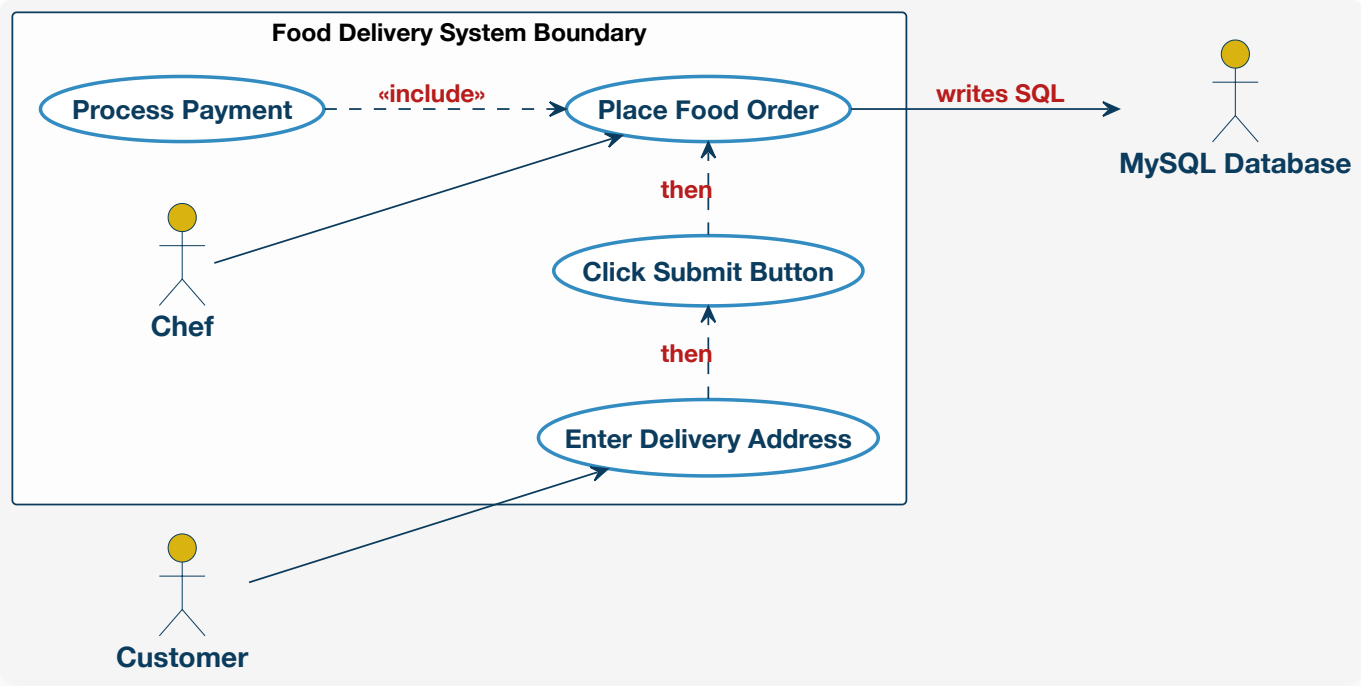


Figure 4.3.13: Draft Use Case Diagram with 3 Critical Modeling Anti-Patterns.

Audit Findings & Architectural Corrections

1. **Granularity & Workflow Trap (Anti-pattern: Flowcharting):**
   - *The Flaw:* The student modeled `Enter Delivery Address` and `Click Submit Button` as distinct use cases connected sequentially with arrows.
   - *Correction:* Use cases represent complete, observable user goals yielding measurable business value, not micro UI clicks or procedural flowchart steps. Entering an address and clicking submit are trivial interactions encapsulated within the single use case `Place Food Order`.
2. **Inverted Include Arrow Direction:**
   - *The Flaw:* The dashed arrow points from `Process Payment` toward `Place Food Order`.
   - *Correction:* In UML, the base use case points *toward* the included subroutine (`Place Food Order ..> Process Payment : <<include>>`). The base case initiates and depends upon the included behavior, never the reverse.
3. **Boundary & Actor Violations:**
   - *The Flaw:* `Chef` is an actor but is drawn inside the system boundary box. Meanwhile, `MySQL Database` is drawn outside the boundary as an external actor.
   - *Correction:* Actors represent external entities (human roles or external systems) and must reside strictly *outside* the system boundary rectangle. Conversely, an internal database is part of the system's software persistence tier, not an external actor.

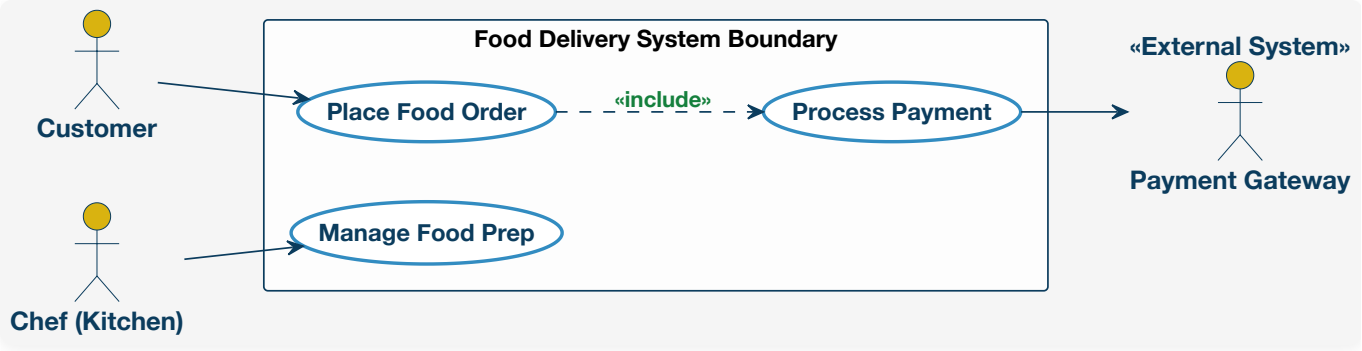


Figure 4.3.14: Corrected Use Case Model. Granularity restored to user goals, include arrow properly directed, and boundary discipline strictly enforced.

4.3.13 Interactive Hands-On Activity: Evolving Use Cases with AI

To practice continuous architectural evolution, students complete the following pair hands-on exercise:

- **Experiment Goal:** Mutate requirements by introducing one new business feature into the Food Delivery prompt and audit the AI's architectural output:
  i. *Option A (Scheduled Ordering):* Allow `Customer` to schedule delivery for a future date/time.
  ii. *Option B (Live Chat Support):* Allow `Customer` to initiate real-time chat with a `Support Agent` or `Courier`.
  iii. *Option C (Group Ordering):* Allow multiple team members to pool items into a single shared order.
- **Audit Checklist:**
  - Did AI keep external actors/gateways outside the boundary?
  - Did AI correctly distinguish mandatory subroutines (`<<include>>`) from optional extensions (`<<extend>>`)?
  - Did AI avoid trivial button-click bubbles (e.g., no "Click Confirm")?

🙋 Concept Check (CCQ 3) — Use Case Relationships: Include vs. Extend

Question

In our Food Delivery Use Case Model, why does `Apply Promo Voucher` point to `Place Food Order` with `<<extend>>`, while `Place Food Order` points to `Process Payment` with `<<include>>`?

- A. Voucher application is mandatory for all orders; payment is optional.
- B. Vouchers are conditional optional behavior; payment is mandatory shared execution.
- C. Vouchers are executed by supporting actors; payment is executed by primary actors.
- D. Vouchers represent class inheritance; payment represents object composition.

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.4 Structural & Class Models

While use case models capture external business interactions, **Class Diagrams** form the static architectural backbone of object-oriented systems. They define the vocabulary of the problem domain, map database entities, and establish compile-time contracts between collaborating modules.

4.4.1 Anatomy of a Class: The Three Compartments

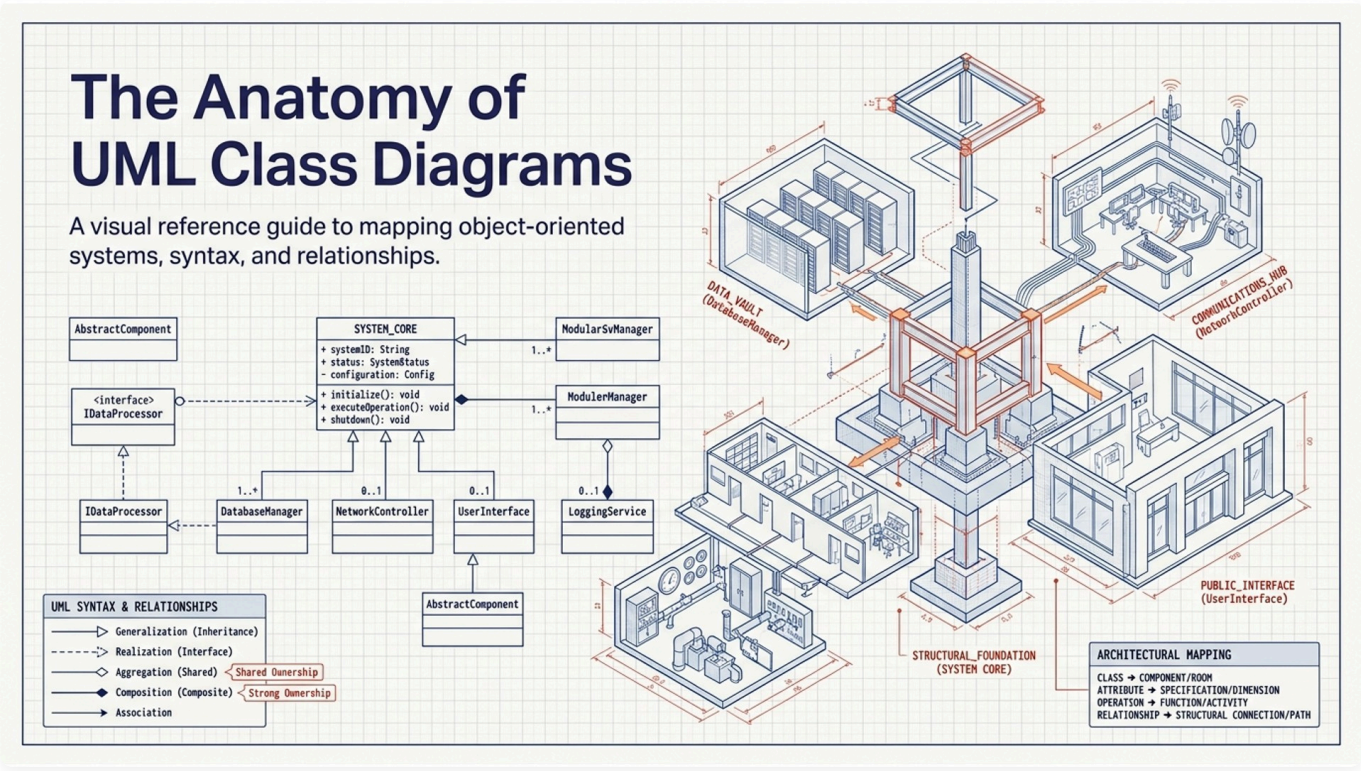


Figure 4.4.1: The Anatomy of UML Class Diagrams. A class box consists of three distinct compartments: Class Name, Attributes, and Operations.

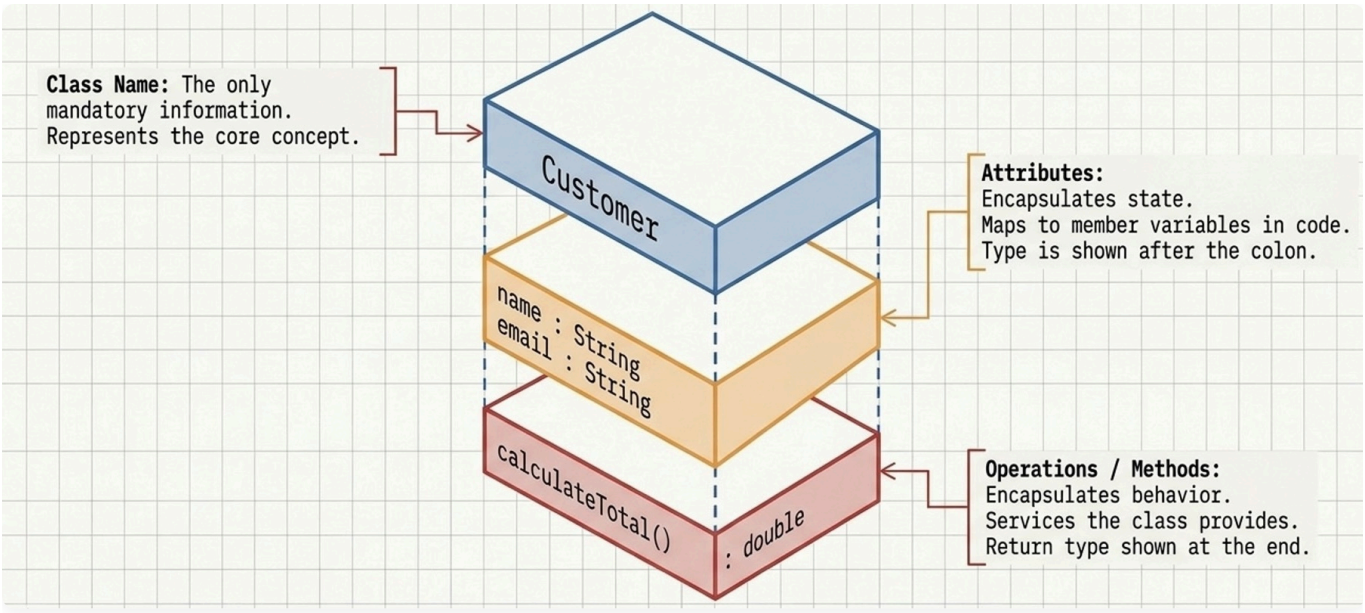


Figure 4.4.2: Internal Class Box Structure. Demonstrating attribute type annotations, default initializers, method signatures, parameter directionality, and visibility specifiers.

Every standard UML class box is partitioned into three horizontal compartments:

1. **Top Compartment (Class Name):** Identifies the concept. Abstract class names are rendered in *italics* or decorated with `{abstract}`. Interfaces are styled with `<<interface>>`.
2. **Middle Compartment (Attributes):** Defines data state fields using the standard syntax: `[visibility] name : type [multiplicity] [= default] [{property-modifiers}]` *(Example:* `- balance: Money = 0.00 {readOnly}` *)*
3. **Bottom Compartment (Operations / Methods):** Defines behavior contracts using the standard syntax: `[visibility] name ( [direction] paramName : paramType , ... ) : returnType` *(Example:* `+ processPayment(in amount: Decimal, in token: String): PaymentResult` *)*

4.4.2 Visibility Specifiers and Information Hiding

Symbol	Label	Meaning	Code Snippet
+	Public	Accessible from anywhere.	`public String name;`
-	Private	Accessible only within the class itself.	`private int id;`
#	Protected	Accessible within the class and its subclasses.	`protected double balance;`

Figure 4.4.3: Visibility Specifier Matrix. Contrasting public (+), private (-), protected (#), and package (~) access levels across object boundaries.

Symbol	Visibility	OOP Equivalent	Architectural Scope
`+`	**Public**	`public`	Accessible by any external class in the system.
`–`	**Private**	`private`	Encapsulated strictly inside the declaring class.
`#`	**Protected**	`protected`	Accessible by declaring class and its sub-classes.
`~`	**Package**	`default` (package-private)	Accessible only by classes within the same package.

4.4.3 Taxonomy of Class Relationships

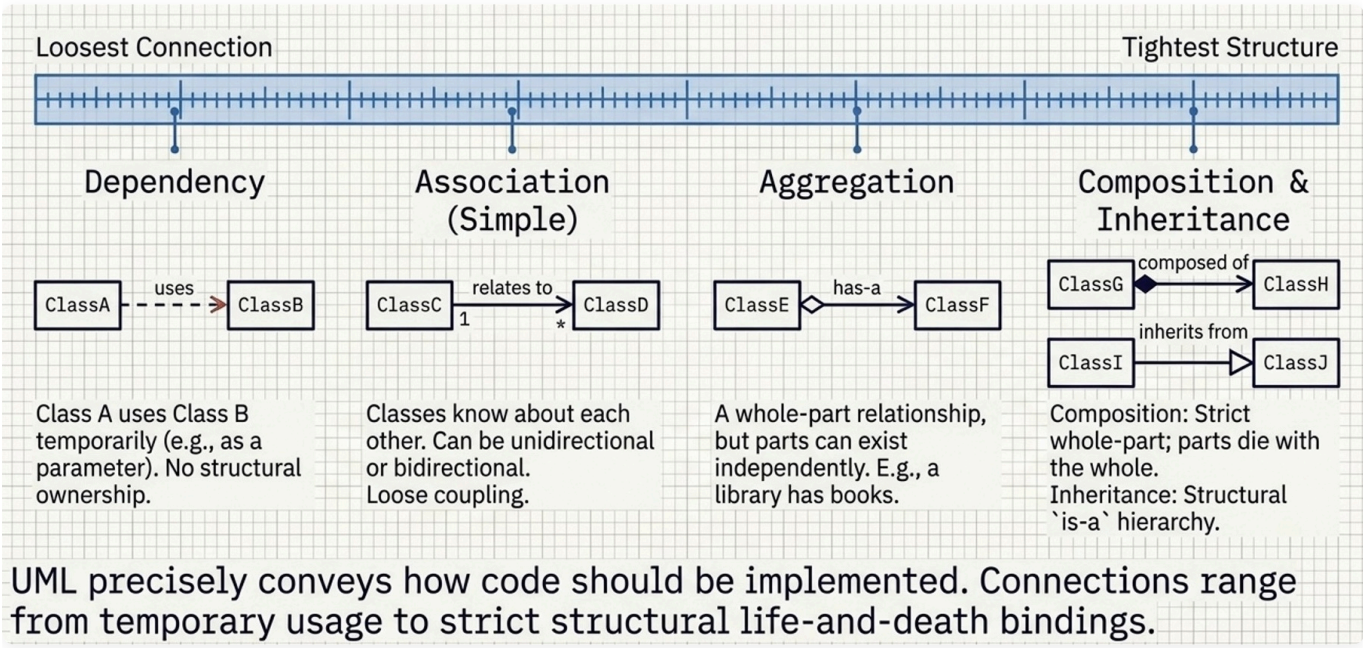


Figure 4.4.4: Taxonomy of Class Relationships. Showing the hierarchy from loose, transient dependencies to tight, permanent lifecycle bindings.

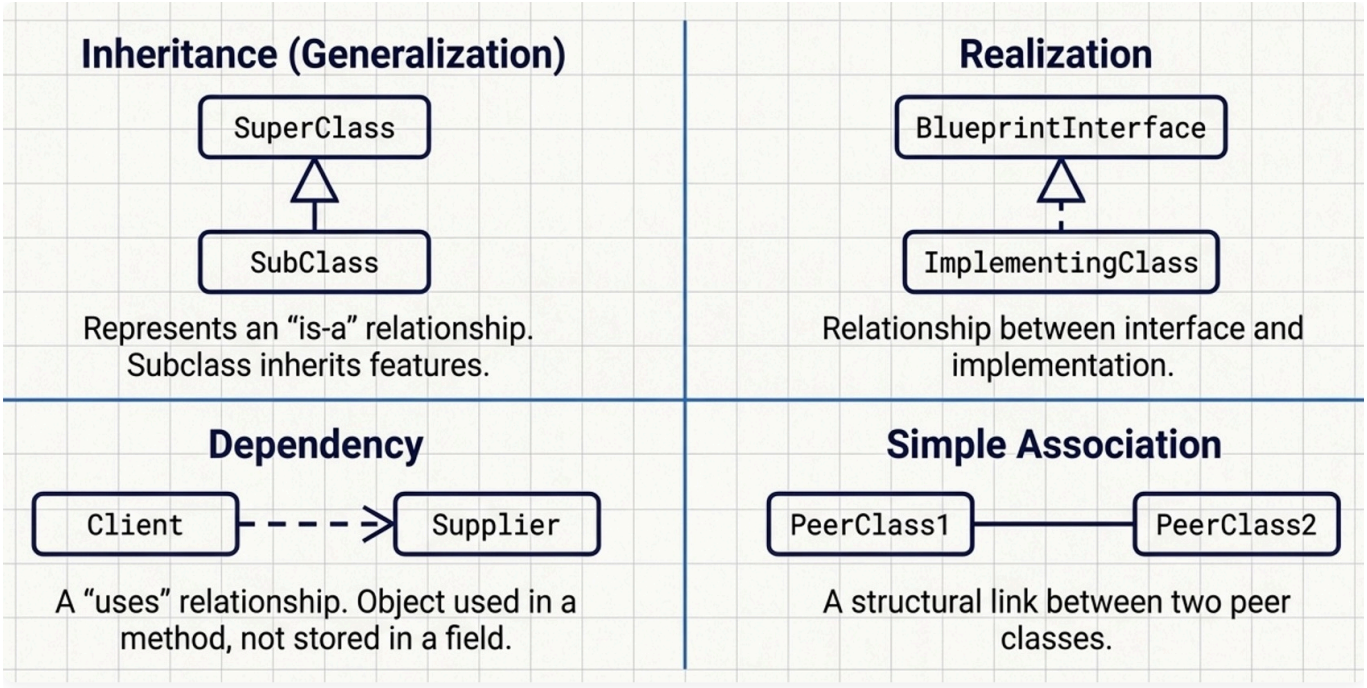


Figure 4.4.5: UML Connector Cheat Sheet. Standard line styles and endpoint ornaments for association, aggregation, composition, generalization, realization, and dependency.

Relationship	Visual Notation	Coupling Level	Lifecycle & Architectural Dependency
Dependency	Dashed arrow `- - >`	Weak / Transient	Temporary usage via method parameter, local variable, or return type.
Association	Solid line `———`	Moderate / Peer	Structural reference via long-lived instance field.
Aggregation	Hollow diamond `———◇`	Moderate / Shared	**Weak ownership.** Child parts can survive independently if container is deleted.
Composition	Filled diamond `———◆`	Strong / Exclusive	**Strict ownership.** Child parts are permanently destroyed when parent is destroyed.
Generalization	Solid + Hollow triangle `—▷`	Inheritance (`is-a`)	Sub-class inherits all attributes, methods, and relationships of super-class.
Realization	Dashed + Hollow triangle `- -▷`	Interface Contract	Concrete class implements the operations declared by an interface.

Aggregation vs. Composition: The Lifecycle Ownership Test

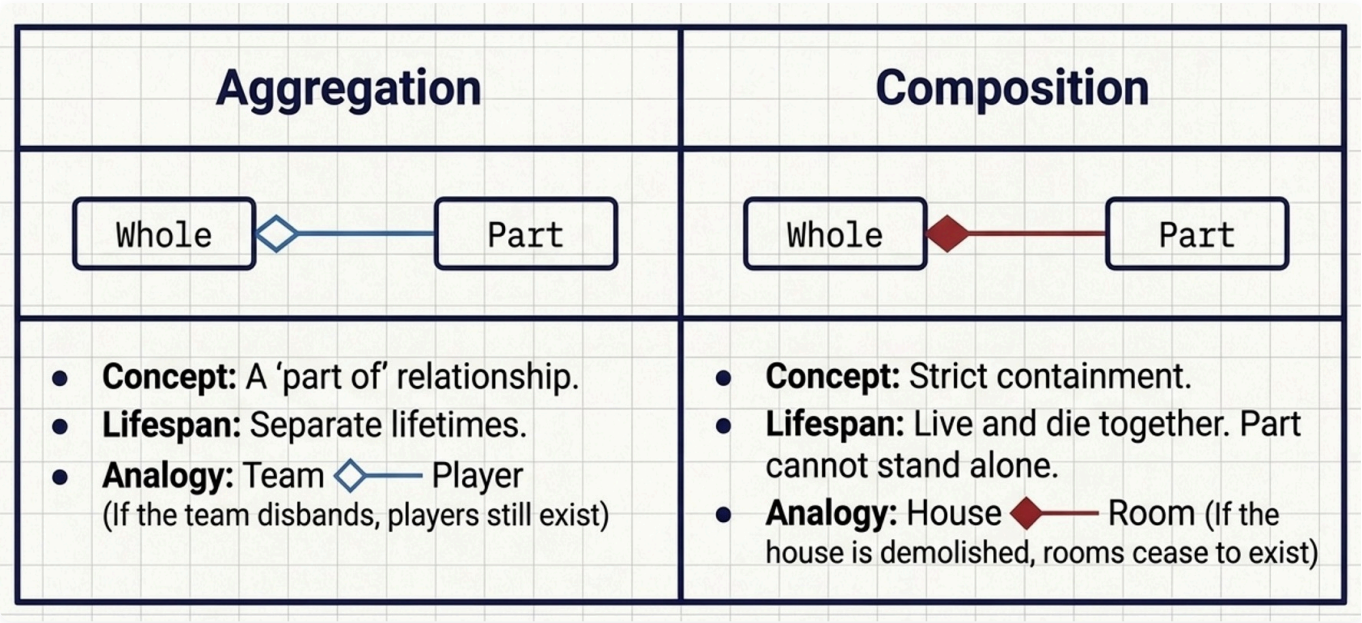


Figure 4.4.6: Aggregation vs. Composition. The definitive test is lifecycle dependency: if the container object is deleted, do the contained objects cease to exist?

- **Composition (Filled Diamond ◆):** Implies exclusive, coincident ownership. A part belongs to *exactly one* whole at a time. If the container is destroyed, all its parts are destroyed.
  - *Example:* An `Order` owns its `OrderItem` line items. Deleting an order from the database cascade-deletes all its order items.
- **Aggregation (Hollow Diamond ◇):** Implies shared, non-exclusive ownership. Parts maintain an independent existence outside the container.
  - *Example:* An `Order` aggregates a `Courier`. If the order is completed or deleted, the courier continues to exist and takes another delivery run.

4.4.4 Multiplicity Constraints and Navigability

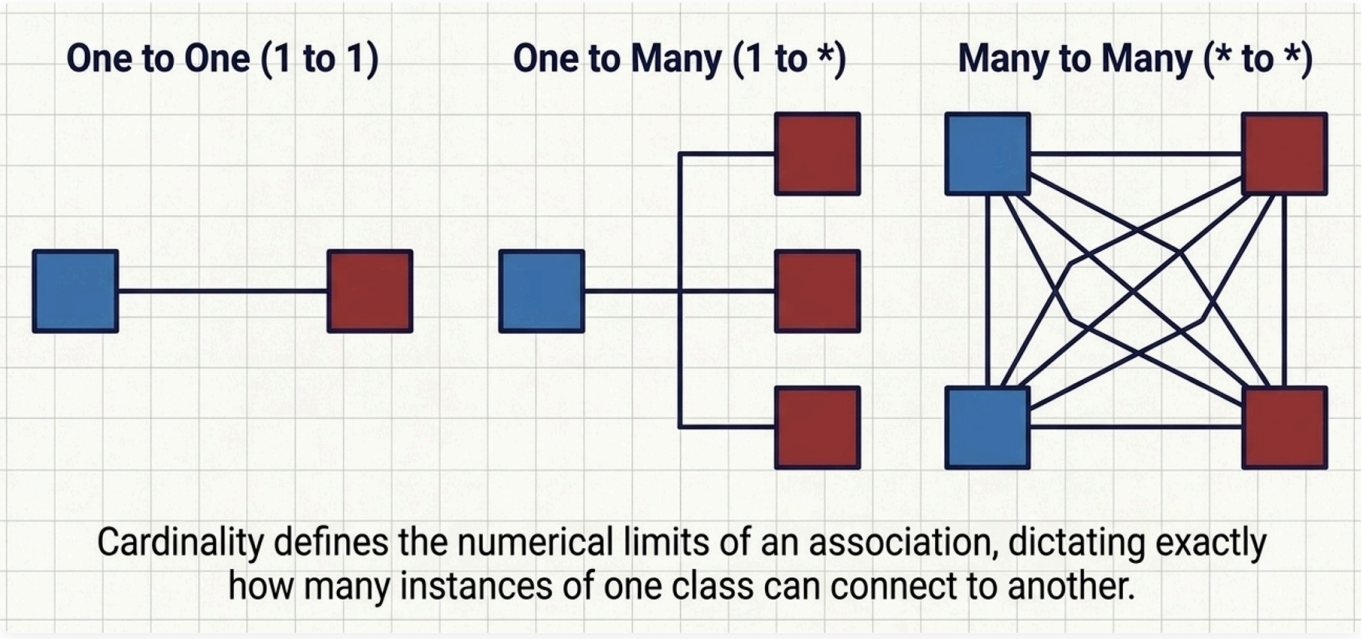


Figure 4.4.7: Multiplicity Constraints and Navigability. Specifying lower and upper bounds (1, , 0..1, 1..) on relationship endpoints.

Multiplicity specifies how many instances of class B can be associated with a single instance of class A:

- `1`: Exactly one (mandatory).
- `0..1`: Zero or one (optional / nullable pointer).
- `*` or `0..*`: Zero or more (unbounded collection / list / set).
- `1..*`: One or more (non-empty collection).
- `m..n`: Specific numeric range (e.g., `2..4`).

4.4.5 Case Study: Food Delivery Platform Domain Model

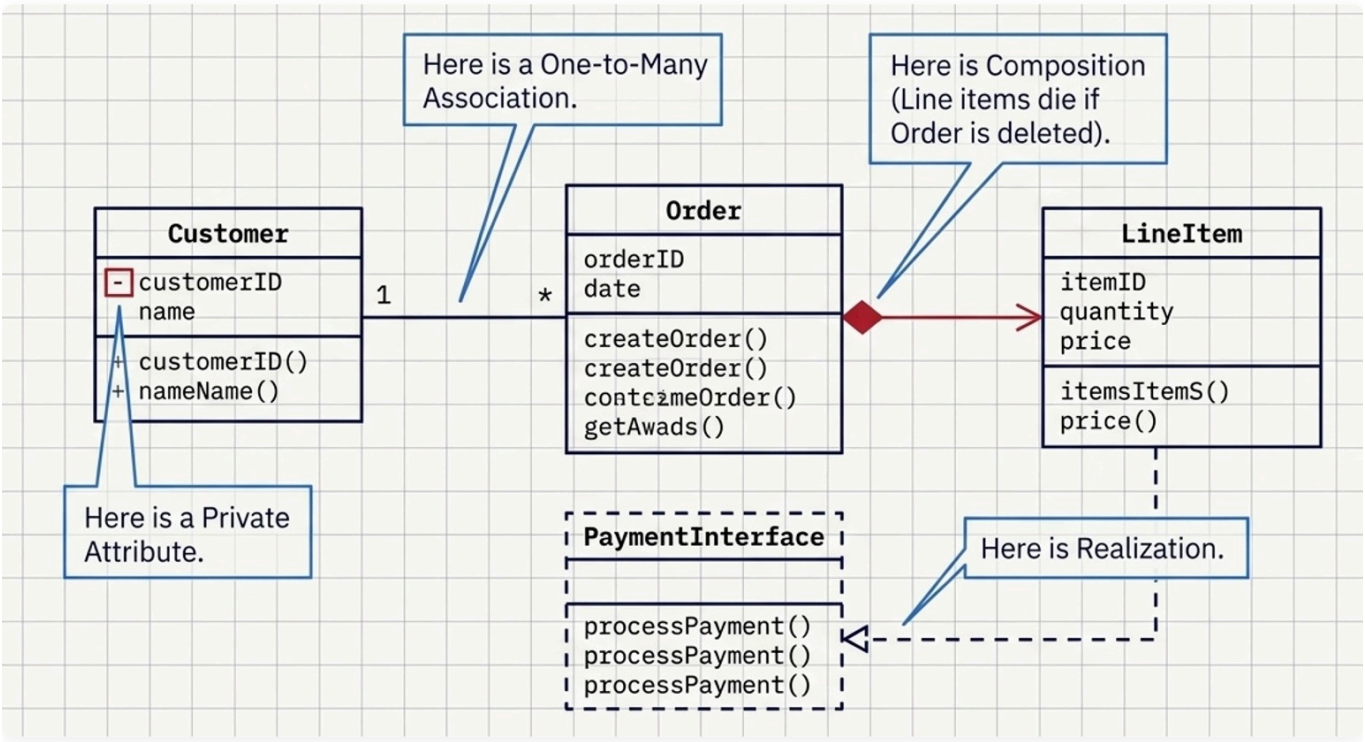


Figure 4.4.8: Food Delivery Domain Model Case Study. Demonstrating Customer, Order, OrderItem, MenuItem, Restaurant, Courier, and Payment entities with proper multiplicity and lifecycle semantics.

In this enterprise domain model:

1. `Customer` has a `1 ── *` association with `Order` (a customer can place multiple historical orders).
2. `Order` has a `1 ◆── 1..*` composition with `OrderItem` (an order must contain at least one item, and items cannot exist without an order).
3. `OrderItem` has a `* ── 1` association with `MenuItem` (many order items reference the same catalog menu item).
4. `Order` has a `* ──◇ 0..1` aggregation with `Courier` (an order is assigned to at most one courier during transit).

5. `Order` has a `1 ── 1` association with `Payment` (each order has an associated transaction record).

4.4.6 Generative AI Assistance for Class Modeling

When using LLMs to synthesize class models from requirements, provide explicit relationship constraints:

```
[System Role: Senior Systems Architect]
Task: Synthesize a PlantUML Class Diagram for Food Delivery.

Design Constraints:
- 3 compartments: Name, Attributes with visibility & types,
  Operations with return types.
- Explicitly apply composition (*--) where lifecycle
  ownership is exclusive.
- Explicitly apply aggregation (o--) where referenced entities
  maintain independent lifecycles.
- Mark multiplicity bounds ('1', '0..1', '1..*') on all ends.
- Apply inheritance (--|>) where specialized entity subtypes
  exist (e.g., CreditCardPayment vs CashPayment).
```

Generated Food Delivery Class Model

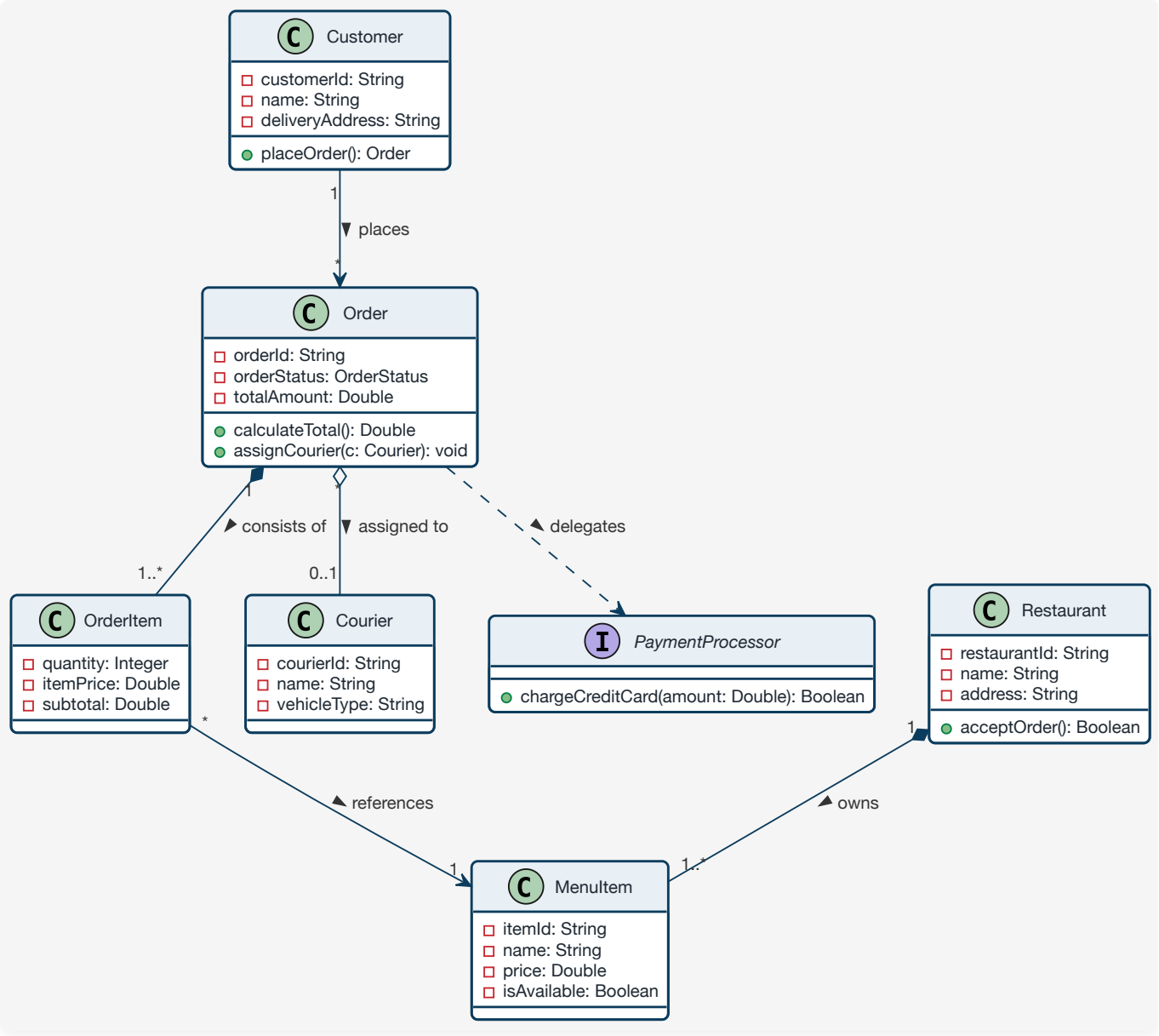


Figure 4.4.9: Generated Food Delivery Class Diagram. Mapping Customer, Order, OrderItem, MenuItem, Restaurant, Courier, and PaymentProcessor entities with precise multiplicities and lifecycle bounds.

```
@startuml
class Customer {
  - customerId: String
  - deliveryAddress: String
  + placeOrder(): Order
}
class Restaurant {
  - restaurantId: String
  - address: String
  + acceptOrder(): Boolean
}
class MenuItem {
  - itemId: String
  - price: Double
}
class Order {
  - orderId: String
  - totalAmount: Double
  + calculateTotal(): Double
  + assignCourier(c: Courier): void
}
class OrderItem {
  - quantity: Integer
  - subtotal: Double
}
class Courier {
  - courierId: String
  - vehicleType: String
}
interface PaymentProcessor {
  + chargeCreditCard(amount: Double): Boolean
}

Restaurant "1" *-- "1..*" MenuItem : owns >
Order "1" *-- "1..*" OrderItem : consists of >
OrderItem "*" --> "1" MenuItem : references >
Customer "1" --> "*" Order : places >
Order "*" o-- "0..1" Courier : assigned to >
Order ..> PaymentProcessor : delegates >
@enduml
```

4.4.7 Diagnostic Case Study: Spotting Flaws in Class Models (The Baseball Matchup)

Designing structural class diagrams requires careful distinction between inheritance (`is-a`), composition (destructive `has-a`), and loose association/aggregation (`uses-a` / shared `has-a`). Consider the following draft class diagram submitted for a baseball tracking engine:

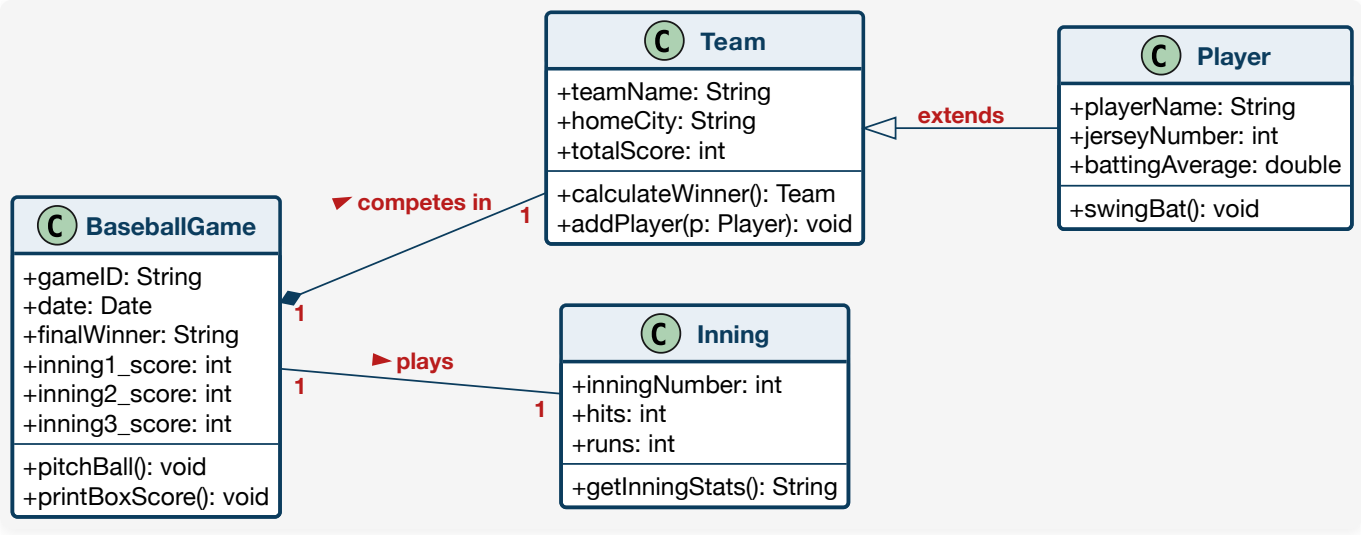


Figure 4.4.10: Draft Baseball Class Model with 4 Major Architectural Flaws.

Audit Findings & Architectural Corrections

1. **Composition vs. Aggregation Disaster (`BaseballGame` & `Team`):**
   - *The Flaw:* `BaseballGame` connects to `Team` with a solid diamond (Composition) and multiplicity `1`.
   - *Correction:* Composition implies that deleting a game record destroys the participating teams! A `Team` (e.g., Yankees) outlives any single game and participates in many games throughout a season; this relationship must be Aggregation (hollow diamond `◇`) or Association. Furthermore, a baseball game requires exactly `2` teams (Home and Away), not 1.
2. **Inheritance Abuse (is-a vs. has-a):**
   - *The Flaw:* `Team <|-- Player` indicates that a Player is a subclass of Team.
   - *Correction:* A Player is a member (part) of a Team, not a specialized kind of Team! Confusing membership with inheritance breaks object polymorphism. The relationship should be modeled as an Aggregation: `Team "1" o-- "9..*" Player`.
3. **Missing Composition and Multiplicity for Inning:**
   - *The Flaw:* `BaseballGame` has a plain association with `Inning` with multiplicity `1`.
   - *Correction:* An inning belongs exclusively to its specific game—deleting the game invalidates its innings. This is true Composition (`BaseballGame "1" *-- "9..*" Inning`).
4. **Broken Encapsulation & Responsibility Misplacement:**
   - *The Flaw:* All fields are marked public (`+`). `pitchBall()` is placed in `BaseballGame` instead of `Player`. `calculateWinner()` is placed in `Team`.
   - *Correction:* Enforce information hiding with private fields (`-`). Operations must belong to the class holding the state: `Player` executes `pitchBall()`, while `BaseballGame` compares team scores to determine `calculateWinner()`.

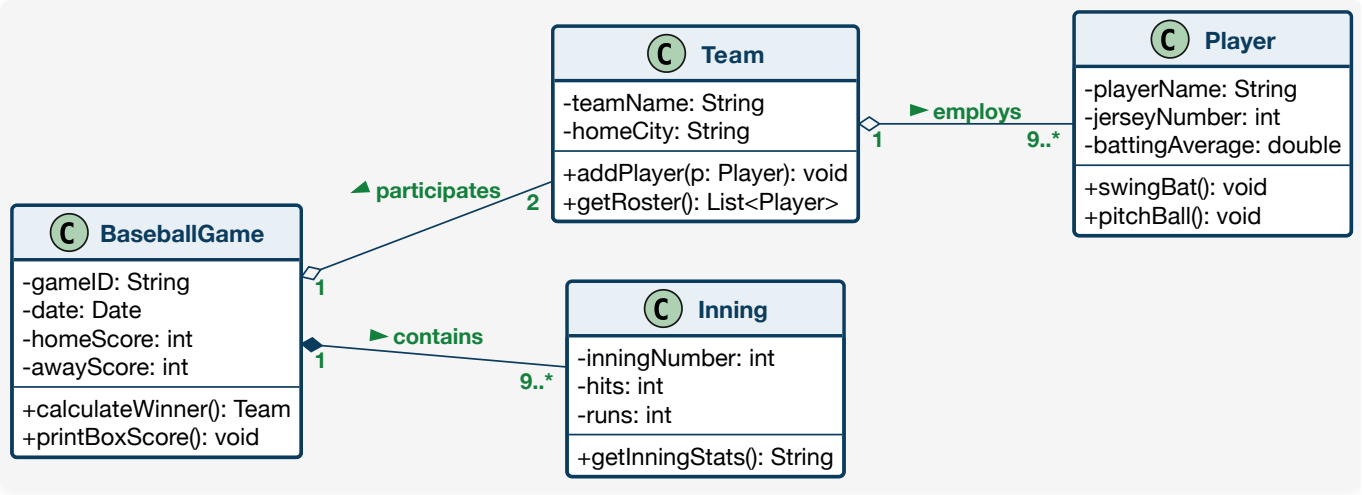


Figure 4.4.11: Corrected Baseball Class Model. Non-destructive aggregation, proper membership, strict private encapsulation, and correct multiplicities.

4.4.8 Interactive Hands-On Activity: Evolving Food Delivery with AI

- **Experiment Goal:** Mutate requirements by introducing one new structural rule into the Food Delivery domain prompt and audit the AI's architectural output:
  i. *Option A (Multi-Vendor Orders):* Allow an `Order` to bundle items from multiple distinct `Restaurant`s in a single checkout.
  ii. *Option B (Courier Tipping & Rating):* Allow `Customer` to tip and rate the assigned `Courier` after delivery.
  iii. *Option C (VIP Membership Tier):* Add subscription membership where `Customer` receives free delivery.
- **Audit Checklist:**
  - Did AI preserve Composition (`*--`) vs. Aggregation (`o--`) semantics?
  - Are multiplicities (`1..*`, `0..1`) logically sound for the new rule?
  - Did AI preserve private encapsulation (`-`) and avoid hallucinating redundant classes?

🤦 Concept Check (CCQ 4) — Class Model Relationships: Composition vs. Aggregation

Question

In our Food Delivery Class Diagram, why is the relationship between `Order` and `OrderItem` modeled as **Composition (◆)**, whereas `Order` and `Courier` is modeled as **Aggregation (◇)**?

- A. An `OrderItem` can exist independently, but a `Courier` cannot exist without an `Order`.
- B. An `OrderItem` is destroyed with its parent `Order`, whereas a `Courier` maintains an independent lifecycle.
- C. Composition represents inheritance between classes, while aggregation represents method invocation.

- D. Composition requires zero-to-one multiplicity, while aggregation requires one-to-many multiplicity.

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.5 Interaction & Sequence Diagrams

While class diagrams reveal what classes *can* communicate, **Sequence Diagrams** illustrate *how* runtime object instances actually collaborate over time to fulfill a specific user story or algorithmic transaction.

4.5.1 The 2D Canvas: Participants vs. Descending Time

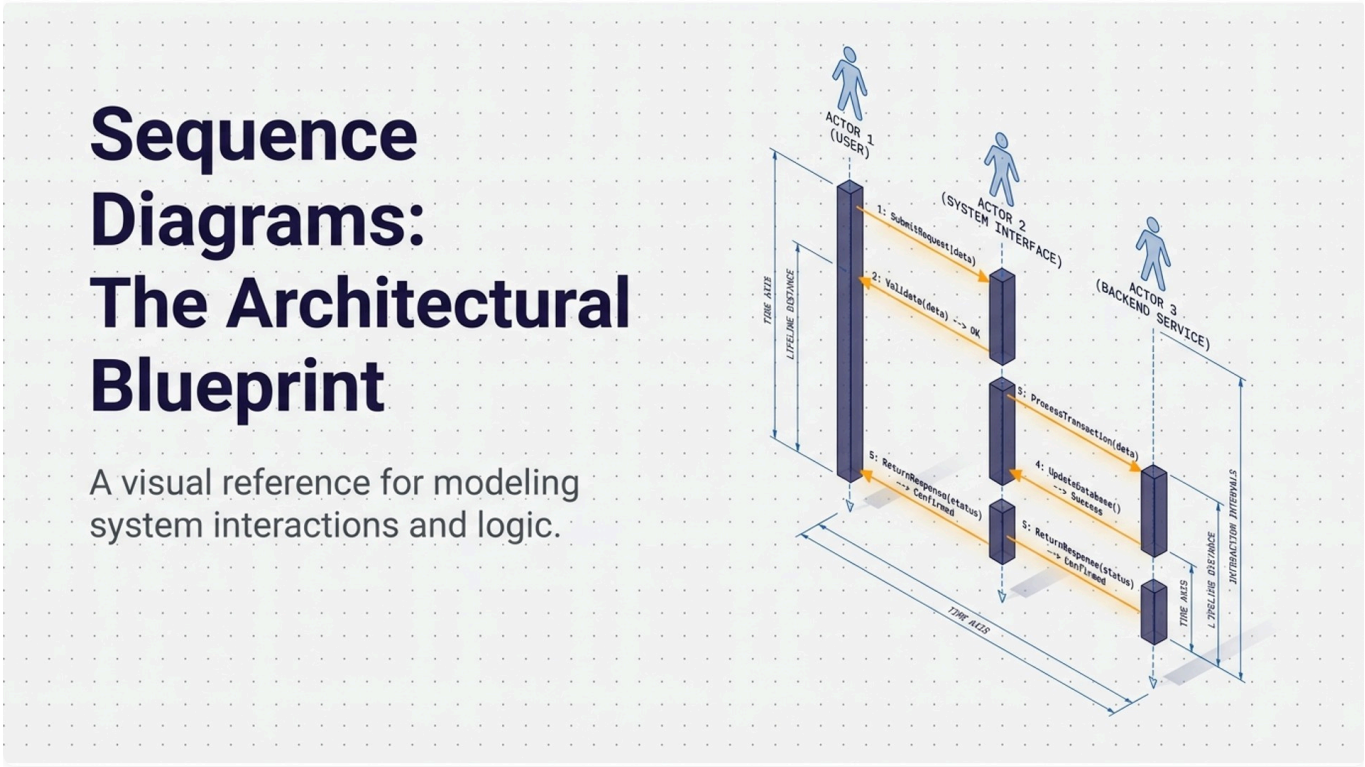


Figure 4.5.1: Dynamic Interaction Overview. Contrasting static class structures with dynamic, time-ordered runtime message exchanges.

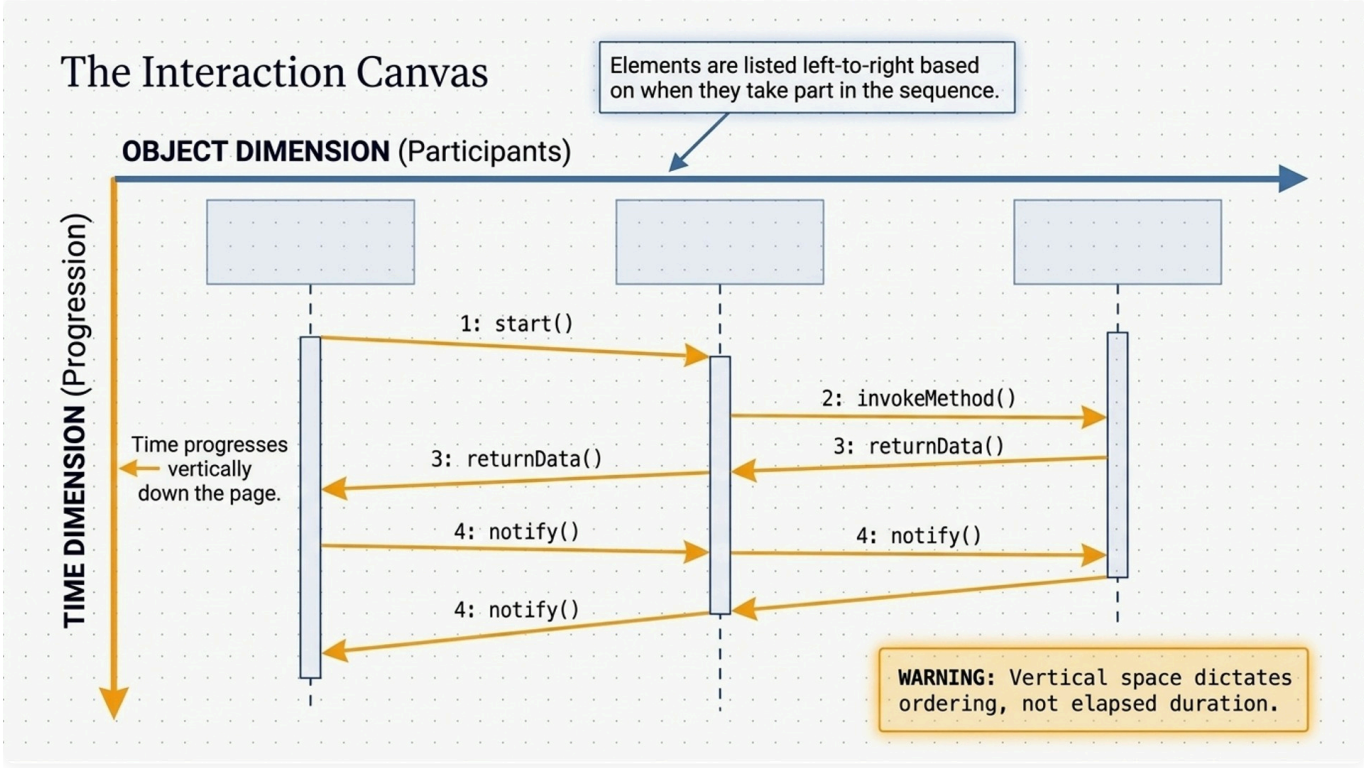


Figure 4.5.2: The Two-Dimensional Interaction Canvas. The horizontal axis organizes participating object lifelines; the vertical axis represents the irreversible passage of time descending downwards.

A sequence diagram organizes runtime interactions across two distinct dimensions:

- **Horizontal Axis (Collaborating Participants):** Each participant is represented at the top by a box labeled `instanceName : ClassName`. Extending downwards is a dashed vertical line known as the **Lifeline**.
- **Vertical Axis (Time):** Time flows strictly **top-to-bottom**. Messages placed lower down occur chronologically later.

4.5.2 Lifelines, Activation Bars, and Message Types

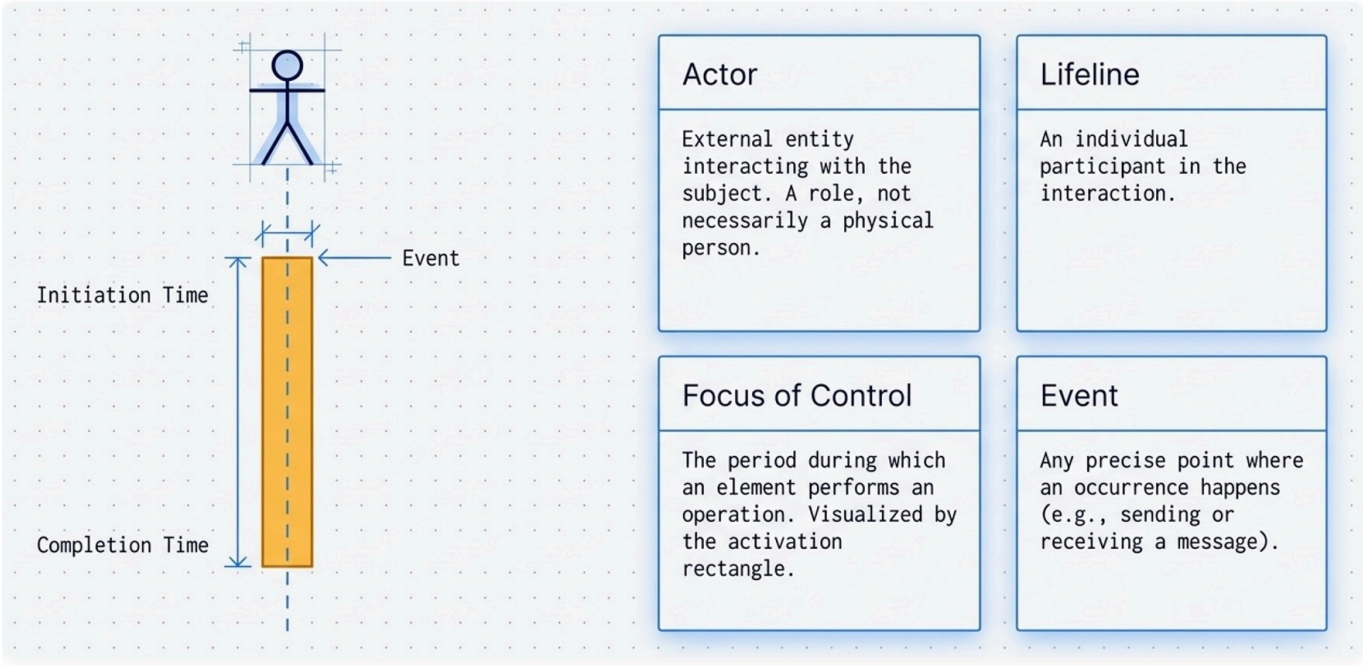


Figure 4.5.3: Lifelines and Activation Focus. The thin vertical rectangle on a lifeline represents the period during which an object is actively executing code or awaiting a subroutine return.

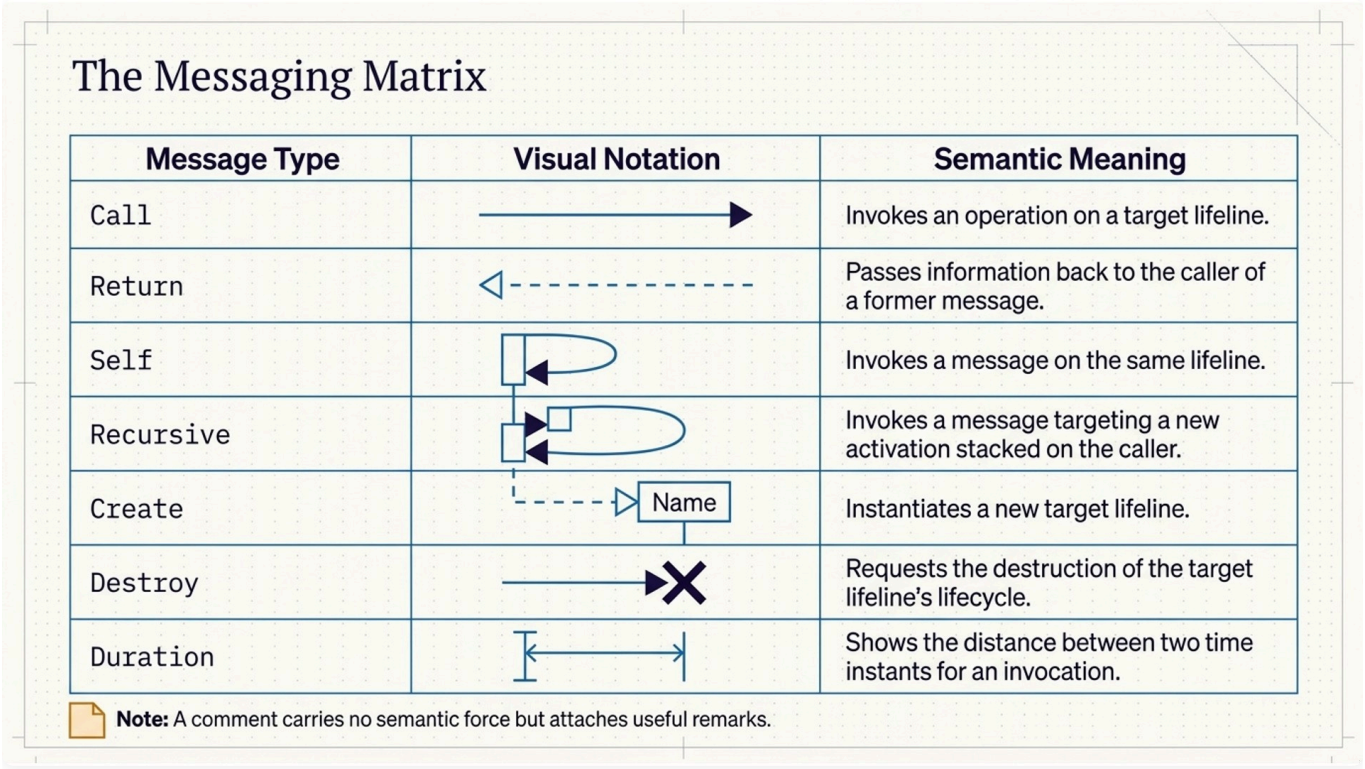


Figure 4.5.4: UML Sequence Messaging Matrix. Visual syntax and execution semantics for synchronous calls, asynchronous signals, return replies, and object creation/destruction.

Message Type	Visual Notation	Execution Semantics
Synchronous Call	Solid line + filled arrow `——►`	**Blocking.** Caller pauses and waits until the callee finishes execution and returns.
Asynchronous Signal	Solid line + open arrow `——>`	**Non-blocking.** Caller dispatches message (e.g. Kafka/queue) and continues immediately.
Return Message	Dashed line + open arrow `◄- - -`	Optional explicit return of control and calculation data back to the caller.
Create Message	Dashed arrow to target box `- - ►`	Instantiates a new runtime object (`new Order()`). Target box drawn at message arrival.
Destroy Message	Large `X` at end of lifeline	Explicit deallocation / garbage collection termination of the runtime instance.

4.5.3 Combined Fragments: Modeling Complex Control Flow

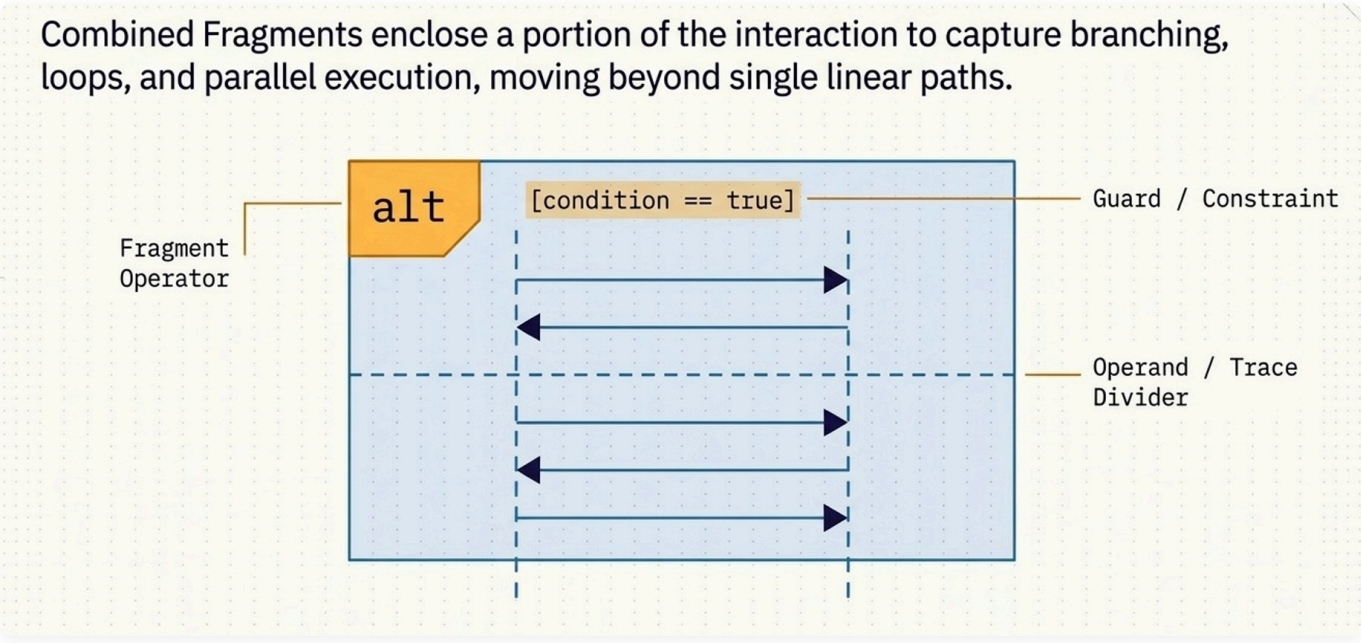


Figure 4.5.5: Combined Fragments Overview. Framing complex logic (branching, loops, concurrency) using standardized UML operator boxes.

Operator	Definition	Code Equivalent
alt	**Alternative:** Multiple fragments; only the one with a true condition executes.	if / else if / else
opt	**Optional:** Executes only if the supplied condition is true. Single trace.	if (without else)
loop	**Loop:** Executes multiple times based on a guard condition.	while / for loop
par	**Parallel:** Each fragment runs concurrently.	multi-threading / async
region	**Critical Region:** Only one thread can execute this fragment at once.	Mutex lock / sync
neg	**Negative:** Shows an explicitly invalid interaction.	Error path / Exception
ref	**Reference:** Refers to an interaction defined on another diagram.	Function / Module call
sd	**Sequence Diagram:** Surrounds an entire root sequence diagram.	Root program

Figure 4.5.6: Mapping Combined Fragments to Code Constructs. alt maps to if-else; opt maps to a single if; loop maps to for/while; par maps to multi-threaded execution.

Operator	Programming Construct	Architectural Meaning
alt	`if (...) { ... } else { ... }`	**Mutually exclusive alternatives.** Only one sub-frame executes based on guard condition.
opt	`if (...) { ... }` *(no else)*	**Optional execution.** Executes if guard condition is true; skipped entirely if false.
loop	`for / while` `loop(min, max)`	**Repetitive execution.** Repeats while condition holds or within specified bounds.
par	`Thread / Promise.all()`	**Parallel concurrent execution.** Sub-frames execute simultaneously across threads.

4.5.4 The BCE (Boundary-Control-Entity) Pattern

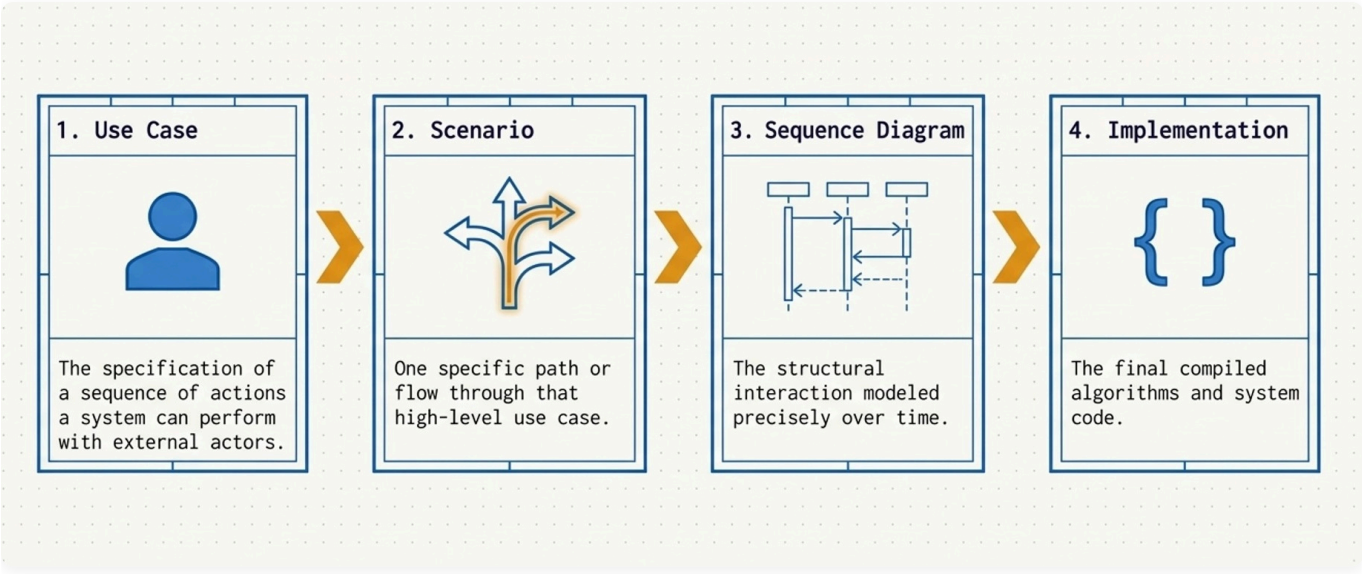


Figure 4.5.7: The BCE Architecture Pipeline. Ensuring robust separation of concerns by routing incoming UI events through dedicated Control orchestrators before touching persistent Entity data.

To prevent sequence diagrams from turning into tightly coupled spaghetti, professional engineers apply the **Boundary-Control-Entity (BCE)** pattern:

- **Boundary (`<<boundary>>`):** Manages communication with external actors (e.g., `CheckoutUI`, `StripePaymentAPI`, `RESTOrderRouter`). Boundaries never contain core business logic.
- **Control (`<<control>>`):** Orchestrates business workflows and transaction boundaries (e.g., `OrderCheckoutController`, `DispatchCoordinator`). Control objects coordinate entities and call boundary adapters.
- **Entity (`<<entity>>`):** Encapsulates persistent business state and domain rules (e.g., `Order`, `Customer`, `OrderItem`). Entities never communicate directly with UI boundaries.

4.5.5 Case Study: Food Delivery Checkout Sequence Flow

1. **User Action:** Customer clicks `submitOrder()` on `CheckoutUI` (Boundary).
2. **Controller Orchestration:** `CheckoutUI` dispatches `checkout(cartId)` to `OrderController` (Control).
3. **Domain Validation:** `OrderController` calls `validateStock()` on `RestaurantCatalog` (Entity).
4. **Third-Party Payment:** `OrderController` invokes synchronous `authorizePayment(amount)` on `PaymentGatewayAPI` (Boundary).
5. **Alternative Branching (`alt`):**
   - `[paymentApproved]`: `OrderController` creates `new Order()` (Entity), updates database state, and asynchronously fires `publishOrderPlaced()` to message broker.
   - `[else / paymentFailed]`: `OrderController` returns error message back to UI; transaction rolled back.

Generated Food Delivery Sequence Model

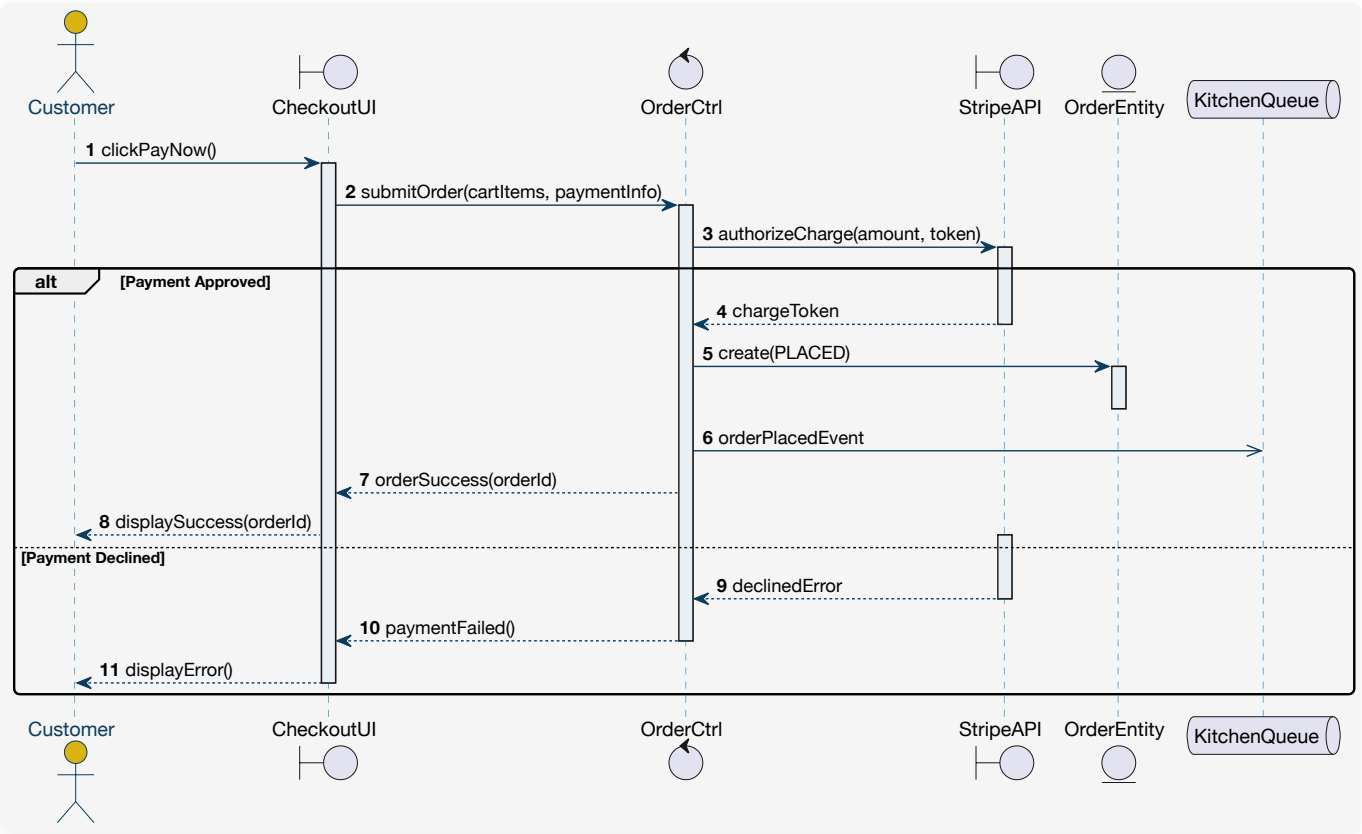


Figure 4.5.8: Generated Food Delivery Sequence Diagram. Tracing the BCE runtime collaboration between Customer, CheckoutUI, OrderCtrl, StripeAPI, OrderEntity, and KitchenQueue.

```
@startuml
autonumber
actor Customer
boundary CheckoutUI
control OrderCtrl
boundary StripeAPI
entity OrderEntity
queue KitchenQueue

Customer -> CheckoutUI: clickPayNow()
activate CheckoutUI
CheckoutUI -> OrderCtrl: submitOrder(cart, payment)
activate OrderCtrl
OrderCtrl -> StripeAPI: authorizeCharge(amount, token)
activate StripeAPI

alt Payment Approved
  StripeAPI --> OrderCtrl: chargeToken
  deactivate StripeAPI
  OrderCtrl -> OrderEntity: create(PLACED)
  OrderCtrl ->> KitchenQueue: orderPlacedEvent
  OrderCtrl --> CheckoutUI: orderSuccess(orderId)
  CheckoutUI --> Customer: displaySuccess()
else Payment Declined
  StripeAPI --> OrderCtrl: declinedError
  OrderCtrl --> CheckoutUI: paymentFailed()
  deactivate OrderCtrl
  CheckoutUI --> Customer: displayError()
  deactivate CheckoutUI
end
@enduml
```

4.5.6 Diagnostic Case Study: Spotting Flaws in Sequence Models (Hotel Reservation Flow)

Just as with structural models, dynamic sequence models frequently suffer from subtle design flaws that compromise architectural integrity, introduce race conditions, or violate component boundaries.

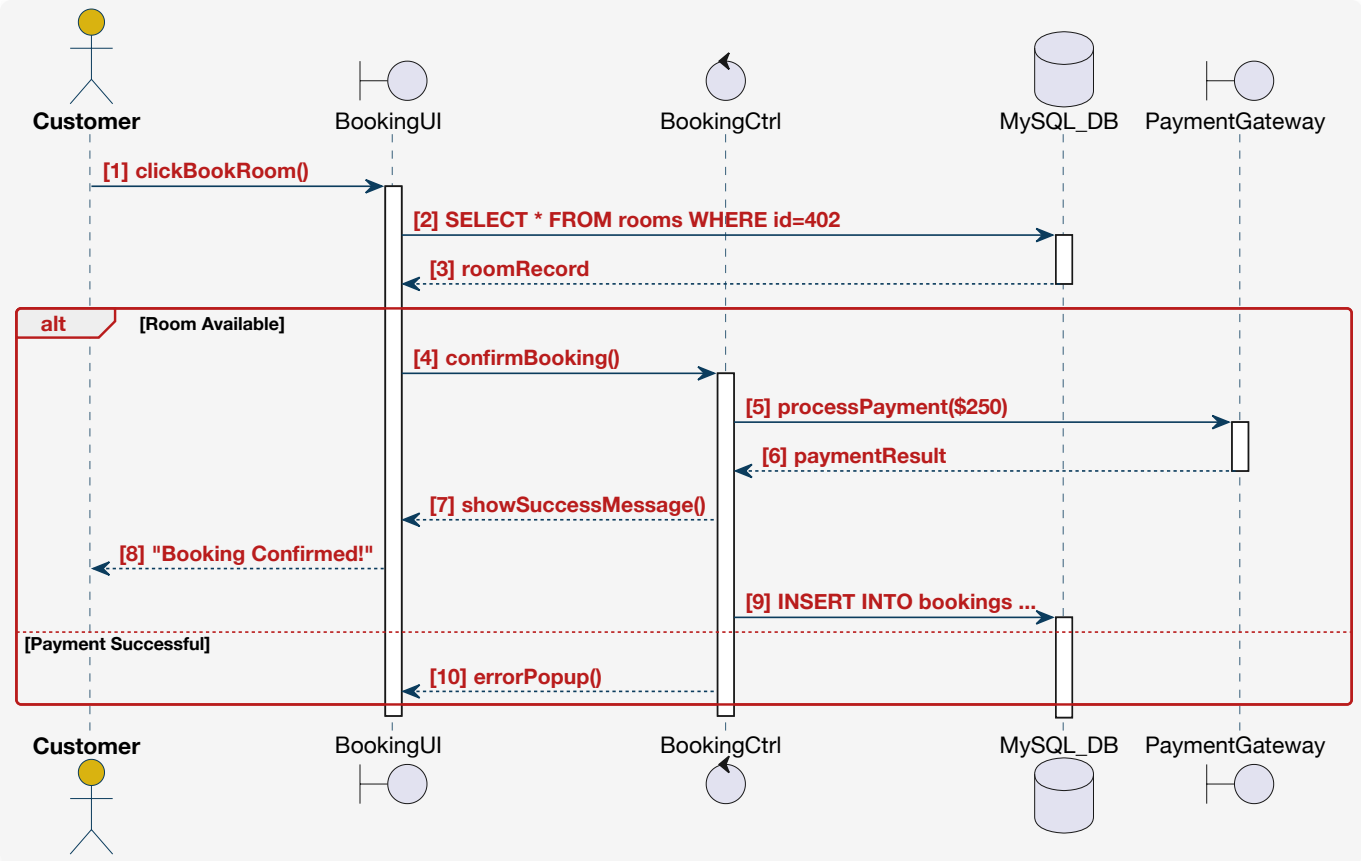


Figure 4.5.9: Flawed Sequence Model Example. A draft sequence diagram exhibiting 4 critical architectural and syntactic anti-patterns.

Diagnostic Architectural Audit: 4 Critical Flaws Identified

1. **Architecture Layering & Encapsulation Violation:**
   - *The Flaw:* In message `[2]`, `BookingUI` sends raw SQL (`SELECT * FROM rooms...`) directly to `MySQL_DB`.
   - *Correction:* In 3-tier and BCE architectures, presentation boundaries must never bypass application controllers to query data stores directly. Such shortcuts create tight coupling and expose database schemas to the UI. Requests must route through a dedicated `BookingController`.
2. **Invalid `alt` Combined Fragment Guards:**
   - *The Flaw:* The first branch of the `alt` block is guarded with `[Room Available]`, but the `else` branch is guarded with `[else Payment Successful]`.
   - *Correction:* Branches within an `alt` fragment must evaluate the same domain condition and be mutually exclusive (e.g., `[Room Available]` vs. `[else / Room Unavailable]`). Mixing room inventory checks with payment processing produces nonsensical execution logic.
3. **Chronological Causality & Premature Confirmation:**
   - *The Flaw:* In messages `[7]` and `[8]`, the UI displays `Booking Confirmed!` to the guest *before* the database write at message `[9]` (`INSERT INTO bookings`) has completed.
   - *Correction:* If the database write fails or crashes, the customer holds an invalid booking confirmation. State changes must be persisted and committed before the presentation layer confirms success.

4. **Dangling Lifeline Activation:**
    - *The Flaw:* The activation bar on `MySQL_DB` starts at message `[9]` but extends downwards indefinitely without a return or deactivation marker.
    - *Correction:* Activation bars represent the active execution focus of a participant. When a query or stored procedure finishes, the activation must terminate cleanly.

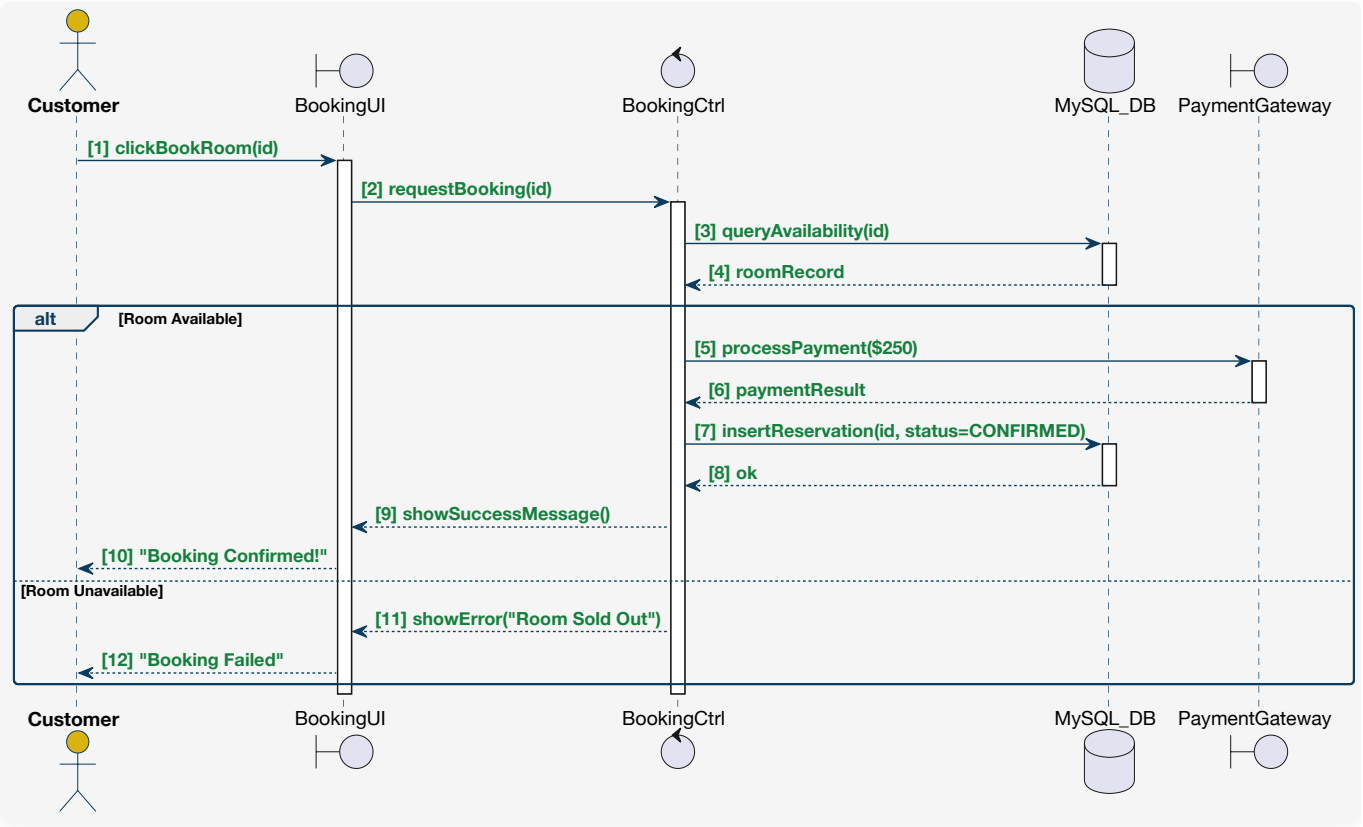


Figure 4.5.10: Corrected Sequence Model. Enforcing BCE layering, mutually exclusive alt guards, state persistence prior to confirmation, and cleanly bounded activations.

4.5.7 Interactive Hands-On Activity: Evolving Sequences with AI

- **Experiment Goal:** Mutate the checkout interaction flow by introducing one edge case into the prompt and audit the AI's architectural output:
    i. *Option A (Gateway Timeout):* Payment gateway takes >5s to respond (trigger retry or graceful fallback).
    ii. *Option B (Split Payment):* Allow customer to pay using Wallet Balance + Credit Card in two coordinated steps.
    iii. *Option C (Out of Stock during Checkout):* Restaurant inventory drops to zero right before payment authorization.
- **Audit Checklist:**
    - Did AI preserve BCE layering without allowing UI boundaries to query databases directly?
    - Are normal success and alternative failure branches cleanly partitioned with `alt` fragments?
    - Are decoupled background tasks (e.g., kitchen alerts) represented as asynchronous messages (`->>`)?

🤔 Concept Check (CCQ 5) — BCE Architecture & Sequence Message Dispatch

Question

In a Boundary-Control-Entity (BCE) sequence diagram, which object should receive the customer's `submitOrder()` event from the checkout UI boundary?

- A. The `Order` entity object to immediately save database state.
- B. The `PaymentGatewayAPI` boundary object to process payment immediately.
- C. The `OrderController` control object to orchestrate business validation and service calls.
- D. The `Restaurant` entity object to confirm kitchen capacity.

Interactive Activity (線上作答)



▶ Click to view Answer & Explanation

4.6 Process & Activity Diagrams

While sequence diagrams excel at tracing messages between specific object instances, **Activity Diagrams** model procedural workflows, business logic, and concurrent token flows across an entire organization or system.

4.6.1 Activity Diagrams vs. Traditional Flowcharts

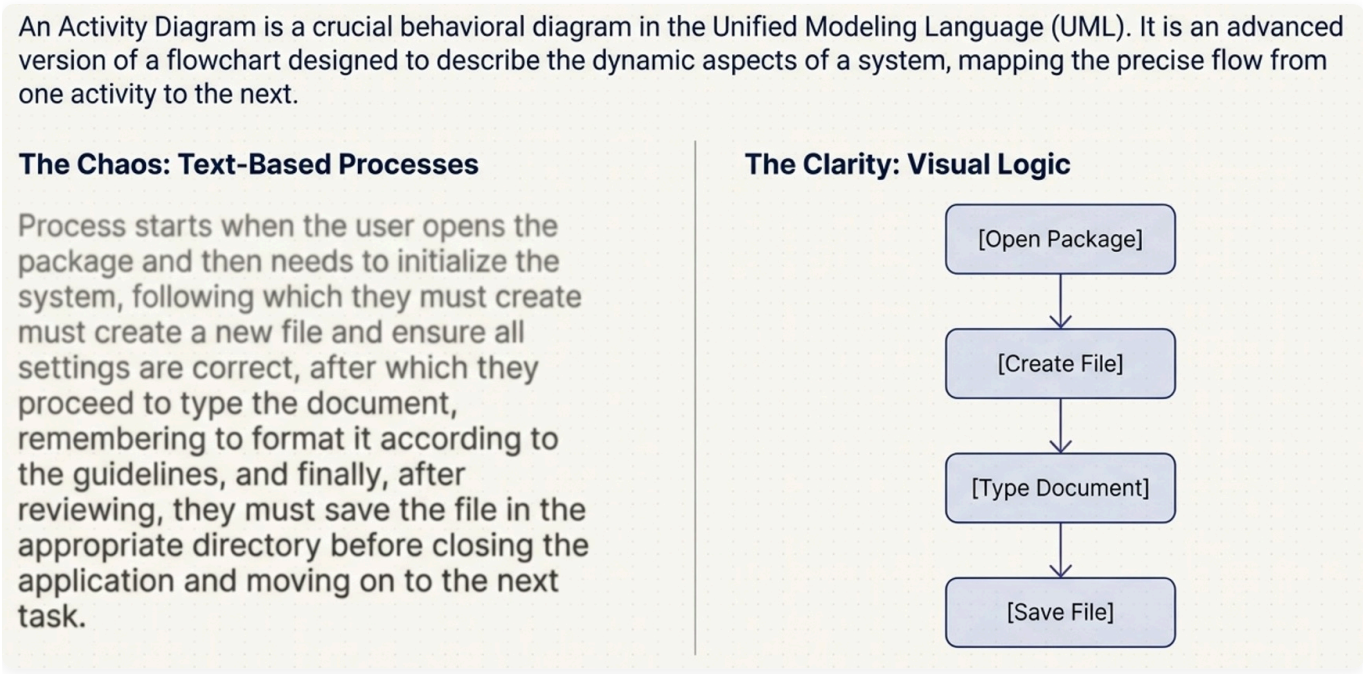


Figure 4.6.1: Visual Logic in Activity Modeling. Activity diagrams extend traditional flowcharts with Petri-net token semantics, parallel fork/join execution, and object state flow.

Characteristic	Traditional Flowchart	UML Activity Diagram
Concurrency Support	None (Strictly single-threaded sequential flow).	**Native Fork/Join bars** for multi-threaded and asynchronous parallel workflows.
Organizational Roles	None (All steps share a single global canvas).	**Swimlanes / Partitions** assign actions explicitly to specific actors, departments, or services.
Data & Object Flow	Bare I/O parallelograms with informal text.	**Explicit Object Nodes** and state pins indicating state changes (`[Paid]`, `[Prepared]`).
Execution Semantics	Informal directional lines.	**Formal Petri-Net Token Semantics** with deterministic join synchronization rules.

4.6.2 Core Syntax: Actions, Decision/Merge, and Fork/Join

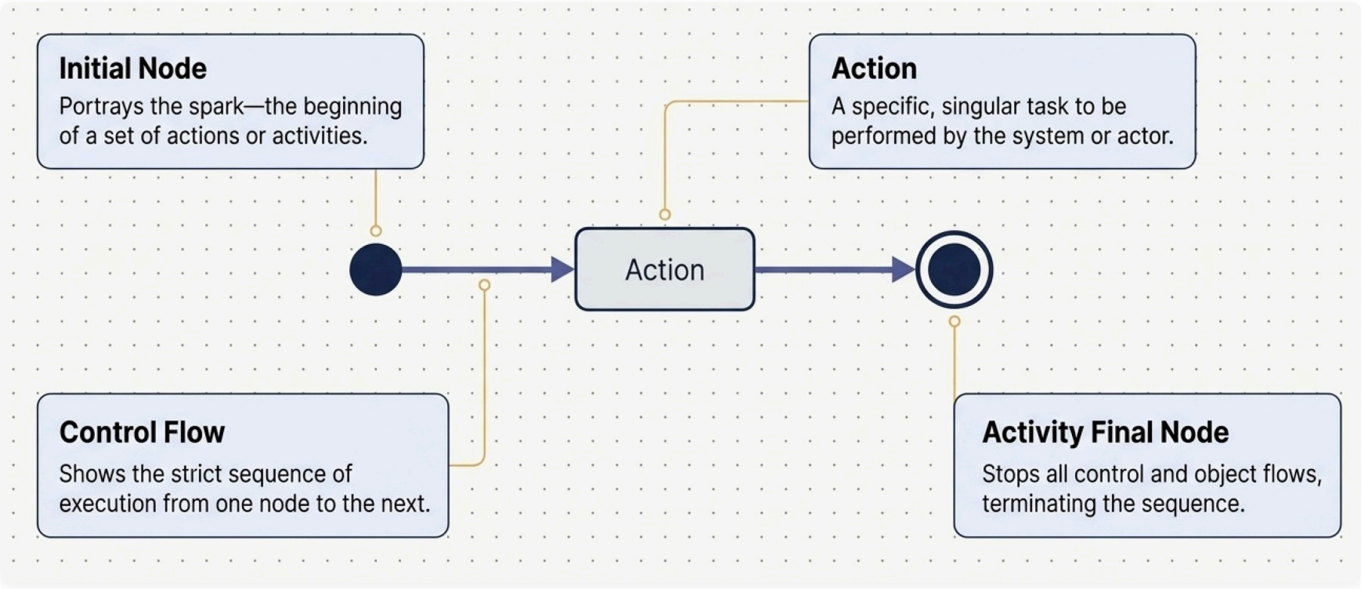


Figure 4.6.2: Core Activity Diagram Nodes. Illustrating Initial node, Action states, Decision/Merge diamonds, and Activity Final nodes.

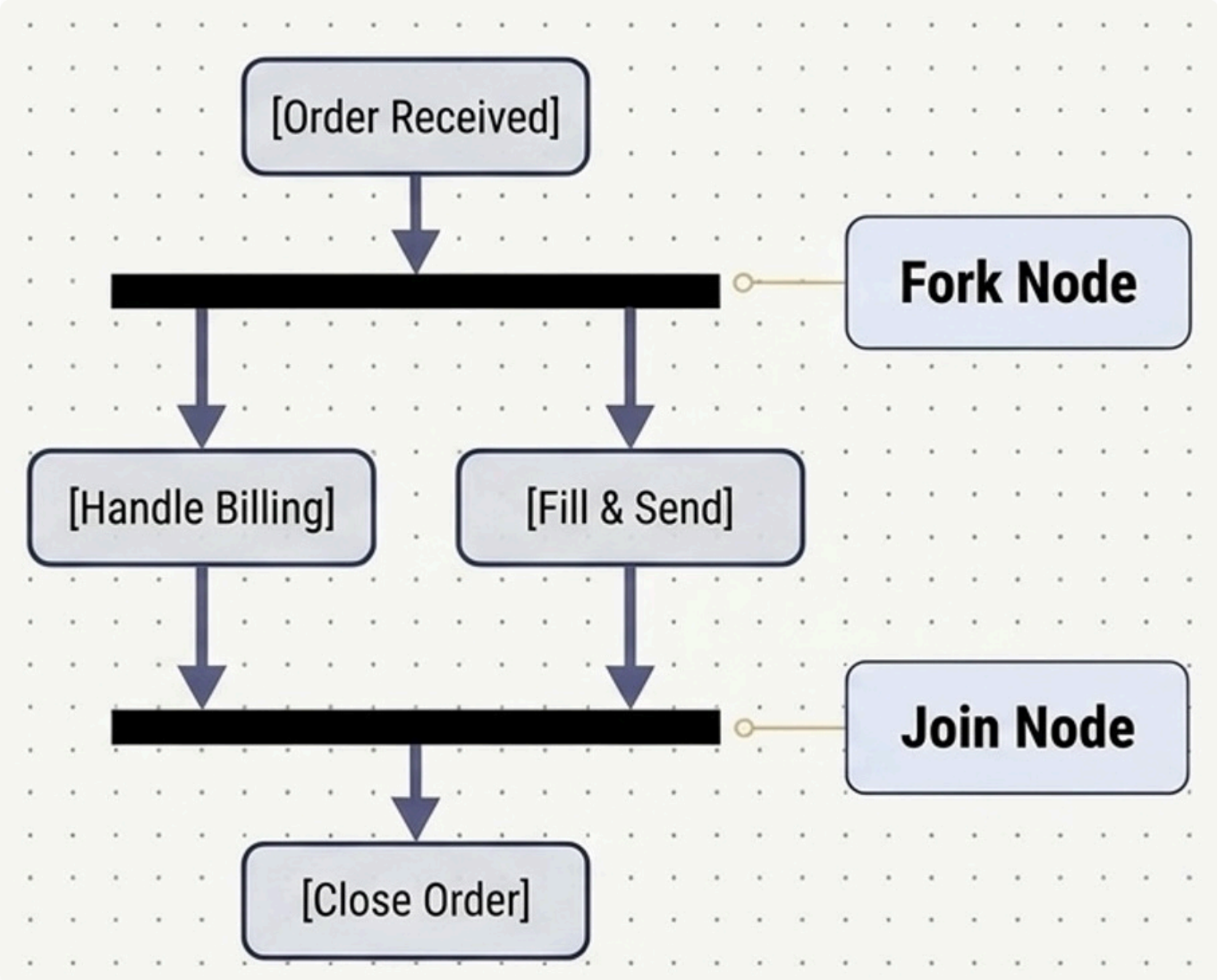


Figure 4.6.3: Concurrency Synchronization with Fork and Join. A Fork node splits a single incoming token into multiple parallel concurrent flows; a Join node synchronizes them back into one.

Critical Distinction: Fork/Join vs. Decision/Merge

- **Decision / Merge (Diamond ◊):** Models **conditional branching**. An incoming token is routed along *exactly one* outgoing branch whose Boolean guard condition evaluates to true (`[stockAvailable]` vs `[outOfStock]`).
- **Fork / Join (Solid Bar ▮):** Models **parallel concurrency**. An incoming token is split into *all* outgoing branches simultaneously. Execution progresses along all parallel threads, and the Join bar blocks until *every* incoming thread has arrived.

4.6.3 Swimlanes (Partitions) and Object Flows

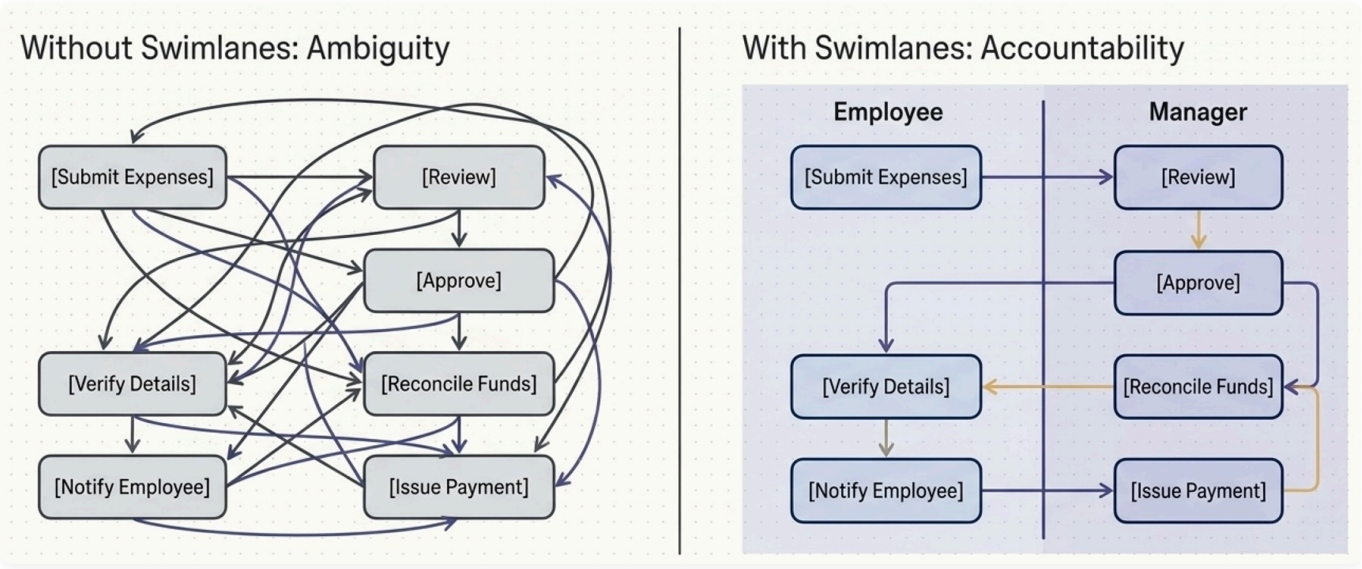


Figure 4.6.4: Swimlanes and Partitions. Columnar divisions allocating responsibility for specific actions to distinct business roles or architectural subsystems.

Swimlanes partition activities into vertical or horizontal columns:

- **Customer Lane:** `Browse Menu` → `Submit Order` → `Eat Meal`.
- **Restaurant Lane:** `Receive Order Ticket` → `Cook Meal` → `Package Food`.
- **Delivery Lane:** `Accept Dispatch Run` → `Pickup from Restaurant` → `Deliver to Doorstep`.

4.6.4 Object Nodes and Stateful Data Flows

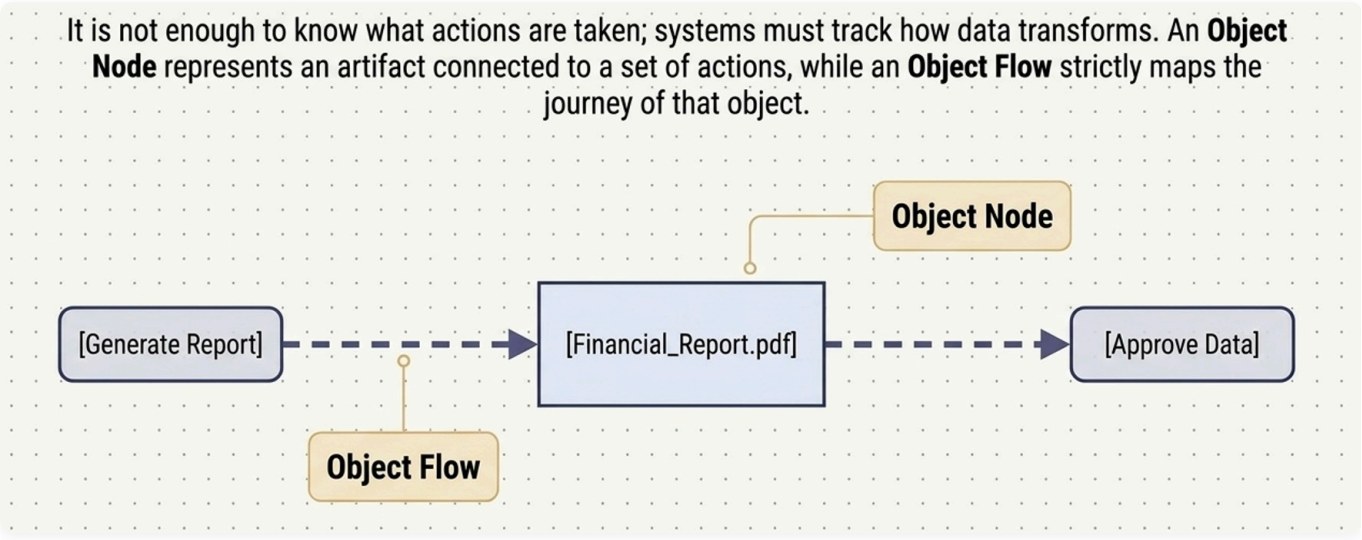


Figure 4.6.5: Object Flows and State Pins. Modeling how business objects change state as they pass between activity actions.

Rather than only showing control arrows, activity diagrams incorporate **Object Nodes** (rectangles displaying `Class [State]`) to illustrate data transformations: `[Order Placed]` ——▶ **Prepare Food** ——▶ `[Meal Cooked]` ——▶ **Package Food** ——▶ `[Meal Ready for Pickup]`.

4.6.5 Case Study: Food Delivery Multi-Actor Activity Workflow

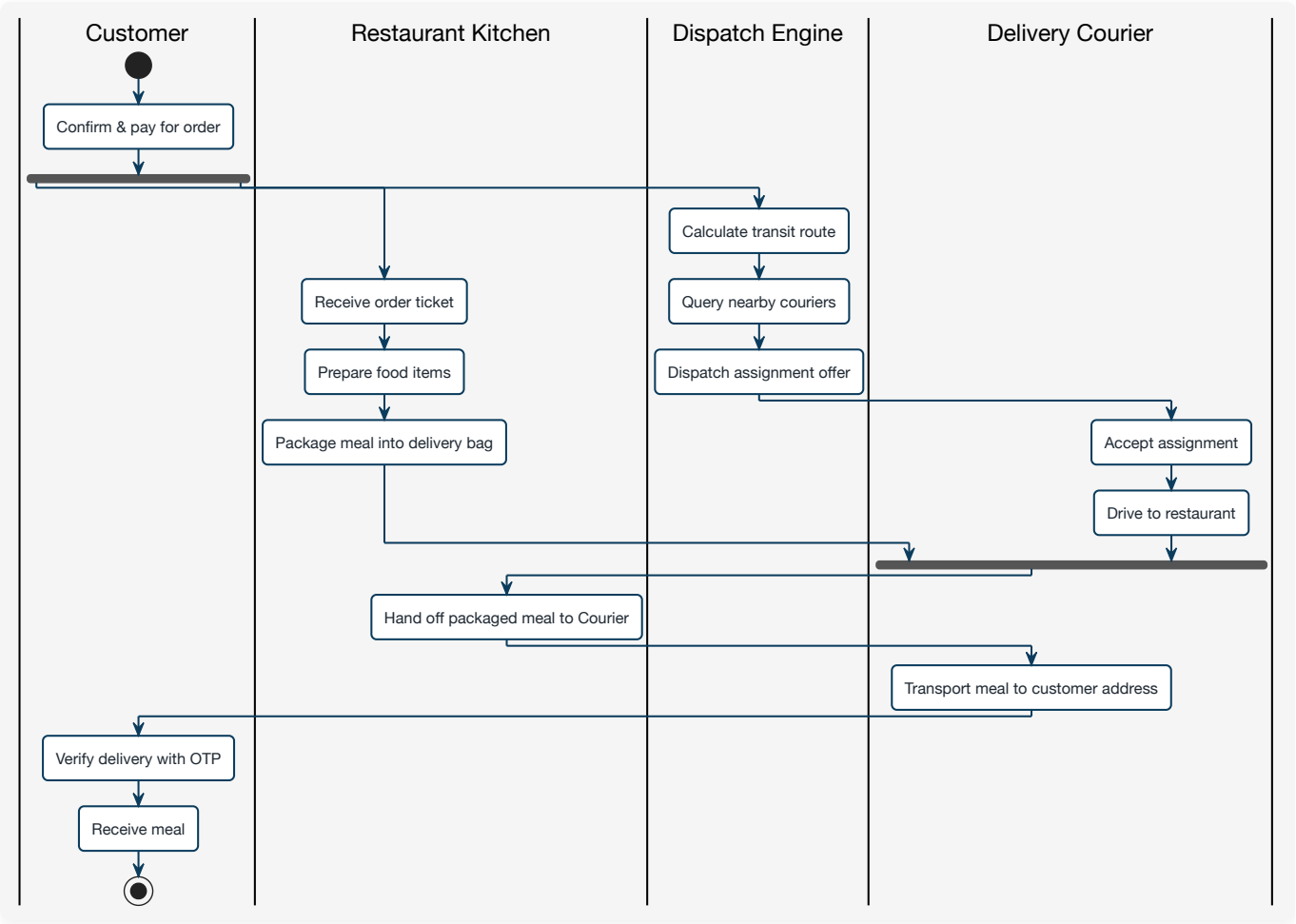


Figure 4.6.6: Generated Food Delivery Activity Diagram. Orchestrating concurrent parallel workflows across Customer, Restaurant Kitchen, Dispatch Engine, and Delivery Courier swimlanes using Fork/Join synchronization.

```
@startuml
|Customer|
start
:Confirm & pay for order;

fork
  |Restaurant Kitchen|
  :Receive order ticket;
  :Prepare food items;
  :Package meal into delivery bag;
fork again
  |Dispatch Engine|
  :Calculate transit route;
  :Query nearby couriers;
  :Dispatch assignment offer;
  |Delivery Courier|
  :Accept assignment;
  :Drive to restaurant;
end fork

|Restaurant Kitchen|
:Hand off packaged meal to Courier;

|Delivery Courier|
:Transport meal to customer address;

|Customer|
:Verify delivery with OTP;
:Receive meal;
stop
@enduml
```

4.6.6 Diagnostic Case Study: Spotting Flaws in Activity Workflows (Food Order Fulfillment)

Designing robust business processes requires careful attention to token semantics, synchronization rules, and actor ownership.

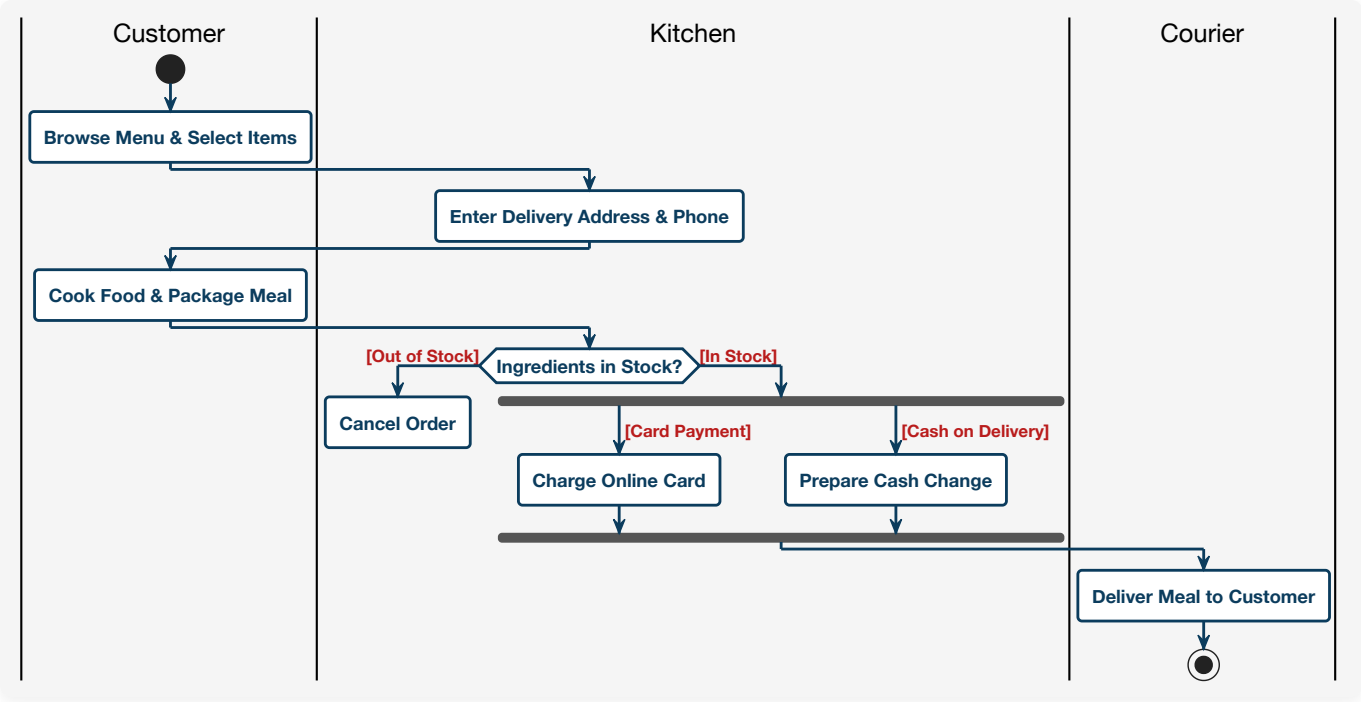


Figure 4.6.7: Flawed Activity Model Example. An order fulfillment workflow exhibiting 4 critical concurrency, branching, and swimlane anti-patterns.

Diagnostic Architectural Audit: 4 Critical Flaws Identified

1. **Fork Bar vs. Decision Diamond:**
   - *The Flaw:* In the Kitchen swimlane, a `fork` bar is drawn with mutually exclusive guards `[Card Payment]` and `[Cash on Delivery]`.
   - *Correction:* A `fork` bar initiates concurrent parallel threads where *all* outgoing branches execute simultaneously. Putting payment options on a fork means the system charges the credit card *and* prepares cash change at the same time. Mutually exclusive choices require a Decision diamond (◇), not a Fork bar (▮).
2. **Swimlane Responsibility Inversion:**
   - *The Flaw:* Roles are absurdly reversed: `Enter Delivery Address & Phone` is assigned to the `Kitchen` lane, while `Cook Food & Package Meal` is placed in the `Customer` lane!
   - *Correction:* Actions must strictly reside within the swimlane of the actor or system responsible for their execution. The customer inputs address details; the kitchen prepares and packages the food.
3. **Dangling Control Flow & Missing Activity Final Node:**
   - *The Flaw:* When ingredients are `[Out of Stock]`, the branch executes `Cancel Order` and abruptly terminates into empty space.
   - *Correction:* In UML activity semantics, every terminating branch must explicitly lead to an Activity Final Node (◉ , solid circle with outer ring) or Flow Final Node (⊗).
4. **Deadlock Hazard on Asymmetric Synchronization:**
   - *The Flaw:* Merging mutually exclusive decision branches into a Join bar.
   - *Correction:* A Join bar blocks until a token arrives on *every* incoming arrow. Feeding alternative branches into a Join creates permanent system deadlock. Merge diamonds must

reunite alternative flows, while Join bars synchronize concurrent threads.

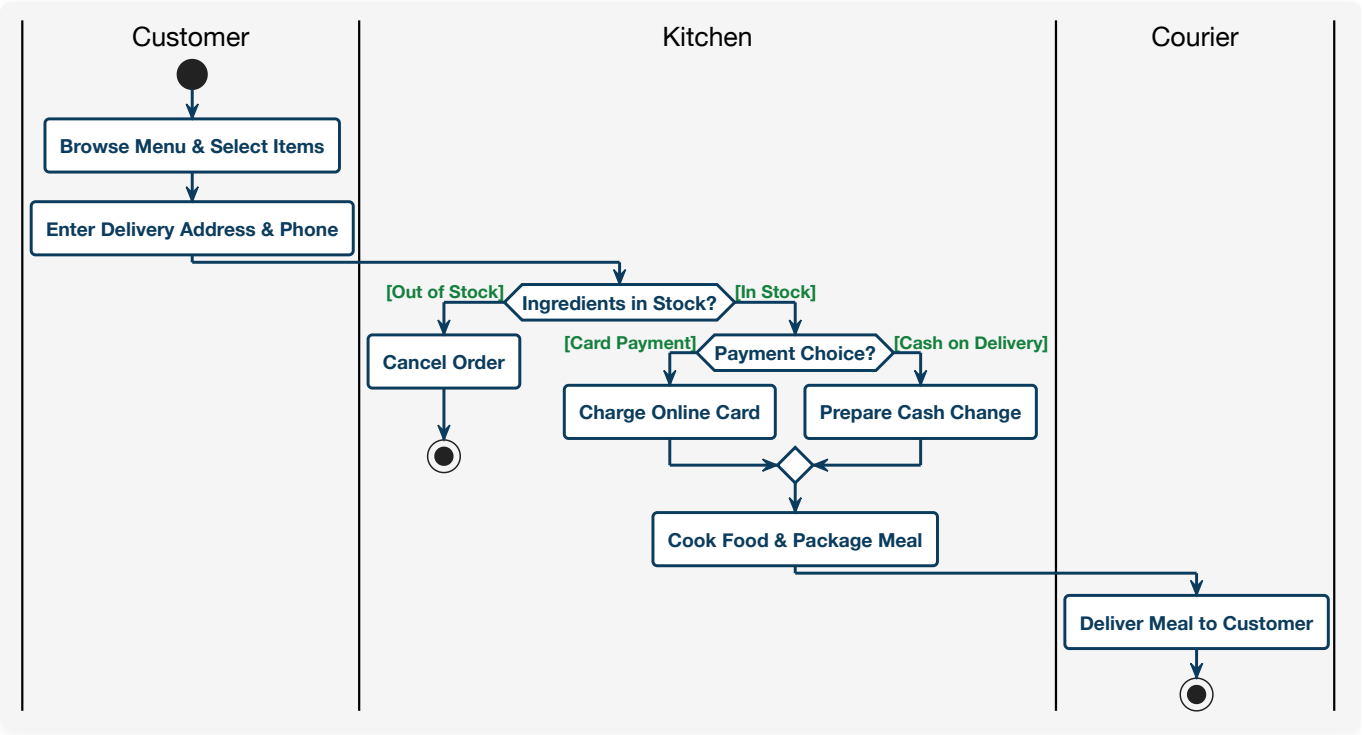


Figure 4.6.8: Corrected Activity Model. Demonstrating proper Decision diamonds, aligned swimlane accountability, explicit final termination, and valid Join synchronization.

4.6.7 Comparative Architectural Matrix: Activity Diagram vs. State Machine

Both activity diagrams and state machines model dynamic behavior, but answer fundamentally different architectural questions:

Comparison Dimension	State Machine Diagram (Statechart)	Activity Diagram (Workflow)
Core Perspective	**Entity-Centric / Reactive:** Models the complete lifecycle of a *single entity* over time.	**Process-Centric / Procedural:** Models the end-to-end *workflow* across multiple actors and systems.
Primary Node	**State:** Quiescent condition where an object waits for events or performs continuous work.	**Action / Activity:** An active computational step or business operation being executed.
Transition Driver	**Event-Driven:** Transitions wait for triggers (`Event [Guard] / Action`).	**Completion-Driven:** Flows advance automatically upon task completion (`Control / Object Flow`).
Concurrency Model	**Orthogonal Regions:** Parallel sub-states within a composite state (`	
Optimal Use Case	Entity lifecycle (Order, Account, TCP socket), IoT devices, status machines.	Business process modeling (checkout pipeline), ETL batch jobs, use case elaboration.

4.6.8 Interactive Hands-On Activity: Evolving Workflows with AI

- **Experiment Goal:** Mutate the food fulfillment process by introducing one operational variation into the prompt and audit the AI's architectural output:
  i. *Option A (Courier Decline & Re-dispatch Loop):* Courier declines assignment; dispatch loops back to query next candidate.
  ii. *Option B (Kitchen Quality Check Loop):* If packaging fails QA inspection, meal is remade before courier hand-off.
  iii. *Option C (Customer Self-Pickup / Takeaway):* Customer picks up meal at counter, completely bypassing courier matching and dispatch swimlanes.
- **Audit Checklist:**
  - Did AI partition actions into the correct swimlanes without role reversal?
  - Are loop constructs (`repeat ... repeat while`) properly terminated with explicit exit branches?
  - Are parallel forks and joins properly paired without causing deadlock?

🤦 Concept Check (CCQ 6) — Activity Diagram Fork/Join vs. Decision/Merge

Question

In a UML Activity Diagram, what is the critical behavioral difference between a **Fork/Join** synchronization bar and a **Decision/Merge** diamond?

- A. Fork/Join splits and synchronizes concurrent parallel threads; Decision/Merge evaluates guards to pick exactly one mutually exclusive branch.
- B. Fork/Join is used for sequential database transactions; Decision/Merge is used for swimlane partitioning.
- C. Fork/Join models class inheritance; Decision/Merge models object creation.
- D. Fork/Join requires human operator approval; Decision/Merge is executed by automated timers.

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.7 Behavioral & State Machine Models

Many software components (such as order fulfillment workflows, IoT thermostats, network connections, and shopping carts) are inherently **state-dependent**. A state machine model captures the complete dynamic lifecycle of a single entity.

4.7.1 Same Event, Different Results

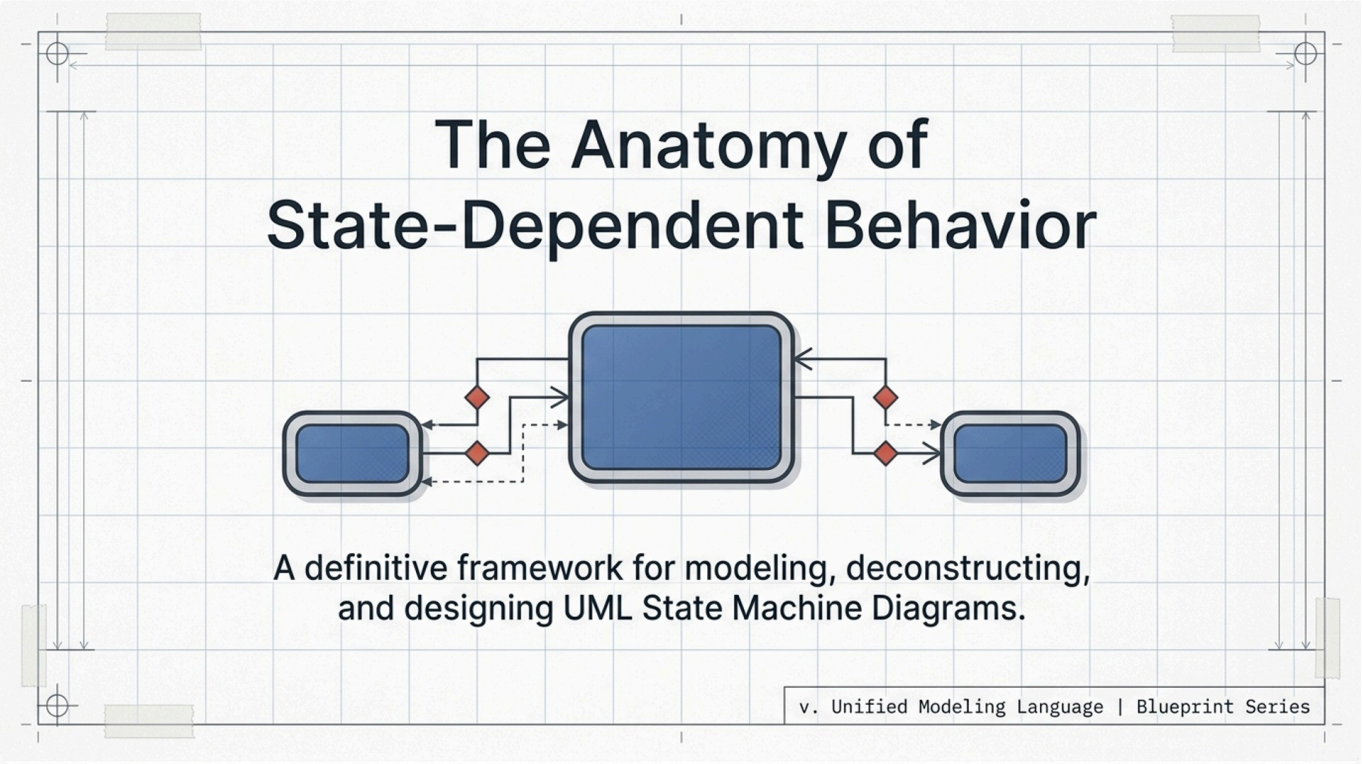


Figure 4.7.1: Anatomy of State-Dependent Behavior. An object's response to an incoming event depends fundamentally on its current internal state.

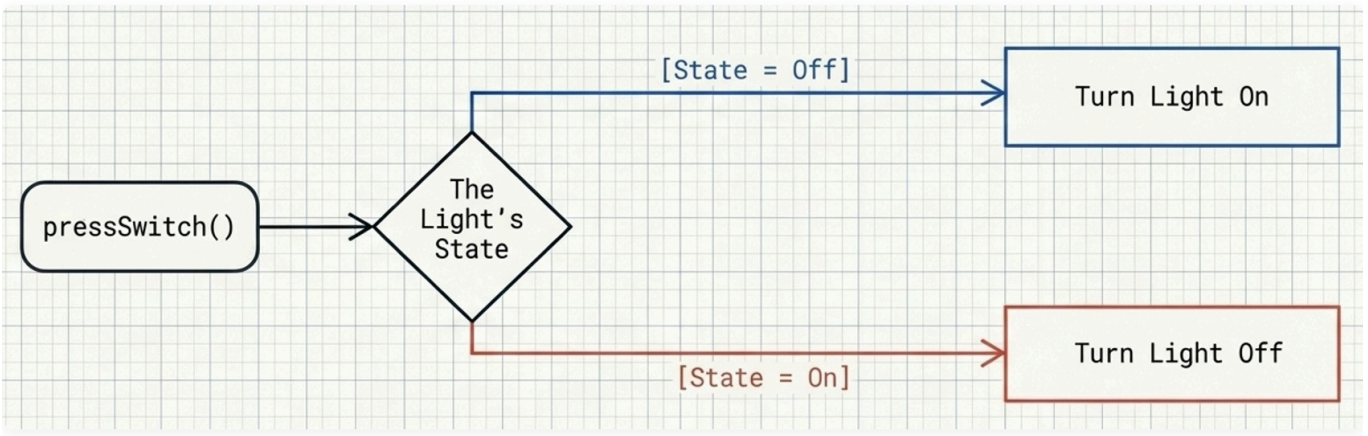


Figure 4.7.2: Same Event, Different Results. Illustrating how submitting payment produces success in the 'Draft' state, but raises an error in the 'Already Paid' state.

Consider an e-commerce `Order` object receiving the event `cancel()`:

- If the order is in the `PaymentPending` state: The order is immediately cancelled and marked `Cancelled`.
- If the order is in the `InTransit` state: Cancellation is rejected; a courier cannot abandon food mid-highway.
- **The fundamental insight:** The behavior of an entity is governed not just by the incoming method call, but by its **current lifecycle state**.

4.7.2 Transition Syntax and Mechanics

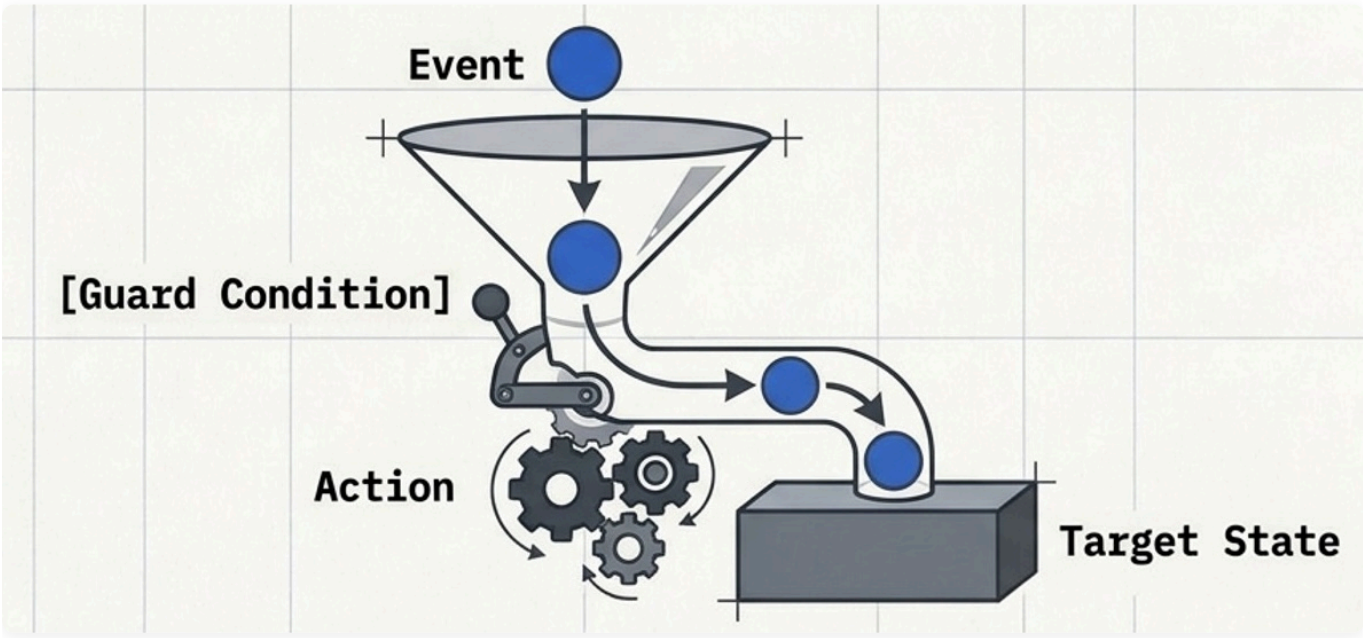


Figure 4.7.3: Transition Syntax. The 5-element transition formula: Trigger [Guard] / Effect.

A UML state transition connects a source state to a target state using the formal syntax:

```
Trigger [Guard] / Effect
```

- **Trigger (Event):** What causes the transition (e.g., `cardSwiped`, `timerExpired`, `cancelRequested`).
- **Guard (Boolean Condition):** A condition enclosed in square brackets `[balance >= total]` that must be true for the transition to fire.
- **Effect (Action):** An atomic, instantaneous procedure executed during the transition (e.g., `/ deductBalance()`).

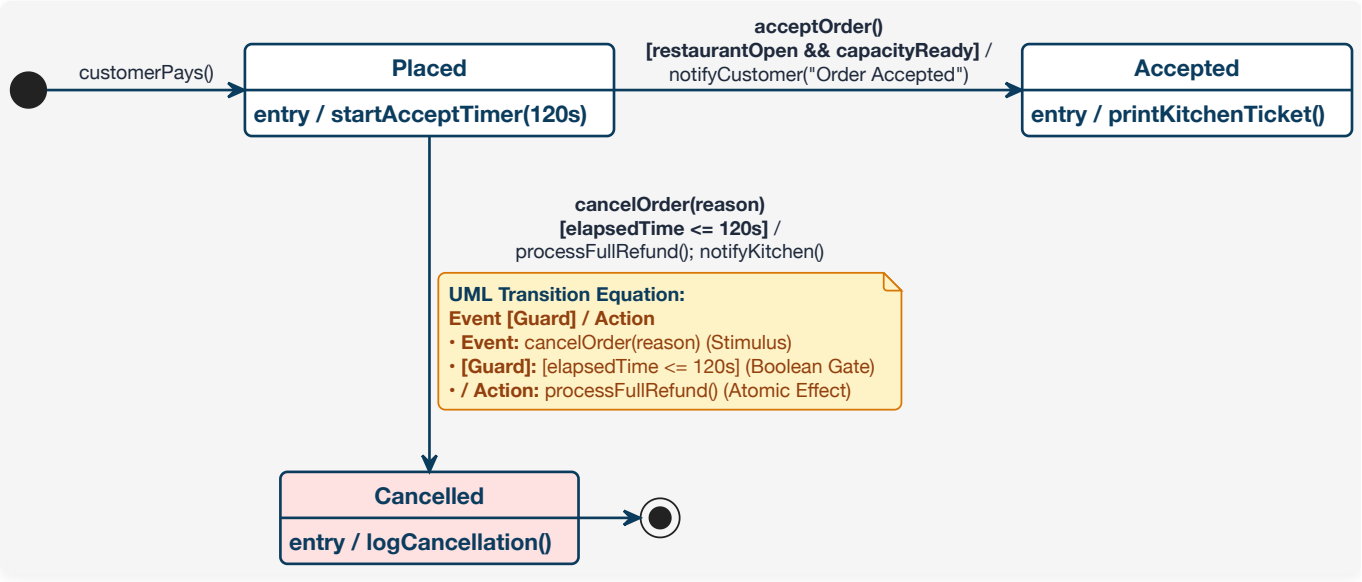


Figure 4.7.4: Transition in Action: Food Delivery Order Example. Applying the 5-element transition formula to enforce business cancellation rules.

In the Food Delivery system:

- **Concrete Transition Syntax:** `cancelOrder(reason) [elapsedTime <= 120s] / processFullRefund(); notifyKitchen()`.
- **Event Stimulus (`cancelOrder`):** The customer taps "Cancel Order" in the mobile client.
- **Guard Condition (`[elapsedTime <= 120s]`):** The system strictly permits free cancellation within two minutes of order placement. If `elapsedTime > 120s`, the guard evaluates to **false**, blocking the transition.
- **Action Effect (`/ processFullRefund()`):** An atomic side-effect executes immediately during the transition, initiating payment reversal.
- **Target State (`Cancelled`):** The order enters `Cancelled` and executes `entry / logCancellation()`.

4.7.3 Actions vs. Activities: Instantaneous vs. Ongoing

Feature	Action	Activity
Nature	Atomic Computation (Operation call, destroy object)	Non-atomic computation
Duration	Instantaneous	Ongoing (Runs to completion or indefinitely)
Interruptibility	NEVER interruptible	YES, can be interrupted by events
Triggers	On Entry, Do, On Event, On Exit	Sustained state presence

Figure 4.7.5: Actions vs. Activities. Actions are instantaneous and atomic; Activities represent continuous, interruptible background work.

Architectural Dimension	Action (`entry /` , `exit /` , transition `/`)	Activity (`do / ...`)
Time Duration	**Zero (Conceptually instantaneous)**	Takes significant real-time duration
Interruptibility	**Non-interruptible (Atomic)**	**Interruptible**; immediately aborted if an outgoing transition triggers
Execution Context	Executes at entry, exit, or along transition line	Runs continuously in background while remaining inside state

Boundary Executions: Entry and Exit Actions

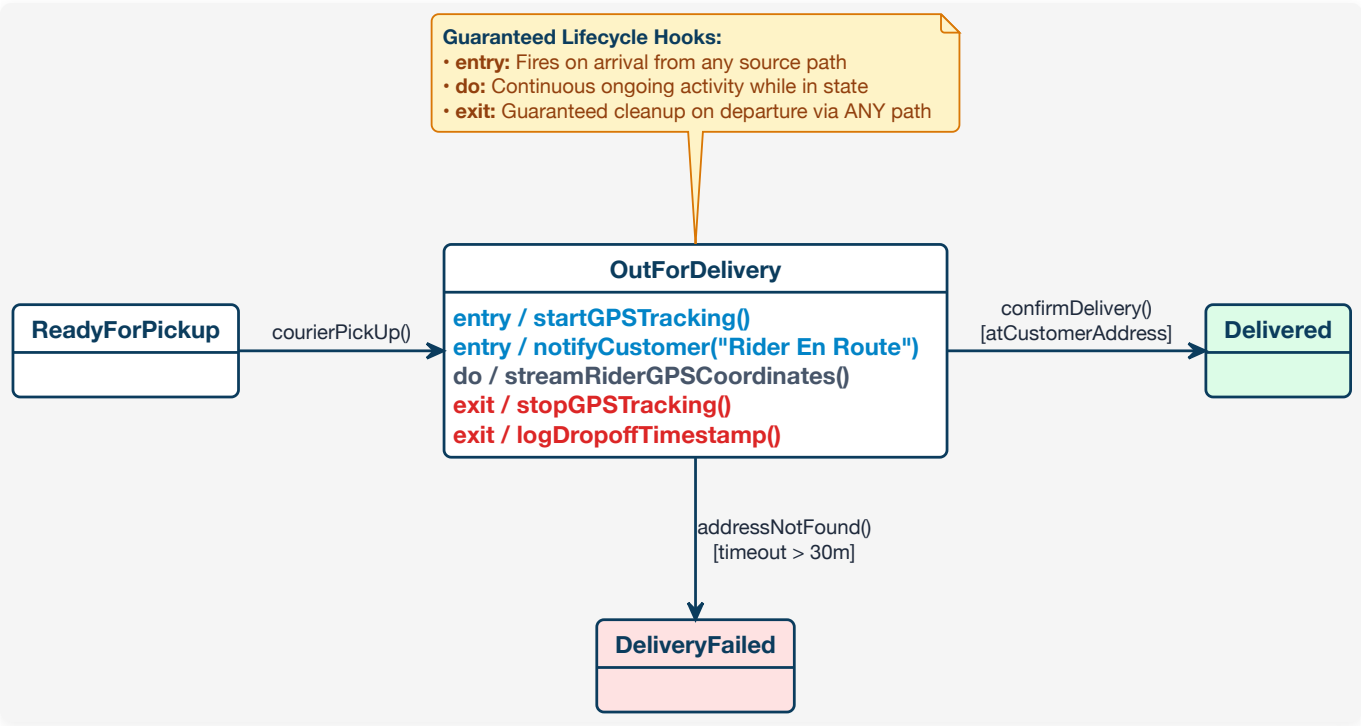


Figure 4.7.6: Boundary Executions: Entry and Exit Actions in Food Delivery. Encapsulating guaranteed lifecycle hooks across state boundaries.

By binding setup and teardown routines to state boundaries, architects prevent duplicate cleanup code across multiple exit transitions:

- **Entry Action (`entry / startGPSTracking()`):** Guaranteed to execute immediately whenever `OutForDelivery` is entered, initializing rider geolocation tracking and notifying the customer.
- **Ongoing Activity (`do / streamRiderGPSCoordinates()`):** Continues streaming real-time coordinates throughout the trip; immediately aborted if arrival or cancellation triggers.
- **Exit Action (`exit / stopGPSTracking()`):** Guaranteed to execute regardless of whether delivery succeeds (`confirmDropoff()`) or fails (`addressNotFound()`), shutting down GPS hardware and logging timestamps.

4.7.4 Composite States, Orthogonal Regions, and History Pseudostates

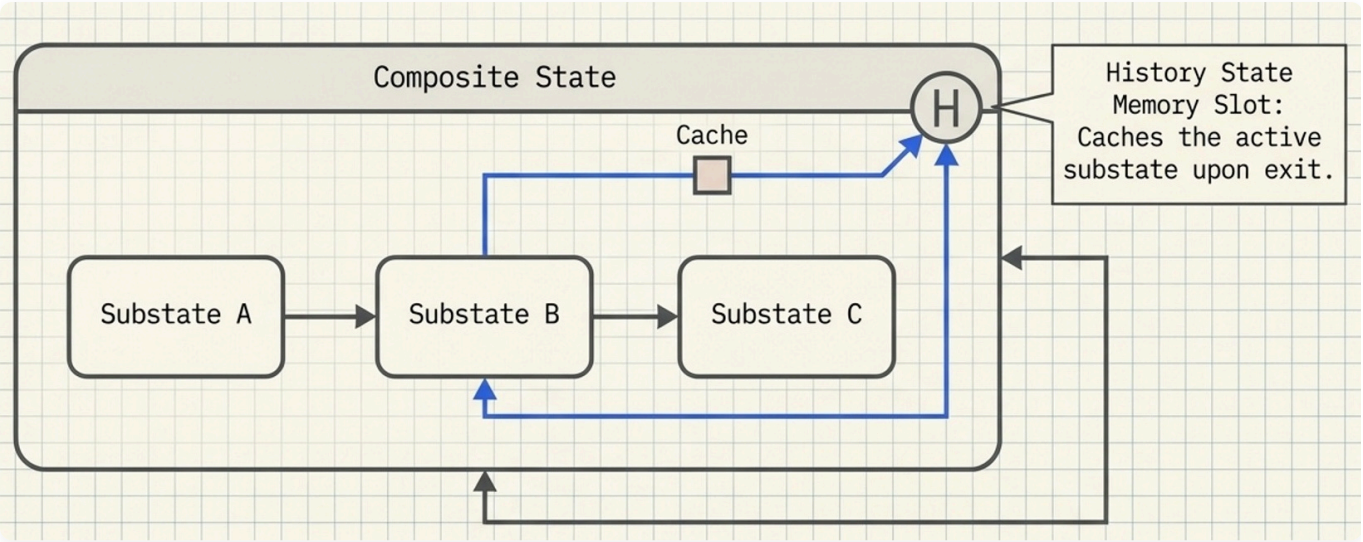


Figure 4.7.7: Composite States and History Pseudostates. Grouping sub-states into a composite parent prevents transition explosion and supports state restoration upon interruption.

- **Composite (Nested) States:** Group related sub-states inside a parent container. Any transition attached to the super-state applies universally to all sub-states (e.g., `emergencyCancel()` on `OrderFulfillment` cancels the order regardless of whether the kitchen was cooking or packaging).
- **Concurrent Orthogonal Regions (`||`):** Independent parallel sub-states executing simultaneously within a composite state.

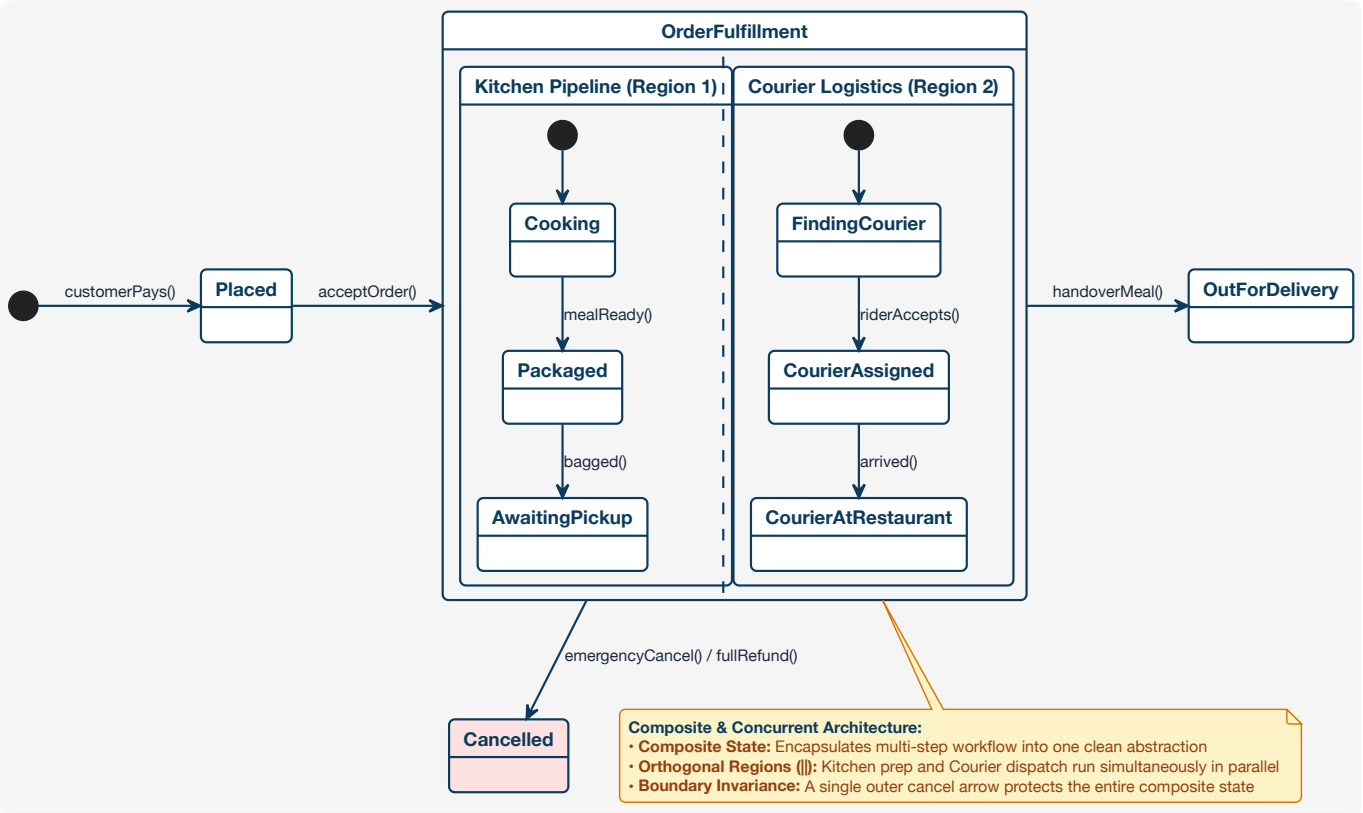


Figure 4.7.8: Composite & Concurrent States in Food Delivery. Orchestrating Kitchen Preparation and Courier Logistics in parallel orthogonal regions.

In `OrderFulfillment`, execution splits into two concurrent tracks:

1. **Kitchen Pipeline (Region 1):** `Cooking` → `Packaged` → `AwaitingPickup`.
2. **Courier Logistics (Region 2):** `FindingCourier` → `CourierAssigned` → `CourierAtRestaurant`.

Both regions progress concurrently without blocking each other.

Preserving Progress with History States (`H` / `H*`)

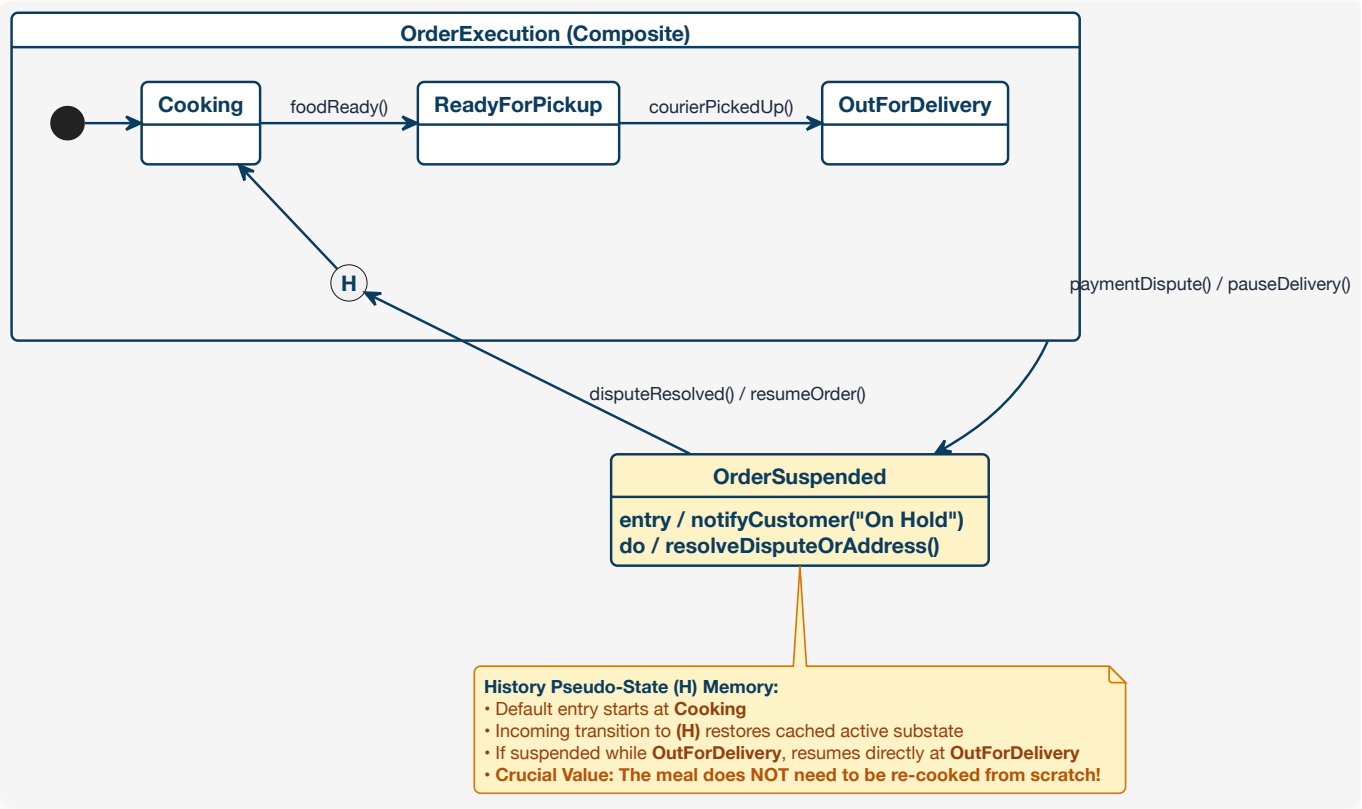


Figure 4.7.9: Preserving Order Progress with History States. Restoring active execution without restarting from scratch.

When an order in `OutForDelivery` is temporarily halted by a payment fraud alert or address dispute, it transitions to `OrderSuspended`.

- **Without History:** Upon dispute resolution, the system would restart at the initial state (`Cooking`), forcing the kitchen to wastefully re-cook the meal!
- **With Deep History (`H*`):** The transition directs execution directly into `(H)`, which restores the exact active substate (`OutForDelivery`), allowing the courier to simply resume transit.

4.7.5 Case Study: Food Delivery Order Lifecycle State Machine

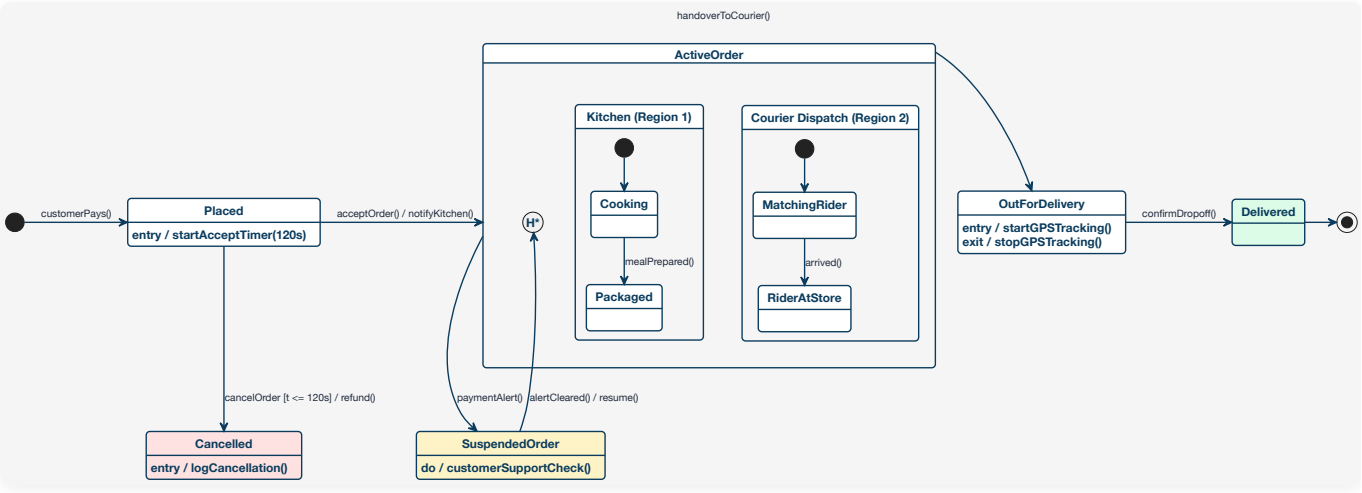


Figure 4.7.10: The Unified Blueprint: Food Delivery Order Architecture. Integrating guarded transitions, concurrent orthogonal regions, history restoration, and boundary actions.

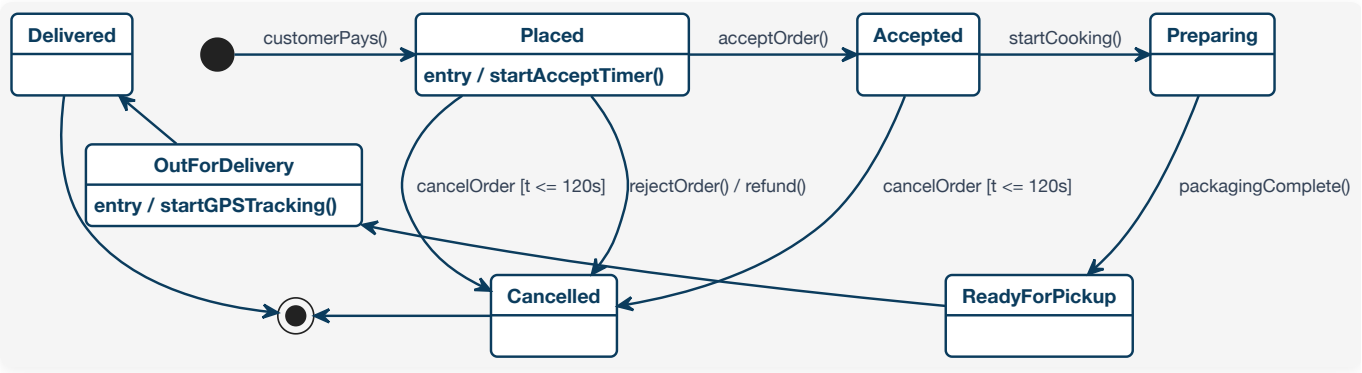


Figure 4.7.11: Generated Food Delivery State Machine Diagram. Baseline order statechart rendered from PlantUML specification.

```
@startuml
[*] -right-> Placed : customerPays()

state Placed {
  Placed : entry / startAcceptTimer()
}
state OutForDelivery {
  OutForDelivery : entry / startGPSTracking()
}

Placed -right-> Accepted : acceptOrder()
Accepted -right-> Preparing : startCooking()
Preparing -down-> ReadyForPickup : packagingComplete()
ReadyForPickup -left-> OutForDelivery : courierPickedUp()
OutForDelivery -left-> Delivered : confirmDropoff()
Delivered -down-> [*]

Placed -down-> Cancelled : rejectOrder() / refund()
Placed -down-> Cancelled : cancelOrder [t <= 120s]
Accepted -down-> Cancelled : cancelOrder [t <= 120s]
Cancelled -left-> [*]
@enduml
```

4.7.6 Diagnostic Case Study: Spotting Flaws in State Machine Models (Order Lifecycle)

State machine diagrams enforce strict behavioral invariants. Even small errors in state definition or transition guards can introduce race conditions or deadlocks.

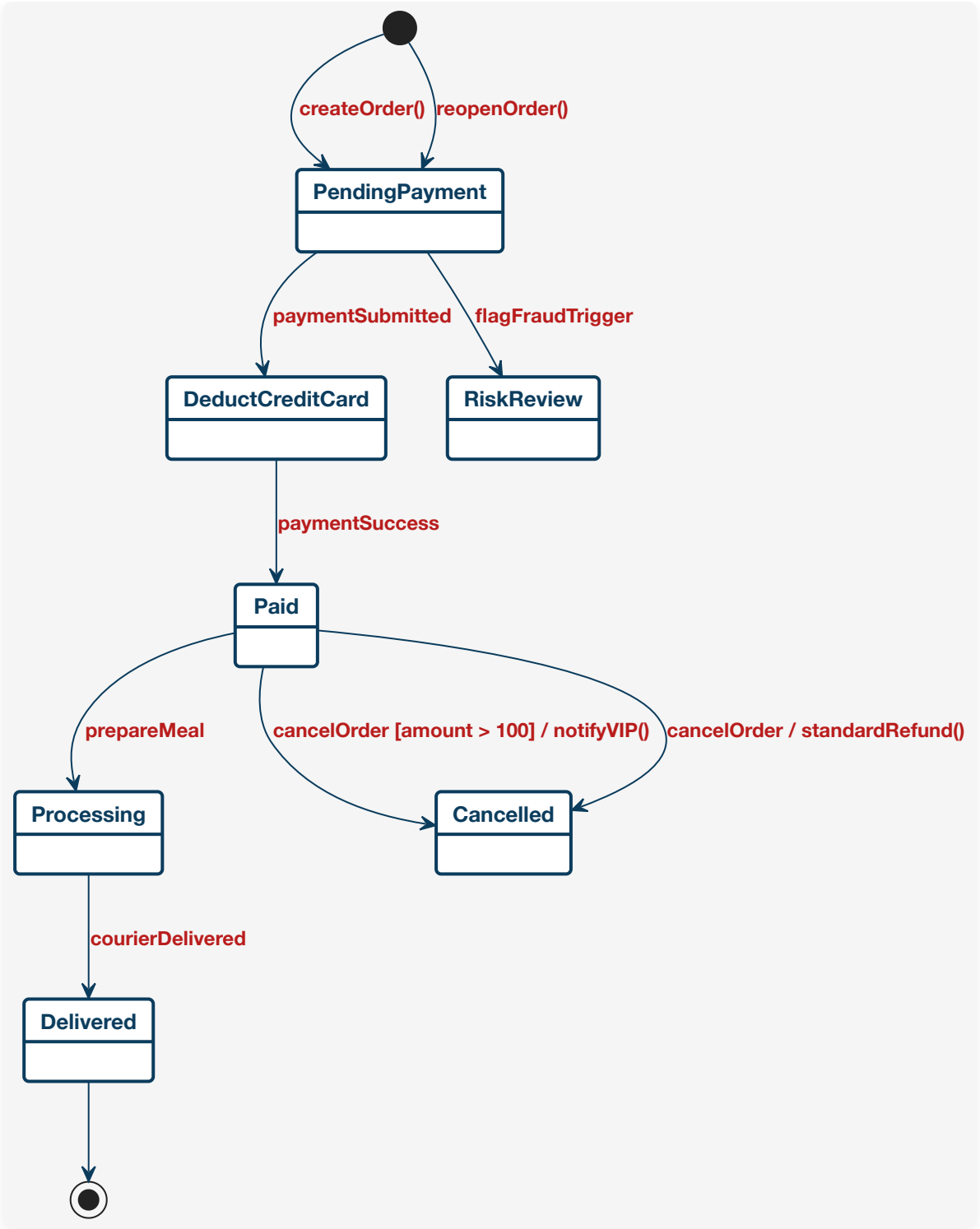


Figure 4.7.12: Flawed State Machine Model Example. A draft order lifecycle exhibiting 4 critical behavioral anti-patterns.

Diagnostic Architectural Audit: 4 Critical Flaws Identified

1. **State vs. Transient Action:**
    - *The Flaw:* A state rectangle is labeled `DeductCreditCard`.
    - *Correction:* A State represents a condition or enduring mode where an object waits for events or performs continuous work (`PendingPayment`, `Paid`). Deducting a card is an

instantaneous, atomic action executing in zero logical time. It must be modeled as a transition effect (`pay / deductCard()`), never as a state box.

2. **Black Hole (Trap) State:**
   - *The Flaw:* `RiskReview` has an incoming transition (`flagFraudTrigger`), but zero outgoing transitions.
   - *Correction:* Once an order enters `RiskReview`, it is trapped forever—unable to be approved, cancelled, or terminated. Every intermediate state must provide at least one exit transition.
3. **Non-Deterministic Transition Conflict:**
   - *The Flaw:* Two transitions leave `Paid` on the trigger `cancelOrder`: one has guard `[amount > 100]`, while the other has no guard.
   - *Correction:* If a cancellation request arrives for an order of $150, both transitions are valid, causing a race conflict. Transitions sharing the same event from a single state must have strictly mutually exclusive guards (e.g., `[amount > 100]` vs. `[amount <= 100]`).
4. **Illegal Outgoing Transition from a Final State:**
   - *The Flaw:* A transition `reopenOrder()` is drawn originating from the Final State `[*]`.
   - *Correction:* A Final State marks the irreversible termination of an entity's lifecycle. UML rules strictly forbid outgoing transitions from final states. If an order can be reopened, it was never truly terminated.

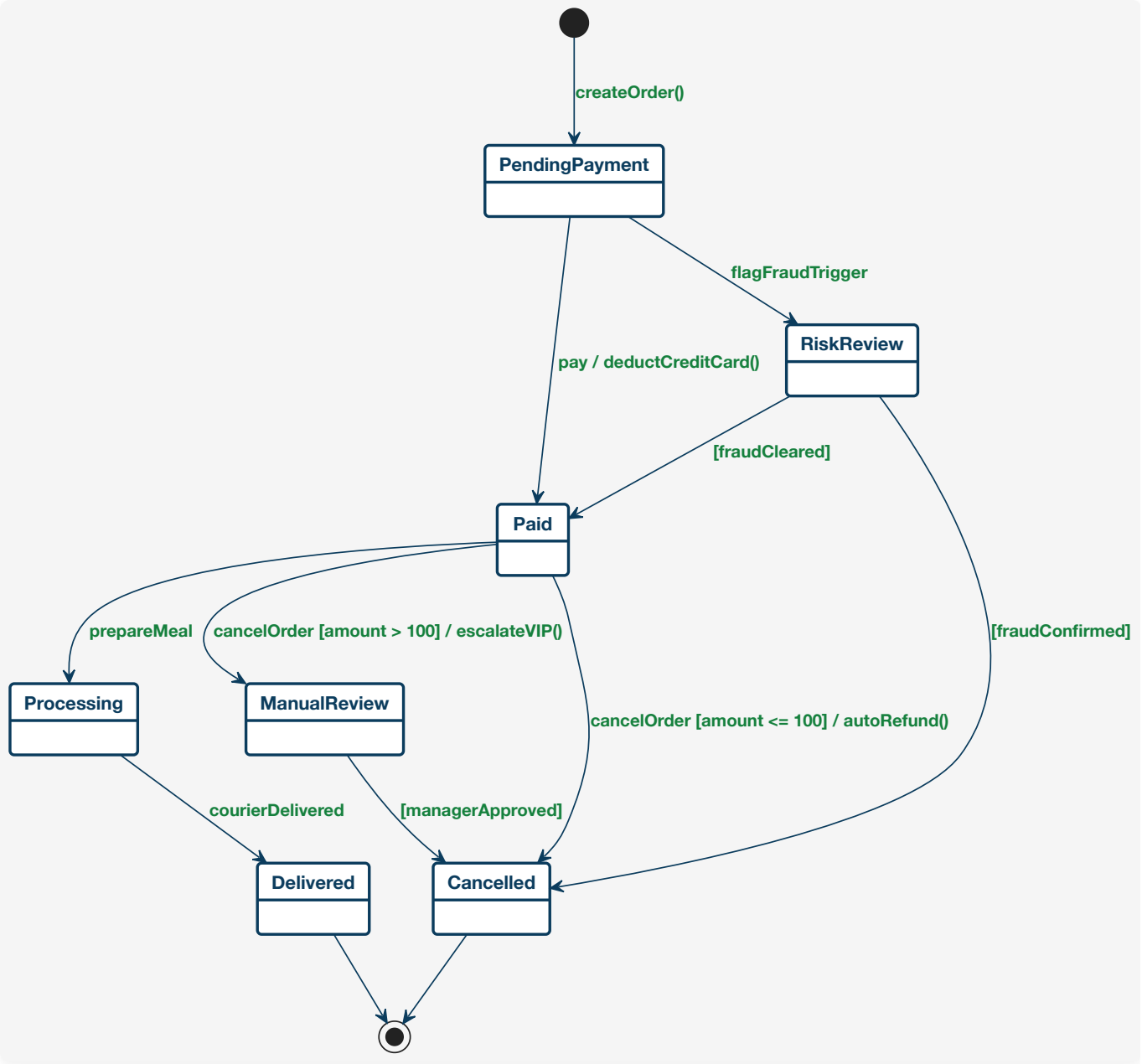


Figure 4.7.13: Corrected State Machine Model. Resolving actions into transition effects, eliminating trap states, enforcing deterministic guards, and respecting final state invariants.

4.7.7 Interactive Hands-On Activity: Evolving State Lifecycles with AI

- **Experiment Goal:** Mutate the order lifecycle by introducing one operational exception into the prompt and audit the AI's architectural output:
  i. *Option A (Courier Abandonment & Reassignment):* Courier abandons delivery; order transitions back to `ReadyForPickup` (`[retries < 3]`) without remaking food.
  ii. *Option B (Customer Reschedule):* Customer requests postponement while in `Preparing` .
  iii. *Option C (Delivery Failed & Return to Store):* Customer unreachable at doorstep; transition to `FailedDelivery` , then `ReturnedToStore` , and terminate.
- **Audit Checklist:**
  - Did AI maintain proper state naming (passive conditions/adjectives, no active verbs)?

- Are illegal transitions (e.g., direct cancellation while `OutForDelivery`) strictly blocked?
- Do exceptional branches cleanly lead to terminal state `[*]`?

👷 Concept Check (CCQ 7) — State Machine Actions vs. Activities

Question

In a UML State Machine, what is the critical architectural difference between an **Action** (such as `/ refundCharge()` or `entry / startTimer()`) and an **Activity** (`do / playAudio()`)?

- A. Actions are instantaneous, atomic, and non-interruptible; Activities are ongoing computations that take time and can be interrupted by incoming events.
- B. Actions are written in Python code; Activities are written in SQL queries.
- C. Actions only apply to class diagrams; Activities only apply to sequence diagrams.
- D. Actions represent manual human tasks; Activities are executed by database servers.

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.8 Text-Based Architecture Modeling with PlantUML

Historically, creating UML diagrams required heavy, proprietary GUI software (Rational Rose, Enterprise Architect, Microsoft Visio). These tools suffered from severe engineering bottlenecks: diagrams were stored in opaque binary blobs, could not be version-controlled in Git, and rapidly became obsolete as code evolved.

4.8.1 The "Architecture-as-Code" Paradigm

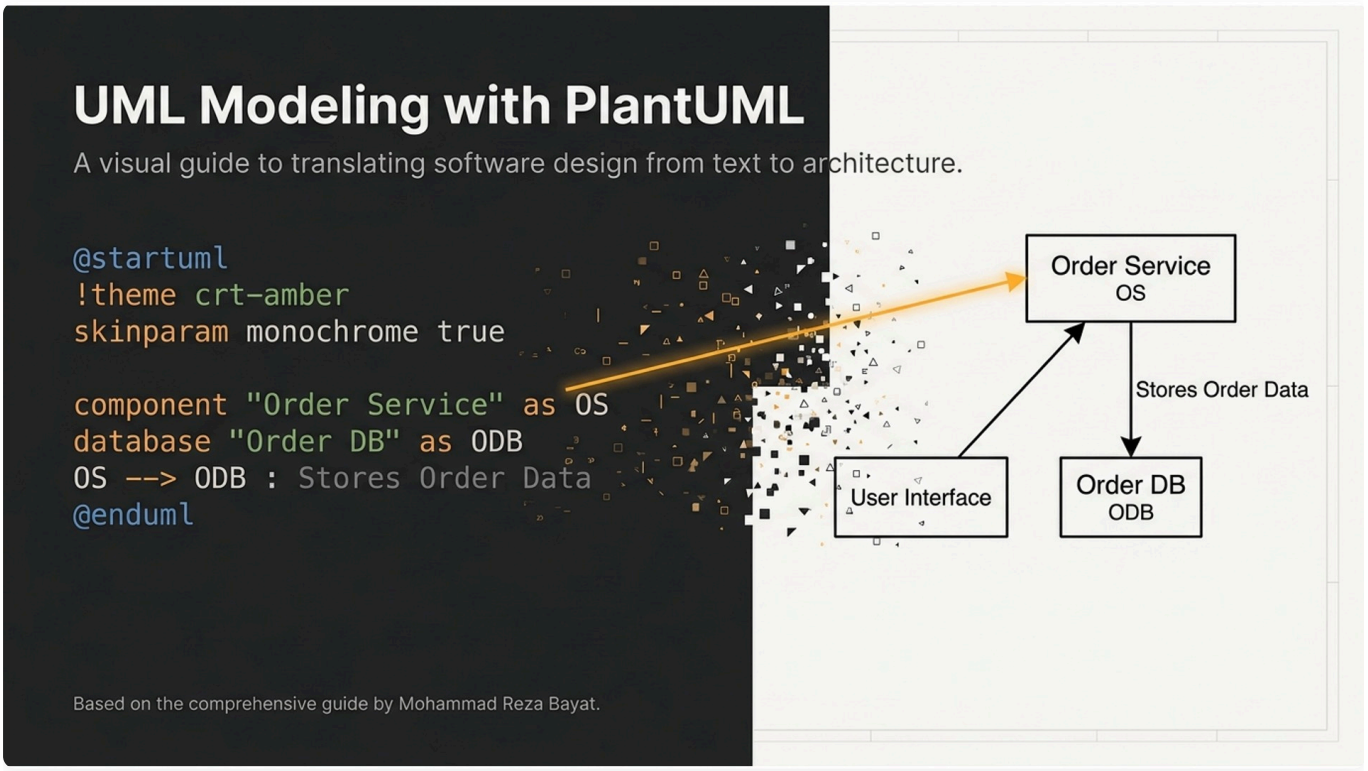


Figure 4.8.1: PlantUML Code-Driven Architecture. Defining visual architecture as plain text source code integrated with Git version control.

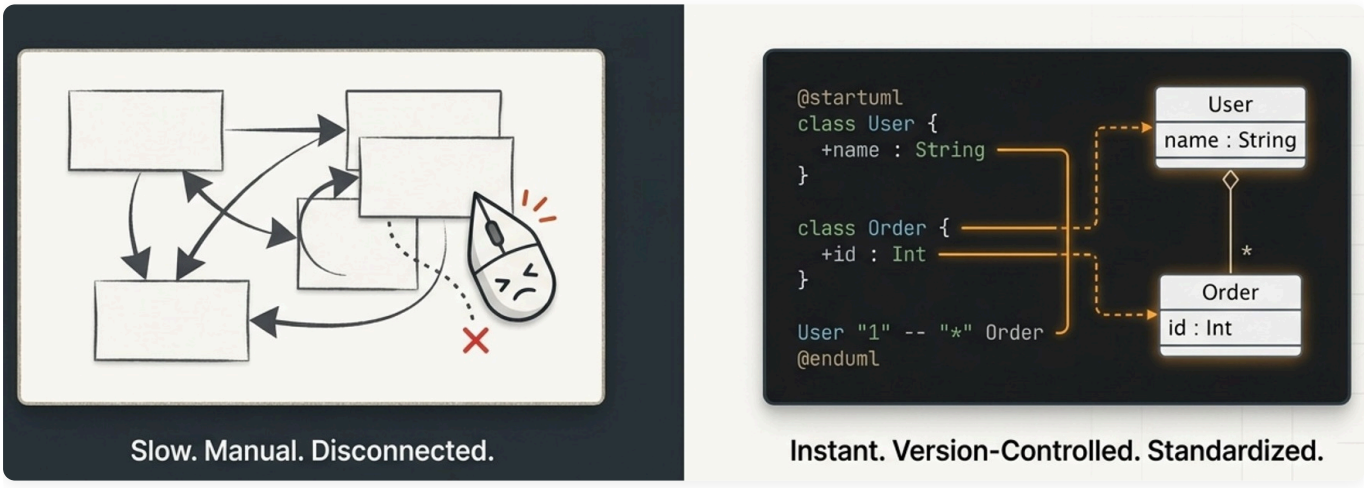


Figure 4.8.2: Stop Dragging, Start Writing. Contrasting fragile GUI drawing with declarative text markup.

Evaluation Dimension	Traditional GUI Drag-and-Drop	PlantUML (Architecture as Code)
Storage Format	Proprietary binary blobs or complex XML	Clean, human-readable plain text (`.puml` / `.md`)
Git Version Control	Constant merge conflicts, impossible diffs	Standard Git branch tracking, readable line diffs, PR reviews
CI/CD Integration	Manual PNG/PDF screenshot export	Automated headless compilation to SVG/PNG in GitHub Actions
LLM Synthesis	Incompatible with text-based AI models	Native direct generation by Claude, GPT, and Gemini

4.8.2 Cross-Diagram Syntax Matrix

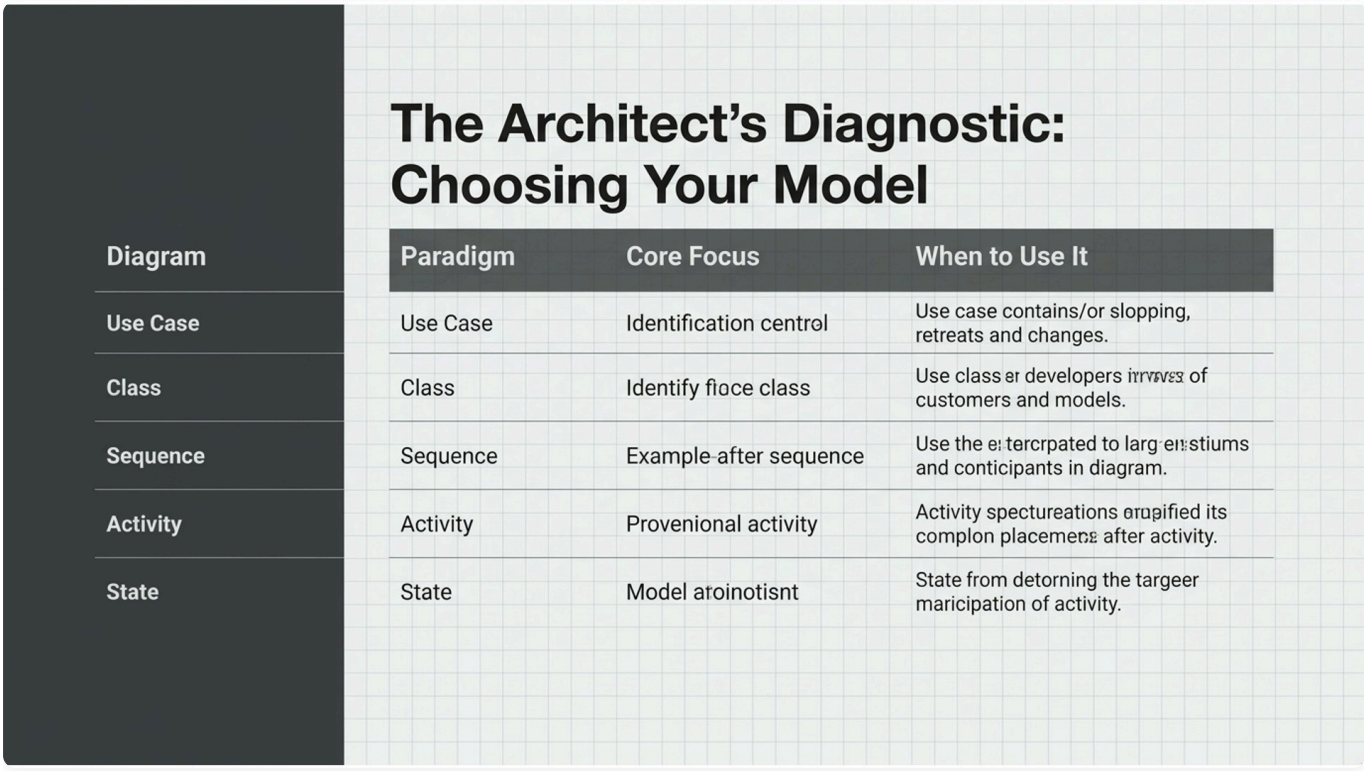


Figure 4.8.3: PlantUML Cross-Diagram Syntax Cheatsheet. Core keywords, arrow styles, and structural markers across all five UML models.

```
@startuml
skinparam handwritten false
skinparam monochrome true

title Food Delivery Interaction Snapshot

actor Customer
participant "CheckoutUI" as UI
participant "OrderController" as Ctrl
database "PostgreSQL" as DB

Customer -> UI : submitOrder()
activate UI
UI -> Ctrl : checkout(cartId)
activate Ctrl
Ctrl -> DB : saveOrder(order)
activate DB
DB --> Ctrl : orderId (201 Created)
deactivate DB
Ctrl --> UI : checkoutSuccess
deactivate Ctrl
UI --> Customer : Display Order Tracking Screen
deactivate UI
@enduml
```

🤦 Concept Check (CCQ 8) — Engineering Advantages of Text-Based Modeling

Question

What is the primary engineering advantage of using text-based diagramming tools like **PlantUML** over traditional proprietary GUI drawing software?

- A. PlantUML eliminates the need to understand software architecture and UML principles.
- B. Text markup can be version-controlled in Git, diffed in Pull Requests, and automated in CI/CD pipelines.
- C. PlantUML automatically generates full production databases and microservices without writing code.
- D. PlantUML is strictly limited to class diagrams and cannot model dynamic workflows.

Interactive Activity (線上作答)

▶ Click to view Answer & Explanation

4.9 Model Selection & Decision Framework

4.9.1 Architectural Decision Matrix

Core Modeling Tool	Primary Architectural Question	Key Visual Syntax Elements	Best Engineering Application
1. Use Case Diagram	**WHAT** business value does the system provide to users?	System Boundary rectangle, Actors, Use Case ovals, `<<include>>`, `<<extend>>`	Defining project scope, contract boundaries, and stakeholder agreements.
2. Class Diagram	**WHAT** static entities, contracts, and associations exist?	3-compartment boxes, Visibility (`+`, `-`, `#`), Multiplicity, Composition (`◆`), Aggregation (`◇`)	Domain modeling, database schema mapping, OOP class architecture.
3. Sequence Diagram	**HOW** do runtime objects collaborate over time?	Lifelines, Activation bars, Synchronous/Asynch calls, BCE roles, Combined fragments (`alt`, `loop`)	Tracing API transactions, microservice call graphs, and control orchestration.
4. Activity Diagram	**WHAT** is the multi-role operational workflow or algorithm?	Rounded Action nodes, Swimlane partitions, Decision/Merge diamonds, Fork/Join bars	Cross-department business processes, background job orchestration, concurrency.
5. State Machine	**HOW** does an entity change state in response to events?	Rounded state boxes, Triggers, Guards `[guard]`, Actions `/action`, Composite states, History `H`	Reactive protocols, IoT lifecycle states, order fulfillment tracking.
6. PlantUML Markup	**HOW** do we maintain models in Git without documentation rot?	Declarative text-to-diagram syntax (`@startuml ... @enduml`)	Architecture-as-Code, Git PR reviews, CI/CD automated documentation pipelines.

4.9.2 Solutions & Architectural Rubrics to All 5 Model Enhancement Activities

Activity Solution 1: Use Case Model Extensions

- **Architectural Solutions & Semantics:**
  - **Scheduled Ordering (`<<extend>>`):** `(Place Food Order) <.. (Schedule Future Delivery) : <<extend>>` with guard `[delivery_type == scheduled]`. An optional feature hooking into

checkout without polluting the base use case.
    - **Live Chat Support (Autonomous Base Case):** Standalone `(Live Order Chat)` where primary actor `Customer` communicates with secondary actors `Support Agent` and `Courier`. External actors remain strictly outside the system boundary.
    - **Group Ordering (`<<extend>>`):** `(Place Food Order)` extended by `(Pool Group Cart)` before entering checkout.
- **PlantUML Delta:**

```
rectangle "Food Delivery Platform" {
  usecase "Place Food Order" as UC_Order
  usecase "Schedule Future Order" as UC_Sched
  usecase "Live Order Chat" as UC_Chat
  UC_Order <.. UC_Sched : <<extend>>
}
Customer --> UC_Order
Customer --> UC_Chat
UC_Chat --> Agent : <<assists>>
```

- **Audit Checklist:**
    - `[Pass]` External partner `Support Agent` resides outside system boundary.
    - `[Pass]` Scheduled delivery uses `<<extend>>`, not mandatory `<<include>>`.
    - `[Pass]` Bubbles represent business user goals, avoiding low-level UI button clicks.

Activity Solution 2: Class Model Refinements

- **Architectural Solutions & Semantics:**
    - **Multi-Vendor Sub-Orders:** Master `Order` decomposes into restaurant-specific sub-orders. `Order` `"1"` `*--` `"1..*"` `SubOrder` (Composition: sub-orders cannot exist without parent order). `SubOrder` `"1..*"` `o--` `"1"` `Restaurant` (Aggregation: restaurant survives order deletion).
    - **Courier Tipping & Rating:** `DeliveryTask` `"1"` `o--` `"0..1"` `CourierReview` (Aggregation; tipping/rating is optional).
    - **VIP Membership Plan:** `Customer` `"1"` `o--` `"0..1"` `MembershipPlan` (Strategy pattern decoupling subscription tiers).
- **PlantUML Delta:**

```
class Order {
  - orderId: UUID
  - total: Money
}
class SubOrder {
  - subTotal: Money
  - status: Status
}
class Restaurant {
  - name: String
  - address: String
}
class CourierReview {
  - rating: int
  - tip: Money
}
Order "1" *-- "1..*" SubOrder : aggregates
SubOrder "1..*" o-- "1" Restaurant : fulfills
SubOrder "1" *-- "1..*" OrderItem
DeliveryTask "1" o-- "0..1" CourierReview
```

- **Audit Checklist:**
  - `[Pass]` Multiplicity logic: 1 master order contains `1..*` sub-orders; 1 sub-order targets 1 restaurant.
  - `[Pass]` Solid diamond (`*--`) for lifecycle dependency; hollow diamond (`o--`) for weak associations.
  - `[Pass]` Strict private encapsulation (`-`) on attributes.

Activity Solution 3: Sequence Interaction Edge Cases

- **Architectural Solutions & Semantics:**
  - **Payment Gateway Timeout (>5s):** `CheckoutUI` calls `OrderController.submitPayment()`. Controller invokes `PaymentGateway.processPayment()`. `alt [Gateway Response < 5s]`: Normal confirmation. `else [Timeout > 5s]`: Controller triggers `cancelTransaction()`, logs audit trail, and returns `PaymentTimeoutError` with retry options.
  - **Split Payment Workflow:** Two-phase execution: debit `Wallet Balance` first; if successful, charge remainder to credit card. If card declines, compensating transaction refunds wallet.
- **PlantUML Delta:**

```
UI -> Controller : submitOrder(cart)
activate Controller
Controller -> Gateway : charge(amount)
activate Gateway
alt #LightGreen Response within 5s
  Gateway --> Controller : 200 OK (txnId)
  Controller -> Order : markPaid()
  Controller --> UI : OrderSuccess
else #Pink Timeout > 5s
  Gateway --> Controller : [Timeout Error]
  Controller -> Gateway : voidTransaction()
  Controller --> UI : PaymentFailed(retryOption)
end
deactivate Gateway
```

- **Audit Checklist:**
  - `[Pass]` UI communicates exclusively with Controller, never directly accessing DB or external Gateway.
  - `[Pass]` Mutually exclusive paths enclosed in `alt` with clear guard expressions.

Activity Solution 4: Activity Workflow Evolution

- **Architectural Solutions & Semantics:**
  - **Courier Decline & Re-dispatch Loop:** Inside `|Dispatch Engine|` lane: query nearest couriers in a `repeat` loop. Dispatch notification passes control token to `|Courier|` lane. Decision diamond: `if (Accept?) then (yes) -> Confirm; else (no) -> Log Decline; repeat while (Couriers Available)`.
  - **Customer Self-Pickup Bypass:** Decision diamond in `|Platform|`: `if (Order Type?) then (Delivery) -> Fork to Kitchen & Courier; else (Self-Pickup) -> Notify Kitchen only; |Customer| :Pick Up at Counter; -> Merge`.
- **PlantUML Delta:**

```
|Dispatch System|
repeat
  :Find Available Courier;
  |Courier|
  :Receive Dispatch Ping;
backward:Log Rejection;
repeat while (Courier accepts?) is (no)
->yes;
:Confirm Courier Assignment;
|Kitchen|
:Hand Off Sealed Package;
|Courier|
:Deliver to Customer;
stop
```

- **Audit Checklist:**
  - `[Pass]` Every action node resides in its responsible actor/system lane.

- `[Pass]` `repeat...repeat while` constructs have definitive termination paths.
- `[Pass]` All terminating branches reach explicit `stop` nodes.

Activity Solution 5: State Machine Lifecycle Exceptions

- **Architectural Solutions & Semantics:**
  - **Courier Abandonment & Reassignment:** `OutForDelivery → ReadyForPickup : courier_abandoned [retries < 3] / unassign()`. Re-triggers courier matching without forcing kitchen to re-cook meal.
  - **Delivery Failed & Return to Store:** `OutForDelivery → FailedDelivery : unreachable [wait >= 10m] / markFailed()`. `FailedDelivery → ReturnedToStore : return_package / recordDisposal()`. Terminal transition: `ReturnedToStore → [*]`.
  - **Strict Invariant Enforcement:** Direct transition `OutForDelivery → Cancelled` is forbidden without compensating penalties.
- **PlantUML Delta:**

```
state OutForDelivery {
  [*] --> InTransit
}
OutForDelivery --> ReadyForPickup : courier_dropped \
  [retries < 3] / reassign()
OutForDelivery --> FailedDelivery : unreachable \
  [wait >= 10m] / logFailed()
FailedDelivery --> ReturnedToStore : return_meal \
  / auditWaste()
ReturnedToStore --> [*]
```

- **Audit Checklist:**
  - `[Pass]` States represent passive conditions (`FailedDelivery`, `ReturnedToStore`), never active verbs.
  - `[Pass]` Transitions declare Event `trigger`, `[guard]`, and `/action`.
  - `[Pass]` Exceptional flows lead to explicit terminal state `[*]`.

4.9.3 Comprehensive Conceptual Review: 14 Fill-in-the-Blank Quizzes

```
📁 Word Bank:
[ Interaction ]
[ Ivar Jacobson ]
[ Boundary-Control-Entity (BCE) ]
[ <<include>> ]
[ <<extend>> ]
[ Composition (◆) ]
[ Aggregation (◇) ]
[ Solid Arrow (→) ]
[ alt fragment ]
[ Swimlanes ]
[ Join Node ]
[ Guard Condition ]
[ Orthogonal Regions (||) ]
[ History State [H*] ]
```

Part 1: Structural & Use Case Foundations

1. The "Three Amigos" who unified UML at Rational Software were Grady Booch, Jim Rumbaugh, and **[Ivar Jacobson]**.
2. In system modeling, the 4 foundational perspectives are External, **[Interaction]**, Structural, and Behavioral.
3. In use case modeling, mandatory shared functionality factored out across use cases uses the **[`<<include>>`]** relationship.
4. Optional or conditional behavior triggered under specific conditions connects via the **[`<<extend>>`]** relationship.
5. The architectural analysis pattern separating user interfaces, business logic, and persistent domain entities is **[Boundary-Control-Entity (BCE)]**.
6. In a UML class diagram, a filled black diamond (◆) represents **[Composition (◆)]**, where part lifetimes strictly depend on the whole.
7. In a UML class diagram, an open hollow diamond (◇) represents **[Aggregation (◇)]**, where parts maintain independent lifecycles.

Part 2: Dynamic Workflows & Statecharts

8. In a sequence diagram, a synchronous call that blocks execution until the target responds is drawn with a **[Solid Arrow (→)]**.
9. In a sequence diagram, conditional branching with mutually exclusive execution alternatives is enclosed in an **[alt fragment]**.
10. In an activity diagram, responsibilities of departments, business roles, or architectural tiers are partitioned by **[Swimlanes]**.
11. In an activity diagram, a thick synchronization bar that merges multiple concurrent execution paths before continuing is a **[Join Node]**.

12. In a UML state machine transition label, the boolean check condition enclosed in square brackets `[ ]` is the **[Guard Condition]**.
13. In a composite state, concurrent parallel sub-states operating side-by-side are modeled inside **[Orthogonal Regions (||)]**.
14. The pseudo-state that caches and restores the most recently active sub-state upon recovery from an interruption is the *[History State [H]]**.

4.9.4 Classic Literature & Further Reading

- **Booch, G., Rumbaugh, J., & Jacobson, I. (2005).** *The Unified Modeling Language User Guide* (2nd ed.). Addison-Wesley.
- **Fowler, M. (2003).** *UML Distilled: A Brief Guide to the Standard Object Modeling Language* (3rd ed.). Addison-Wesley.
- **Cockburn, A. (2000).** *Writing Effective Use Cases*. Addison-Wesley.
- **Object Management Group (OMG). (2017).** *OMG Unified Modeling Language (OMG UML) Specification*, Version 2.5.1.
- **PlantUML Language Reference Manual:** plantuml.com

Appendix: Solutions & Explanations to All Interactive Concept Checks (CCQ 1–8)

CCQ 1: Four System Modeling Perspectives

- **Correct Answer:** C (Structural Perspective)
- **Detailed Explanation:** Defining how domain entities are organized and how classes inherit and associate with each other—completely independent of execution timing—is the exact definition of the Structural Perspective. The External Perspective models context/boundaries; the Interaction Perspective models runtime message exchanges; the Behavioral Perspective models state transitions and dynamic workflow execution.

CCQ 2: UML Pioneers & The Three Amigos

- **Correct Answer:** C (Ivar Jacobson)
- **Detailed Explanation:** Ivar Jacobson invented Use Cases in 1986 as part of Object-Oriented Software Engineering (OOSE) to ground system requirements in user goals. Grady Booch contributed the Booch Method (macro object design), Jim Rumbaugh developed OMT (object modeling technique with rich associations), and Martin Fowler authored the renowned reference *UML Distilled*.

CCQ 3: Use Case Relationships (Include vs. Extend)

- **Correct Answer:** B (Vouchers are conditional optional behavior; payment is mandatory shared execution)
- **Detailed Explanation:** `<<include>>` signifies mandatory shared behavior that must execute for the base use case to finish (an order cannot be placed without payment authorization). In contrast, `<<extend>>` denotes optional, conditional behavior that injects into the base case only when triggered at an extension point (applying a promo voucher is optional).

CCQ 4: Class Model Relationships (Composition vs. Aggregation)

- **Correct Answer:** B (An OrderItem is destroyed with its parent Order, whereas a Courier maintains an independent lifecycle)
- **Detailed Explanation:** Composition (filled diamond ◆) represents strong part-whole ownership where parts cannot exist without the whole; if an `Order` is deleted, its individual `OrderItem` records cease to have meaning and are destroyed. Aggregation (hollow diamond ◇) represents shared association where parts maintain independent lifecycles (a `Courier` exists before and after delivering an order).

CCQ 5: BCE Architecture & Sequence Message Dispatch

- **Correct Answer:** C (The OrderController control object to orchestrate business validation and service calls)
- **Detailed Explanation:** The Boundary-Control-Entity pattern dictates strict separation of concerns. UI boundary objects must never directly manipulate database entities or call external gateways. Instead, they delegate to Control objects (`OrderController`) to orchestrate validation, entity coordination, and external boundary dispatch.

CCQ 6: Activity Diagram Fork/Join vs. Decision/Merge

- **Correct Answer:** A (Fork/Join splits and synchronizes concurrent parallel threads; Decision/Merge evaluates guards to pick exactly one mutually exclusive branch)
- **Detailed Explanation:** A Fork node splits a single control flow into multiple parallel, concurrent threads (e.g. food prep and courier dispatch simultaneously), and Join waits for all threads to arrive. A Decision diamond evaluates Boolean guards `[guard]` to choose exactly one mutually exclusive alternative path.

CCQ 7: State Machine Actions vs. Activities

- **Correct Answer:** A (Actions are instantaneous, atomic, and non-interruptible; Activities are ongoing computations that take time and can be interrupted by incoming events)
- **Detailed Explanation:** In UML state machine semantics, an Action (associated with transitions, `entry/` , or `exit/`) takes conceptually zero time, is atomic, and cannot be interrupted. An Activity (`do/ activity`) represents ongoing, durational execution that is immediately aborted if an outgoing transition event occurs.

CCQ 8: Engineering Advantages of Text-Based Modeling

- **Correct Answer:** B (Text markup can be version-controlled in Git, diffed in Pull Requests, and automated in CI/CD pipelines)
- **Detailed Explanation:** Text-based modeling ("Architecture-as-Code") represents diagrams in plain text markup, enabling seamless Git version control, visual Pull Request diffing, CI/CD automated doc builds, and native LLM code generation without vendor lock-in.